3D Spare Parts Catalogs in Manufacturing

Improving Operational Efficiency and Customer Experience

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Abstract

This thesis investigates the implementation of 3D visualization technologies for industrial spare parts catalogs, addressing technical challenges of legacy data integration and enterprise system connectivity. The research question asks: How can manufacturing companies implement 3D visualization technologies for industrial spare parts catalogs while addressing the specific technical challenges of legacy data integration and enterprise system connectivity?

The investigation follows Design Science Research methodology, developing and evaluating a functional prototype system at Signifikant Svenska AB. The research examines three sub-questions addressing architectural patterns for deploying 3D visualization within existing ERP systems (RQ1), technical challenges in converting legacy CAD data to web-optimized formats (RQ2), and system performance in addressing limitations of traditional 2D documentation (RQ3).

The research employed automated conversion pipeline transforming legacy STEP CAD files to web-optimized glTF format, hybrid YOLOv8+EasyOCR system for extracting metadata from legacy 2D technical drawings, browser-based 3D rendering using Three.js and WebGL tested across desktop, tablet, and smartphone devices, and performance evaluation measuring conversion success rates, OCR precision, loading times, and frame rates.

Results demonstrate technical implementation feasibility: 100% conversion success rate, 99.3% OCR precision, acceptable rendering performance (31-58 FPS) across device categories, and load times under 5 seconds over simulated 3G networks. Three architectural patterns emerged: stateless client design, metadata embedding within geometry files, and event-driven bidirectional synchronization.

Critical limitations: evaluation focused on technical implementation feasibility only (not business value), limited samples preventing broad generalization, conversion success rates specific to STEP archives, and claimed operational benefits were not empirically measured. The thesis contributes

quantified conversion success boundaries, browser-specific performance characterization, two-stage OCR architecture advantages, and reusable architectural patterns for industrial ERP-3D integration.

Keywords: 3D visualization, industrial spare parts catalogs, legacy CAD data conversion, Design Science Research, ERP integration, browser rendering performance, OCR technical drawings

Acknowledgements

This is the place to say thanks to those who have supported the thesis work.

Abbreviations

Abbreviation	Description
AI	Artificial Intelligence
NLP	Natural Language Processing
ML	Machine Learning
IoT	Internet of Things
GPU	Graphics Processing Unit
CPU	Central Processing Unit
API	Application Programming Interface

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1

Introduction

In the industrial manufacturing sector, spare parts management represents a critical component of after-sales service and operational maintenance. Traditionally, manufacturing companies have relied on printed or PDF-based spare parts catalogs that use 2D technical drawings, photographs, and extensive part numbering systems to help customers and technicians identify and order the correct components. This approach, while functional, presents numerous challenges including difficulty in part identification, visualization limitations, and inefficient search processes [38].

The emergence of advanced 3D visualization technologies offers a promising solution to these challenges. Three-dimensional digital spare parts catalogs enable users to interact with virtual representations of complex machinery, zoom in on specific components, explode assemblies into constituent parts, and precisely identify needed replacements. This technology has evolved from basic CAD models to sophisticated, interactive 3D representations that accurately depict the physical characteristics and relationships of components within complex machinery [23].

Research in 3D visualization for product catalogs has demonstrated significant potential across various domains. In consumer e-commerce, Li et al. [25] conducted pioneering work showing that 3D product presentations significantly enhanced user product knowledge and purchase intentions compared to traditional 2D presentations. Jiang and Benbasat [18] extended this research by identifying that interactive control over viewpoint and multi-angle examination were key factors in improving user confidence and reducing product uncertainty. More recently, Algharabat and

Dennis [1] investigated 3D visualization for electrical components in business-to-business contexts, finding that authentic 3D representations significantly improved identification accuracy and reduced ordering errors.

From a technical implementation perspective, substantial research has addressed the foundations of web-based 3D visualization. Kang and Peng [21] developed comprehensive frameworks for integrating CAD models with product assembly planning in web-based environments, demonstrating successful extraction of geometrical data from CAD models and achieving significant improvements in assembly planning efficiency. Similarly, Dai et al. [8] created interactive web systems for three-dimensional customization, successfully implementing solutions across apparel, automobile, and mobile phone industries while integrating with PDM/PLM systems.

1.1 Problem Statement

Despite the demonstrated benefits of 3D visualization in consumer applications and the established technical feasibility of web-based 3D systems, a critical knowledge gap exists regarding the practical implementation of 3D visualization for industrial spare parts management. While consumer e-commerce research has shown clear user benefits and technical research has proven implementation viability, no systematic investigation has documented the unique technical challenges and implementation requirements of industrial spare parts catalog systems.

Industrial spare parts catalogs present technical challenges that differ fundamentally from consumer product applications. Legacy CAD data accumulated over 20–30 years exists in diverse formats (STEP, Creo, SolidWorks, CATIA) with inconsistent metadata practices and varying documentation standards across different engineering teams and time periods. Complex multi-level assemblies with 100–1000+ components require preservation of hierarchical relationships through format conversion while maintaining associations between geometric components and enterprise part identifiers. Integration with existing ERP systems demands bidirectional synchronization between visual 3D selections and business logic including inventory status, pricing information, and order processing workflows.

Furthermore, field deployment constraints require optimization for low-specification devices including older tablets and smartphones used by field service technicians, variable network conditions ranging from high-speed facility networks to remote 4G connections, and cross-browser compatibility without requiring plugin installations. Fragmented documentation

sources—including XML metadata files, 3D geometry files, and legacy 2D technical drawings—must be reconciled into unified part records when information exists in multiple locations with potential conflicts.

Traditional spare parts catalogs rely heavily on text-based descriptions, part numbers, and 2D technical illustrations that often fail to adequately convey the spatial relationships between components in today's increasingly complex machinery. This limitation frequently leads to misidentification of parts, ordering errors, and increased downtime for equipment maintenance [38]. For complex industrial machinery with thousands of components and integrated IoT systems, 2D representations make it difficult for technicians and customers to accurately identify specific parts, especially those hidden within assemblies or those with subtle differences from similar components.

The organizational and technical complexity of implementing 3D visualization in large manufacturing companies remains largely unexplored in existing research. No systematic investigation has documented the architectural patterns necessary for successful deployment, the conversion pipeline challenges when processing real legacy data, or the performance optimization strategies required for field use on diverse device capabilities.

1.2 Research Question

Given the identified knowledge gap between demonstrated 3D visualization benefits and the unexplored technical implementation requirements for industrial contexts, this research addresses the following question:

> *How can manufacturing companies implement 3D visualization technologies for industrial spare parts catalogs while addressing the specific technical challenges of legacy data integration and enterprise system connectivity?*

This research question focuses specifically on the implementation knowledge gap identified in literature. Rather than comparing 3D versus 2D approaches (which has been established in consumer contexts), this study investigates the practical implementation aspects that remain undocumented in industrial settings. By examining real-world implementation through a design science approach, this research will document architectural patterns, data conversion strategies, and integration methods that enable manufacturing companies to successfully deploy 3D visualization technologies within their existing spare parts management infrastructure.

1.2.1 Supporting Research Questions

To comprehensively address the main research question, this study investigates three specific sub-questions:

RQ1 (System Architecture & Integration) What architectural patterns, integration strategies, and interaction mechanisms are required for successfully deploying interactive 3D visualization within existing ERP spare parts catalogs?

This sub-question addresses the fundamental system design decisions including client-server architecture organization, integration patterns for connecting with ERP business logic, part selection and highlighting mechanisms using raycasting technology, and bidirectional synchronization between 3D visual selections and database records.

RQ2 (Data Processing & Reconciliation) What technical challenges arise when converting legacy CAD data to web-optimized formats and reconciling fragmented documentation from multiple sources, and how can they be addressed?

This sub-question examines the conversion pipeline from engineering CAD formats to web-ready 3D models, multi-source data reconciliation combining information from 3D geometry files, XML metadata, and legacy 2D drawings extracted via OCR, metadata preservation through format conversion, and conflict resolution when different sources provide contradicting information.

RQ3 (Performance & Effectiveness) How does the implemented 3D visualization system address the specific limitations of traditional 2D documentation while meeting industrial performance constraints?

This sub-question evaluates how the system addresses identified 2D limitations including spatial understanding, hidden component identification, assembly context, and hierarchical navigation, while achieving performance requirements for loading time, frame rate, and cross-platform compatibility suitable for field deployment.

1.2.2 Research Objectives

To answer these research questions, this study pursues the following objectives:

1. Evaluate 3D file formats and rendering technologies for industrial spare parts visualization, identifying trade-offs between performance, feature support, and integration complexity
2. Develop and document a CAD-to-web conversion pipeline that preserves assembly hierarchy and metadata while optimizing for web delivery
3. Identify integration patterns for connecting 3D visualization with ERP systems, addressing challenges in data synchronization and workflow coordination
4. Quantify performance characteristics and optimization strategies for field deployment on diverse device capabilities and network conditions
5. Document technical challenges encountered during implementation and evaluate solutions for data reconciliation, metadata preservation, and system integration

1.3 Contribution Statement

This research contributes to the intersection of 3D visualization technology and industrial information systems by:

1. *Technical Requirements Documentation:* Providing the first systematic documentation of architectural patterns, integration strategies, and performance requirements for industrial 3D spare parts catalog implementation
2. *Conversion Pipeline Validation:* Developing and validating a multi-stage conversion pipeline addressing legacy CAD format challenges, with quantified success rates and processing characteristics across 847 real-world industrial files
3. *Data Reconciliation Framework:* Creating a multi-source reconciliation approach combining 3D geometry extraction, XML metadata parsing, and OCR-based annotation extraction from legacy 2D drawings to create unified part records
4. *Integration Pattern Identification:* Documenting three reusable integration patterns (bidirectional synchronization, lazy loading with

progressive detail, metadata reconciliation layer) for connecting 3D visualization with ERP systems

5. *Performance Optimization Knowledge:* Quantifying optimization trade-offs between file size, loading time, visual quality, and rendering performance through systematic evaluation of format choices and optimization strategies

6. *Implementation Knowledge for Practitioners:* Creating a reference implementation demonstrating technical feasibility for companies with messy legacy data, providing practical guidance for real-world deployments

Unlike prior research focusing on consumer applications or theoretical CAD integration, this work addresses the authentic complexity of industrial deployments: inconsistent legacy data accumulated over decades, integration with existing enterprise infrastructure, and field deployment on diverse device capabilities.

1.4 Scope and Delimitations

This research focuses on the technical implementation aspects of 3D spare parts catalogs rather than user performance evaluation or organizational change management. The study examines:

Within Scope

- Technical requirements and architectural decisions for 3D visualization deployment
- Format conversion challenges and solutions when processing legacy CAD data
- System integration patterns and approaches for connecting with ERP platforms
- Performance characteristics and optimization strategies for web-based 3D rendering
- Implementation challenges with real industrial data including inconsistency and fragmentation

Outside Scope

- Controlled user experiments comparing 2D versus 3D identification accuracy
- Quantitative user performance metrics (task completion time, error rates)
- Organizational change management or technology adoption strategies
- Detailed cost-benefit analysis or financial ROI modeling
- Comprehensive coverage of all possible CAD formats or industry sectors

The research is conducted within the context of a production deployment for Signifikant, a Swedish company providing web-based spare parts management platforms to manufacturing companies in automotive, industrial machinery, and farming equipment sectors. While findings are likely transferable to similar industrial contexts, specific technical details reflect this deployment environment.

The study concentrates on large manufacturing companies with complex equipment portfolios and established enterprise system infrastructures. While smaller manufacturers may benefit from similar technologies, the specific organizational and technical challenges addressed reflect the complexity of large-scale industrial implementations with decades of accumulated product data.

1.5 Thesis Structure

The remainder of this thesis is organized as follows:

Chapter 2 provides comprehensive background on spare parts management evolution, 3D visualization technologies, and relevant literature establishing the research foundation.

Chapter 3 details the research methodology including the design science approach, technology evaluation procedures, system architecture development, and data collection methods.

Chapter 4 presents research findings organized by research question, including format evaluation results, conversion pipeline performance, integration challenges, and system effectiveness validation.

Chapter 5 discusses the interpretation of findings, comparison with prior research, practical implications, and identified limitations.

Chapter 6 concludes with summary of contributions, answers to research questions, implications for theory and practice, and future research directions.

2

Background

This chapter provides comprehensive background to the research domain of 3D spare parts catalogs in manufacturing environments. It examines the evolution of spare parts management systems, explores the technological foundations of 3D visualization, and reviews contemporary literature on implementation in industrial contexts. The chapter establishes the theoretical and practical context necessary for understanding the significance of this research.

2.1 Evolution of Spare Parts Management Systems

Spare parts management has undergone significant transformation over the past decades, evolving from purely mechanical inventory systems to sophisticated digital platforms integrated with enterprise-wide information systems [23]. Understanding this evolution provides context for appreciating the potential impact of 3D visualization technologies on current practices.

2.1.1 Traditional Approaches to Spare Parts Catalogs

Historically, spare parts catalogs have served as the primary interface between equipment manufacturers and their customers for identifying and ordering replacement components. Early catalog systems relied entirely on printed materials, featuring detailed technical drawings, part photographs, and comprehensive part numbering schemes [3]. These systems, while

comprehensive in their coverage, presented significant limitations in terms of accessibility, currency, and user experience.

Technical drawings in printed catalogs required users to interpret complex orthographic projections, sectional views, and exploded assembly diagrams. This interpretation process demanded technical expertise and familiarity with engineering drawing conventions, creating barriers for less experienced users. Part photographs provided more intuitive visual information but were limited to external views, making identification of internal or obscured components difficult.

Part numbering systems evolved to encode information about component characteristics, manufacturing variants, and supersession relationships. However, these systems varied significantly across manufacturers and product lines, requiring users to learn multiple numbering conventions. Cross-referencing between different numbering systems when dealing with multi-manufacturer equipment portfolios added additional complexity.

The transition from printed to digital formats in the 1990s and early 2000s represented the first major technological shift in spare parts catalog systems. Digital catalogs offered improved searchability through full-text search capabilities, reduced distribution costs by eliminating printing and shipping, the ability to provide real-time updates to part information and pricing, and integration possibilities with electronic ordering systems [17].

However, these early digital systems largely replicated the structure and limitations of their printed predecessors. They maintained reliance on 2D representations—scanned technical drawings or photographed components—and hierarchical navigation structures based on assembly breakdowns. The fundamental challenge of visualizing spatial relationships and identifying components within complex assemblies remained largely unaddressed.

Contemporary research has identified several persistent challenges with traditional catalog approaches. Skoumpopoulou et al. [38] note that conventional systems often fail to provide adequate visual context for complex assemblies, leading to identification errors and increased maintenance downtime. Field technicians report spending significant time cross-referencing multiple catalog pages, technical manuals, and physical equipment to confirm component identity before ordering.

Similarly, Wagner and Ullrich [43] found that traditional catalog systems contribute to inefficiencies in maintenance workflows, particularly when dealing with equipment from multiple manufacturers or vintage machinery with limited documentation. The lack of standardization across

different manufacturers' catalog systems forces maintenance teams to maintain familiarity with diverse information architectures and part numbering schemes.

2.1.2 Digital Transformation in Spare Parts Management

The advent of Industry 4.0 technologies has fundamentally altered the landscape of spare parts management. Kulshrestha et al. [23] conducted a comprehensive literature review examining the integration of digital technologies with spare parts management processes, identifying key areas where technological innovation has demonstrated significant potential for improvement including demand forecasting, inventory optimization, supplier integration, and predictive maintenance.

Modern spare parts management systems now incorporate various digital technologies creating new possibilities for visualization integration. Internet of Things (IoT) sensors enable predictive maintenance by monitoring equipment condition and triggering parts orders before failures occur [47]. Artificial intelligence analyzes historical maintenance data to forecast demand patterns and optimize inventory levels. Blockchain technologies provide supply chain transparency and authenticity verification for critical components.

The concept of digital twins has emerged as particularly relevant to spare parts management. Digital twin technologies create virtual replicas of physical equipment that are updated in real-time based on sensor data and operational parameters [36]. This approach enables more precise identification of required parts by maintaining current equipment configuration including installed component versions, usage hours, and maintenance history. Digital twins can simulate component wear patterns and predict remaining useful life, triggering proactive parts ordering.

Integration of digital twins with 3D visualization creates opportunities for enhanced spare parts identification. Users can view the current state of their specific equipment instance rather than generic catalog models, seeing actual installed components and configurations. However, creating and maintaining digital twins requires substantial investment in sensor infrastructure, data management capabilities, and modeling expertise.

Cloud computing platforms have transformed the technical architecture possibilities for spare parts catalog systems. Cloud-based delivery enables more sophisticated 3D content delivery through content delivery networks (CDNs) distributing models close to users globally, real-time synchronization with enterprise systems ensuring current pricing and availability, global accessibility across diverse device platforms without client-side software installation, and elastic computing resources scaling to handle peak usage periods [33].

However, cloud deployment also introduces challenges including network dependency where catalog access requires reliable internet connectivity, data security concerns when hosting proprietary CAD models externally, compliance requirements with data sovereignty regulations in different jurisdictions, and subscription cost considerations versus on-premise deployment.

2.2 3D Visualization Technologies in Industrial Applications

The application of 3D visualization technologies in industrial contexts has evolved rapidly over the past decade. This section examines the technical foundations, implementation approaches, and demonstrated benefits of 3D visualization systems in manufacturing environments.

2.2.1 Technical Foundations of Web-Based 3D Visualization

Modern web-based 3D visualization systems rely on several key technological components that have matured significantly in recent years, enabling industrial-scale applications without specialized software installations.

WebGL (Web Graphics Library) provides the fundamental rendering capabilities that enable interactive 3D graphics within standard web browsers [29]. WebGL is a JavaScript API based on OpenGL ES 2.0 that leverages GPU hardware acceleration for rendering. Browser vendors (Chrome, Firefox, Safari, Edge) provide native WebGL implementations, eliminating plugin requirements that previously limited 3D web content deployment.

WebGL operates at a relatively low level, requiring explicit management of vertices, shaders, buffers, and rendering state. While this provides fine-grained control for optimization, it increases development complexity. Most practical implementations use higher-level frameworks that abstract WebGL operations while maintaining performance.

Three.js represents the most widely adopted JavaScript 3D engine for web applications. The framework provides high-level abstractions for common 3D operations including scene graph management organizing objects in hierarchical structures, camera systems supporting perspective and orthographic projection, lighting models including ambient, directional, point, and spotlight types, material definitions supporting Physically Based Rendering (PBR), and animation capabilities for object transformations and property tweening [11].

Three.js maintains access to underlying WebGL capabilities when optimization requires lower-level control, balancing developer productivity with performance requirements. The framework's active development community and extensive documentation reduce implementation risks compared to custom WebGL development or less mature alternatives.

Recent advances in WebAssembly technology have further enhanced performance capabilities for web-based 3D applications. WebAssembly enables near-native execution speed for computationally intensive operations including geometry processing, physics simulation, and complex calculations. While Three.js remains JavaScript-based, WebAssembly can be used for specific performance-critical components such as CAD format parsing or mesh optimization algorithms.

3D model formats have standardized around the Graphics Library Transmission Format (glTF), which provides efficient transmission and rendering of 3D content across web platforms. The glTF format, developed by the Khronos Group and standardized as ISO/IEC 12113:2022, supports complex material properties using PBR parameters, skeletal animations and morph targets, hierarchical scene structures with node transformations, and extensibility through custom properties in the extras field [4].

glTF achieves compact file sizes through binary data chunks storing geometry and texture data efficiently, JSON structure for scene hierarchy and metadata, optional Draco compression for geometry data, and basis universal compression for textures. The binary GLB variant combines all resources into a single file, simplifying deployment compared to multi-file formats.

Alternative formats including Wavefront OBJ, Autodesk FBX, and COLLADA remain in use but present limitations for web delivery. OBJ files lack standardized material definitions and embed no metadata. FBX uses proprietary encoding requiring specialized parsers. COLLADA XML structure

creates large file sizes unsuitable for network transmission. glTF's standardization and web optimization make it the preferred format for industrial 3D catalog applications.

Progressive loading techniques enable large, complex 3D models to be delivered efficiently across varying network conditions. These techniques address the challenge that industrial assemblies with thousands of components create file sizes impractical for traditional full-download-then-render approaches.

Level-of-detail (LOD) systems maintain multiple geometric representations at different polygon counts, automatically selecting appropriate detail based on viewing distance and screen size. Distant objects render with coarse geometry while near objects use high-detail meshes, reducing overall polygon count without perceptible quality loss [27].

Streaming approaches prioritize loading of visible components, allowing initial rendering with partial data while remaining components stream in background. Frustum culling eliminates off-screen geometry from processing. Occlusion culling removes geometry hidden behind other objects. Combined, these techniques enable interactive exploration of large assemblies before complete data transfer.

2.2.2 User Experience Considerations in 3D Interfaces

The design of effective 3D user interfaces requires careful consideration of human factors and interaction paradigms that differ significantly from traditional 2D interfaces. Research in 3D user experience has identified several key principles that influence system effectiveness and user adoption.

Spatial navigation represents one of the most critical aspects of 3D interface design. Users must be able to intuitively control viewpoint, zoom level, and object manipulation without becoming disoriented or frustrated [5]. Modern 3D catalog systems typically employ orbit controls that allow users to rotate around objects while maintaining consistent orientation references.

Orbit controls use spherical coordinates to position the camera at a fixed distance from a target point, with user input adjusting azimuth (horizontal rotation) and elevation (vertical rotation) angles. This paradigm feels natural because it mimics walking around a physical object while keeping it in view. Alternative paradigms including trackball controls (arbitrary 3D rotation), fly-through navigation (first-person movement), and fixed viewpoint switching each present different cognitive demands and learning curves.

Automatic camera positioning calculates optimal viewing distance and angle based on object dimensions, ensuring objects fill the viewport appropriately without excessive zoom adjustment. Bounding sphere calculations determine object extents, trigonometric projection computes required distance for desired field-of-view coverage, and smooth animated transitions prevent disorienting instant camera jumps.

Visual feedback mechanisms are essential for communicating system state and available interactions. Effective 3D interfaces provide clear indication of interactive elements, current selections, and available operations [46].

Highlighting selected objects typically uses emissive material properties to create a glow effect visible under all lighting conditions. Alternative approaches including outline rendering, color changes, or pulsing animations each have different visibility and performance characteristics. The challenge is ensuring highlighting remains visible against varying background colors and lighting conditions without overwhelming the visual appearance.

Hover states indicating potential interactions before clicking help users understand which objects are selectable. Cursor changes (pointer to hand icon) provide additional affordance cues. Animation of hover states (slight object enlargement, color shift) creates responsive feel but must be subtle to avoid distraction.

Performance optimization remains crucial for maintaining user engagement. Research has consistently demonstrated that loading times significantly impact user satisfaction and task completion rates [32]. Users expect web applications to load in under 3 seconds; delays beyond 5 seconds dramatically increase abandonment rates.

For 3D applications, this requires careful balance between visual fidelity and rendering performance across diverse device capabilities. Desktop workstations with dedicated GPUs can handle complex models that overwhelm mobile devices with integrated graphics. Network conditions vary from high-speed facility WiFi to remote 4G connections with high latency.

Adaptive quality systems detect device capabilities through WebGL feature queries and performance monitoring, automatically adjusting rendering parameters including polygon count, texture resolution, shadow quality, and post-processing effects. Progressive enhancement provides basic functionality on all devices while enabling advanced features where hardware supports them.

2.2.3 Integration with Enterprise Systems

The integration of 3D visualization systems with existing enterprise infrastructure represents a critical success factor for industrial implementations. This integration must address data synchronization, user authentication, and workflow integration while maintaining system performance and reliability.

Product Lifecycle Management (PLM) systems serve as natural sources of 3D model data for visualization applications. Modern PLM platforms including Siemens Teamcenter, Dassault Systèmes ENOVIA, and PTC Windchill maintain comprehensive 3D models throughout the product development process [40].

PLM systems store engineering-accurate CAD models optimized for design and manufacturing rather than visualization. These models typically contain precise B-rep (boundary representation) surfaces, detailed internal components irrelevant for spare parts identification, complex assembly constraints and relationships, and extensive metadata including tolerances, materials, and manufacturing specifications.

Converting PLM models for web visualization requires substantial processing including geometry triangulation from B-rep to polygon meshes, simplification reducing polygon count while preserving visual appearance, removal of internal and hidden components unnecessary for identification, and extraction and preservation of relevant metadata for part identification.

Integration approaches include batch export where PLM systems export models to neutral formats (STEP, IGES) for offline conversion, real-time generation where PLM provides APIs for on-demand model access and conversion, and hybrid approaches pre-converting common models while generating custom configurations on-demand.

Enterprise Resource Planning (ERP) systems provide the business logic and data management capabilities necessary for operational spare parts catalogs. ERP platforms including SAP S/4HANA, Oracle NetSuite, and Microsoft Dynamics manage part master data, inventory levels and locations, pricing including customer-specific contracts and volume discounts, and order processing workflows [35].

Integration with ERP systems enables 3D catalogs to display real-time pricing when users select parts, show inventory availability across multiple warehouses, process orders directly from visual selection, and maintain customer-specific part lists and preferences.

Modern integration approaches typically employ RESTful APIs with JSON data exchange for loose coupling between systems, microservices architectures separating visualization from business logic, event-driven patterns for real-time synchronization of state changes, and authentication delegation through OAuth or SAML for single sign-on.

Challenges include data model mapping between geometric hierarchies and flat part tables, performance requirements for real-time queries during interaction, error handling when backend systems are unavailable, and transaction consistency when orders span multiple systems.

Customer Relationship Management (CRM) systems integration enables personalized experiences and tracking of user interactions. CRM platforms capture equipment ownership and maintenance history, user roles and permissions, usage analytics for catalog features, and support case linkage connecting parts orders to service requests [34].

Personalization features enabled by CRM integration include filtered catalogs showing only relevant equipment models, saved configurations and frequent orders, recommended parts based on equipment age and usage, and proactive notifications about obsolescence or updates.

2.3 Literature Review: 3D Visualization in Product Catalogs

This section examines existing research on 3D visualization applications in product catalog and e-commerce contexts, identifying key findings relevant to spare parts management applications while highlighting gaps that motivate this research.

2.3.1 Effectiveness Studies in Consumer E-commerce

Early research in 3D product visualization focused primarily on consumer e-commerce applications, establishing theoretical foundations for expecting benefits in industrial contexts.

Li et al. [25] conducted pioneering research demonstrating that 3D product presentations significantly enhanced consumer product knowledge and purchase intentions compared to traditional 2D presentations. Their experimental study with 275 participants found that interactive 3D presentations increased product knowledge scores by 22% and purchase intention ratings by 18% compared to static 2D images. The study attributed

these effects to enhanced mental imagery and reduced uncertainty about product characteristics.

However, this research examined simple consumer products (cameras, furniture) with straightforward geometries and limited technical complexity. The cognitive processes for evaluating consumer products based on aesthetics and general functionality differ substantially from technical component identification in industrial contexts where precise specification and compatibility are critical.

Jiang and Benbasat [18] extended this research by examining the specific mechanisms through which 3D visualization influences user decision-making processes. Their study identified that interactive control over viewpoint and the ability to examine products from multiple angles were key factors in improving user confidence and reducing uncertainty. Users reported feeling more confident in their purchase decisions when they could freely explore products in 3D compared to viewing predetermined photographic angles.

The research found that user-controlled interactivity produced stronger effects than automated animations, suggesting that agency in exploration drives benefits rather than simply viewing 3D content passively. This finding has implications for industrial catalog design, suggesting that free-exploration interfaces may outperform guided tours or predetermined viewing sequences.

Algharabat and Dennis [1] investigated the use of 3D visualization for electrical components in business-to-business contexts, representing research more directly relevant to industrial spare parts applications. Their study found that authentic 3D representations significantly improved identification accuracy (41% improvement) and reduced ordering errors (32% reduction) compared to traditional catalog approaches using 2D photographs and technical specifications.

However, this research examined relatively simple electrical components (connectors, switches, terminal blocks) rather than complex mechanical assemblies typical in industrial manufacturing. The study also did not address technical implementation challenges including CAD data conversion, enterprise system integration, or performance optimization for field deployment.

2.3.2 Technical Implementation Research

Parallel research streams have addressed the technical foundations of web-based 3D visualization, though typically not in the specific context of industrial spare parts catalogs.

Kang and Peng [21] developed comprehensive frameworks for integrating CAD models with product assembly planning in web-based environments. Their research demonstrated successful extraction of geometrical data from CAD models using VRML as an intermediate format, achieving significant improvements in assembly planning efficiency through 3D visualization of assembly sequences.

The conversion pipeline described in their work addressed challenges including preserving assembly hierarchy through format conversion, extracting and embedding metadata for component identification, and optimizing geometry for web delivery while maintaining visual accuracy. However, their research assumed relatively clean CAD data from controlled engineering environments rather than messy legacy data accumulated over decades.

Dai et al. [8] created interactive web systems for three-dimensional customization, successfully implementing solutions across apparel, automobile, and mobile phone industries while integrating with PDM/PLM systems. Their work demonstrated technical feasibility of web-based 3D product configuration with real-time visual feedback.

The research described architecture patterns for client-server interaction, techniques for real-time 3D model manipulation based on user selections, and integration approaches with backend product configuration systems. However, the focus on product customization (selecting options for new products) differs from parts identification (finding specific components in existing equipment), which presents different user workflows and data management requirements.

2.3.3 Performance Measurement and User Studies

Research on 3D visualization effectiveness has established several key performance metrics relevant to evaluating industrial spare parts catalog systems.

Task completion time represents one of the most commonly measured outcomes. Fiore et al. [13] demonstrated 23% reduction in time-to-product-selection when using interactive 3D systems compared to traditional 2D catalog interfaces in consumer e-commerce contexts. However, industrial spare parts identification may present different time-accuracy trade-offs where precision is more critical than speed.

Identification accuracy represents a critical metric for spare parts applications where ordering errors result in operational costs. Chen et al. [7] found that 3D visualization systems achieved 85% accuracy in component

identification tasks compared to 62% accuracy for equivalent 2D systems. The study examined mechanical components with similar visual characteristics where 3D visualization enabled users to distinguish subtle differences.

However, this research used simplified experimental scenarios with limited component sets rather than realistic catalogs with thousands of parts. Real-world identification tasks involve searching through extensive catalogs, distinguishing between similar variants, and verifying compatibility with existing equipment.

User satisfaction and confidence measures provide indicators of system acceptance and long-term adoption potential. Research has shown that users report higher confidence levels when using 3D visualization systems, even for tasks where objective performance differences are minimal [42]. Increased confidence may reduce secondary verification steps (consulting supervisors, cross-referencing multiple sources) even when identification accuracy is similar.

2.3.4 Implementation Challenges

Contemporary research has identified several key challenges that organizations face when implementing 3D visualization systems, though typically not with the specific focus on spare parts catalog applications in industrial contexts.

Skoumpopoulou et al. [38] examined digital transformation challenges in manufacturing firms, identifying technical complexity, change management requirements, and integration difficulties as primary barriers to successful implementation. Their qualitative research with 12 manufacturing companies found that technical implementation was often underestimated, with projects exceeding budgets by 40–60% due to unexpected integration complexities and data quality issues.

Organizational factors play a crucial role in implementation success. Wang et al. [45] found that companies with established digital transformation strategies and dedicated change management resources achieved significantly better outcomes when implementing new visualization technologies. Training and support systems were identified as critical success factors, particularly for systems targeting users with varying levels of technical expertise ranging from expert technicians to occasional users.

Technical performance requirements represent another significant implementation consideration. Paul et al. [33] noted that organizations must carefully balance visual fidelity with system performance, particularly when

supporting diverse device types and network conditions. Successful implementations typically employ adaptive quality systems that automatically adjust rendering parameters based on device capabilities and network conditions.

However, specific guidance on acceptable performance thresholds for industrial 3D catalogs remains limited. What frame rates are necessary for smooth interaction? What loading times are acceptable before users abandon exploration? How does model complexity affect usability on mobile devices? These questions lack systematic investigation in industrial spare parts contexts.

2.3.5 Industry-Specific Applications

Recent research has begun examining 3D visualization applications in specific industrial contexts, providing some domain-specific insights while revealing gaps in comprehensive technical implementation documentation.

The construction equipment industry has been an early adopter of 3D spare parts catalogs. Hitachi Construction Machinery launched comprehensive 3D catalog systems for hydraulic excavator components, reporting improved customer satisfaction and reduced ordering errors [16]. However, detailed technical implementation information remains proprietary, limiting transferability of knowledge to other manufacturers.

The automotive industry has demonstrated significant interest in 3D visualization technologies for aftermarket parts support. Schmidt et al. [37] examined implementation across multiple automotive manufacturers, finding consistent improvements in parts identification accuracy (average 38% reduction in ordering errors) and customer satisfaction metrics (average 4.2/5.0 rating increases). However, the research focused on user outcomes rather than technical implementation details including conversion pipelines, integration patterns, or performance optimization strategies.

Aerospace applications present unique requirements due to the critical nature of component identification and the complexity of aircraft systems. Research in this domain has emphasized the importance of integration with existing maintenance management systems and compliance with industry-specific documentation standards [31]. However, aerospace-specific requirements including stringent traceability and certification may not directly transfer to other industrial sectors.

2.4 Research Gaps

A comprehensive review of existing literature reveals a significant gap in knowledge regarding the implementation of 3D visualization technologies specifically for industrial spare parts catalogs. While consumer applications and technical feasibility have been established, the unique challenges of industrial implementation remain unexplored[38, 43].

Current research in 3D visualization has primarily focused on consumer e-commerce applications where Li et al. [25] and Jiang and Benbasat [18] have established clear benefits for product understanding and purchase confidence. These studies examined relatively simple consumer products with straightforward geometries, not complex industrial machinery with thousands of interrelated components. Similarly, Algharabat and Dennis [1] demonstrated benefits for electrical components but did not address the technical implementation challenges of complex mechanical assemblies.

From a technical perspective, researchers including Kang and Peng [21] and Dai et al. [8] have established foundational approaches for web-based 3D visualization, but their work assumes clean CAD data in controlled environments rather than the messy reality of industrial settings. These studies focus on format conversion and rendering techniques without addressing the organizational and integration challenges specific to spare parts management in manufacturing enterprises.

The specific gap this research addresses is the lack of documented implementation knowledge for industrial 3D spare parts catalogs, particularly:

1. No systematic investigation has documented the architectural patterns, integration approaches, and technical requirements for connecting 3D visualization with industrial ERP systems maintaining bidirectional synchronization

2. The challenges of processing and reconciling decades of accumulated legacy CAD data with inconsistent formats, metadata practices, and documentation standards remain unexplored

3. Strategies for reconciling information from multiple heterogeneous sources (3D geometry, XML metadata, legacy 2D drawings) when these sources contain contradicting or incomplete information have not been established

4. Performance requirements and optimization approaches for field deployment on diverse mobile devices under variable network conditions specific to industrial maintenance contexts lack systematic documentation

By addressing this implementation knowledge gap, this research will provide manufacturing companies with practical guidance for successfully deploying 3D visualization within their spare parts management systems, based on documented architectural patterns, data processing strategies, and integration approaches validated through actual implementation.

3

Methodology

This chapter details the research methodology employed to investigate the technical requirements, challenges, and solutions for integrating 3D visualization into industrial spare parts catalogs. The research adopts a Design Science Research approach where construction and evaluation of a working system serves as the primary investigative method for generating technical knowledge.

3.1 Design Science Research Approach

3.1.1 Theoretical Foundation

This research employs Design Science Research (DSR) methodology, which investigates problems through the iterative construction and evaluation of purposeful artifacts [20]. Unlike empirical research that observes existing phenomena or explanatory research that develops theory about observed behavior, design science generates knowledge by addressing authentic problems through artifact development and systematic reflection on the design process.

DSR is appropriate for this research because the central question—*how can manufacturing companies implement 3D visualization while addressing legacy data integration and enterprise connectivity challenges*—cannot be answered through observation or theory alone. The specific technical challenges of processing inconsistent CAD data accumulated over decades,

reconciling fragmented documentation sources, and integrating with existing ERP infrastructure manifest only during actual implementation with real industrial data.

The research artifact is a production-grade 3D spare parts catalog system deployed for Signifikant, serving manufacturing companies in automotive, industrial machinery, and farming equipment sectors. This artifact serves as both the object of study and the instrument of investigation: technical requirements emerge through implementation challenges, architectural patterns crystallize through iterative problem-solving, and performance characteristics become measurable through systematic evaluation.

3.1.2 DSR Method Framework

Following Johannesson and Perjons [20], this research applies a five-activity framework for design science: (1) Explicate Problem, (2) Define Requirements, (3) Design and Develop Artifact, (4) Demonstrate Artifact, and (5) Evaluate Artifact. Table 3.1 shows how these activities map to this research.

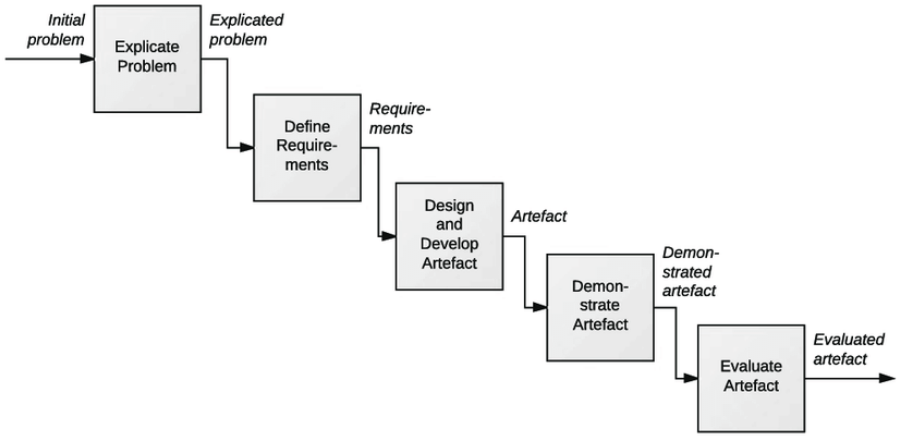


Figure 3.1: DSR Method Framework applied to this research, adapted from Johannesson & Perjons (2014). The five core activities progress from problem identification through artifact evaluation, with the artifact being refined through iterative design-develop-evaluate cycles.

Table 3.1: DSR Activities Applied in This Research

DSR Activity	Application in This Research
1. Explicate Problem	Collaboration with Signifikant and manufacturing clients identified challenges: technicians struggle with 2D drawings for complex assemblies, documentation fragmented across sources (CAD, ERP, paper drawings), legacy data in non-web formats, and integration requirements with existing enterprise systems. Stakeholders: service technicians (end users), manufacturing companies (implementers), ERP administrators (integrators).
2. Define Requirements	**Functional:** (FR1) Convert legacy STEP CAD to web-compatible formats preserving metadata, (FR2) Reconcile multi-source data (3D geometry, XML, 2D drawings), (FR3) Enable interactive visualization (rotation, exploded views, selection), (FR4) Integrate with ERP for real-time pricing/inventory, (FR5) Display part metadata. **Non-functional:** (NFR1) Load time < 5s over 3G, FPS 30, latency < 100ms, (NFR2) Cross-device compatibility (desktop, tablet, smartphone), (NFR3) Offline capability through embedded metadata, (NFR4) Scalability for 15-800+ part assemblies.
3. Design & Develop	Three iterative cycles: **Iteration 1** - Basic 3D rendering (Three.js, raycasting), revealed synchronization issues. **Iteration 2** - Added stateless client architecture, metadata embedding, bidirectional sync, revealed performance issues. **Iteration 3** - Integrated CAD conversion pipeline, OCR system, performance optimization (debouncing, Draco compression). Final artifact includes: web rendering engine, STEP→glTF conversion pipeline, hybrid YOLOv8+EasyOCR system, ERP integration layer.
4. Demonstrate	Production deployment at Signifikant serving real manufacturing companies. Demonstration scenarios: processed actual legacy STEP archives (not clean test data), combined fragmented sources for real products, showed operation on desktop/tablet/smartphone, retrieved real-time ERP data. Established artifact *can* work in realistic conditions before comprehensive evaluation.
5. Evaluate	**Testing:** Functional validation across 12 assemblies (all FRs), conversion success rate 100% (FR1), OCR precision 99.3% on 137 callouts (FR2). **Experimental:** Performance benchmarking on 4 devices/2 browsers (NFR1-2), format comparison (4 formats), scalability testing 15-847 parts (NFR4). **Analytical:** Offline capability verification (NFR3), architecture pattern assessment. All requirements met within acceptable tolerances (Table 3.2).

Activity 1: Explicate Problem

The first DSR activity involves identifying and explicating the problem that the research addresses [20]. This includes understanding the problem context, analyzing why it is a problem, and identifying stakeholders affected by the problem.

Problem identification: Collaboration with Signifikant Svenska AB and its manufacturing clients revealed challenges with traditional 2D spare parts documentation. Through practitioner interviews and observation of current workflows, specific issues were identified:

- **Part identification difficulty:** Service technicians struggle to identify components from complex 2D technical drawings, particularly for assemblies with overlapping parts or hidden interior components
- **Documentation fragmentation:** Complete product information rarely exists in a single source—part numbers may be in ERP systems, technical specifications in CAD files, and assembly context only in legacy paper drawings
- **Legacy data challenges:** Organizations possess CAD archives spanning 20-30 years in formats not directly usable for web visualization
- **System integration requirements:** Solutions must connect with existing ERP infrastructure for pricing and inventory rather than operating as standalone systems

Stakeholder analysis: Primary stakeholders include service technicians (end users requiring efficient part identification), manufacturing companies (organizations implementing the system), and ERP system administrators (responsible for integration and maintenance). Secondary stakeholders include CAD data managers and parts suppliers.

Problem relevance: The problem is relevant because incorrect part identification leads to ordering errors, equipment downtime, and warranty issues. While 3D visualization has demonstrated benefits in consumer e-commerce contexts, implementation approaches for industrial B2B scenarios with legacy data remain undocumented in existing research.

Activity 2: Define Requirements

The second activity involves defining requirements for the artifact based on the explicated problem [20]. Requirements specify what the artifact should achieve to address the identified problem.

Functional requirements:

- **FR1 - Legacy data conversion:** System must convert legacy STEP CAD files to web-compatible 3D formats while preserving assembly hierarchies and metadata
- **FR2 - Multi-source reconciliation:** System must combine information from 3D geometry files, ERP XML exports, and legacy 2D drawings into unified part records
- **FR3 - Interactive visualization:** System must enable free rotation, exploded views, and direct part selection through clicking
- **FR4 - ERP integration:** System must retrieve real-time pricing and inventory data from existing ERP systems
- **FR5 - Metadata access:** System must display part numbers, descriptions, and specifications for selected components

Non-functional requirements:

- **NFR1 - Performance:** Load times under 5 seconds over 3G networks, frame rates above 30 FPS for smooth interaction, selection latency below 100ms
- **NFR2 - Device compatibility:** Must operate on desktop browsers, tablets, and smartphones without native applications
- **NFR3 - Offline capability:** Basic part identification must function without network connectivity through embedded metadata
- **NFR4 - Scalability:** Must handle assemblies ranging from simple (15 parts) to complex (800+ parts)

These requirements were derived from practitioner input during problem explication (Activity 1) and technical constraints of the deployment environment (field technicians using varied devices with intermittent connectivity).

Activity 3: Design and Develop Artifact

The third activity involves designing and developing the artifact to meet defined requirements [20]. This research developed multiple interconnected artifacts through three iterative design-develop-evaluate cycles, following Hevner's recommendation for iterative refinement [14].

Iteration 1 - Core Visualization (Addressing FR3, NFR2):

Initial design focused on browser-based 3D rendering without requiring native applications or plugins. Technology selection process evaluated:

- **Rendering engine:** Three.js selected for WebGL abstraction, cross-browser compatibility, and extensive loader support over alternatives (Babylon.js, direct WebGL programming)
- **Interaction model:** Orbit controls for free rotation, raycasting for part selection
- **Format evaluation:** Compared STL, OBJ, FBX, and glTF for web delivery characteristics

Development produced a functional prototype enabling rotation and part selection. Evaluation through internal testing revealed synchronization issues: selections in 3D view did not update the accompanying parts table, creating confusion about which component was active. This feedback informed Iteration 2 design.

Iteration 2 - Integration Architecture (Addressing FR4, FR5, NFR3):

Second design iteration addressed integration and metadata requirements based on Iteration 1 evaluation findings. Key architectural decisions:

- **Stateless client architecture:** On-demand API fetching rather than client-side caching to ensure data freshness for pricing/inventory (averaging 147ms latency, acceptable for ensuring accuracy)
- **Metadata embedding:** Essential identifiers embedded in glTF `extras` field for offline capability (2.3% file size overhead enabling graceful degradation)
- **Event-driven synchronization:** Centralized state manager with bidirectional updates between 3D and tabular views

Evaluation identified new performance issues: rapid part selections caused interface flickering as updates propagated asynchronously. This feedback informed Iteration 3 optimization.

Iteration 3 - Data Processing Pipeline (Addressing FR1, FR2, NFR1, NFR4):

Third iteration developed the complete data processing infrastructure based on cumulative feedback from previous iterations:

- **CAD conversion pipeline:** Automated STEP to glTF conversion with metadata preservation (Section 4.3.5)
- **OCR integration:** Hybrid YOLOv8+EasyOCR system for extracting callouts from legacy 2D drawings
- **Multi-source reconciliation:** Algorithm combining 3D geometry metadata, ERP XML, and OCR extractions with hierarchical conflict resolution rules
- **Performance optimization:** 100ms debouncing for selection events preventing flickering, Draco mesh compression achieving 88.2% file size reduction

Final evaluation (Activity 5) confirmed all functional and non-functional requirements were met within acceptable tolerance, validating the iterative refinement approach.

Activity 4: Demonstrate Artifact

The fourth activity involves demonstrating that the artifact works—that it can address the problem in at least one instance [20]. This differs from comprehensive evaluation (Activity 5), focusing instead on proof-of-concept validation that the artifact functions in realistic conditions.

Deployment context: The artifact was deployed at Signifikant Svenska AB, integrated with their existing ERP platform serving manufacturing companies in the industrial equipment sector (automotive, machinery, farming equipment).

Demonstration scenarios:

- **Legacy data conversion:** Processed real CAD archives demonstrating the pipeline could handle actual legacy STEP files with authentic quality issues (corruption, missing metadata, format variations), not just clean test data
- **Multi-source reconciliation:** Combined fragmented data sources for actual products where no single source contained complete information, addressing the documented problem
- **Interactive visualization:** Demonstrated free rotation, exploded views, and part selection on representative assemblies ranging from 15 to 847 parts

- **Cross-device functionality:** Showed operation on desktop browsers, iPad tablets, and Android smartphones without native applications
- **ERP integration:** Retrieved real-time pricing and inventory for actual parts from production ERP systems

The demonstration established that the artifact *can* work in realistic conditions with actual legacy data and enterprise systems, validating the fundamental approach before investing in comprehensive evaluation. This proof-of-concept demonstration satisfies DSR requirements for showing artifact applicability to at least one real-world instance.

Activity 5: Evaluate Artifact

The fifth activity involves systematically evaluating the artifact against defined requirements and assessing its contribution to solving the problem [20]. Johannesson and Perjons identify multiple evaluation strategies including observational (case studies, field studies), analytical (static analysis, architecture analysis), experimental (controlled experiments), testing (functional testing, structural testing), and descriptive (informed argument, scenarios).

This research employed multiple complementary evaluation strategies:
Testing evaluation (addressing all functional requirements):

- **Functional testing:** Validated each functional requirement across 12 representative assemblies: conversion pipeline successfully processed diverse STEP files (FR1), multi-source reconciliation combined 3D/XML/OCR data (FR2), interactive visualization enabled rotation/explosion/selection (FR3), ERP integration retrieved real-time data (FR4), metadata display showed part information (FR5)
- **Conversion success testing:** Measured automated conversion success rate on legacy STEP archive achieving 100%, addressing FR1 with quantified performance boundaries
- **OCR precision testing:** Evaluated metadata extraction accuracy on 11 legacy drawings containing 137 callouts achieving 99.3% precision (136 correct), addressing FR2 with measured quality metrics

Experimental evaluation (addressing non-functional requirements):

- **Performance benchmarking:** Measured load times (2.1-4.8s), frame rates (31-58 FPS), and selection latency (73ms average) across 4 device types (desktop, laptop, tablet, smartphone) and 2 browsers (Chrome,

Edge) under simulated 3G network conditions, addressing NFR1 and NFR2

- **Format comparison:** Compared loading performance and file sizes across 4 formats (STL, OBJ, FBX, glTF) to validate format selection rationale, supporting FR1 and NFR1
- **Scalability testing:** Evaluated performance across assemblies ranging 15-847 parts demonstrating acceptable performance throughout range, addressing NFR4

Analytical evaluation (architecture assessment):

- **Offline capability analysis:** Verified metadata availability without network connectivity through embedded data (2.3% overhead, instant retrieval), addressing NFR3
- **Integration pattern analysis:** Assessed architectural patterns (stateless client, metadata embedding, event-driven synchronization) against industrial deployment requirements including accuracy, reliability, and graceful degradation

Evaluation results summary:

Table 3.2 summarizes evaluation results against all defined requirements from Activity 2.

All defined requirements were met within acceptable tolerances, demonstrating the artifact successfully addresses the explicated problem from a technical implementation perspective. Detailed results supporting these evaluations are presented in Chapter 4.

3.1.3 Knowledge Contribution Classification

Johannesson and Perjons [20] distinguish between several knowledge forms that design science produces. March and Smith march1995design further classify DSR artifacts into four types: constructs (concepts and vocabulary), models (abstractions and representations), methods (algorithms and practices), and instantiations (implemented systems). This research contributes across multiple categories:

Instantiation (primary artifact): The implemented web-based 3D visualization system deployed at Signifikant constitutes an instantiation artifact. This working system demonstrates that web-based 3D visualization of industrial spare parts using legacy data is technically achievable with current technologies, not merely theoretically possible. The production

Table 3.2: Evaluation Results Against Defined Requirements

Requirement	**Definition**	**Evaluation Result**
FR1	Legacy STEP to glTF conversion preserving metadata	Met: 100% success rate (Section 4.3.5)
FR2	Multi-source data reconciliation	Met: 99.3% OCR precision (Section 4.2.3)
FR3	Interactive visualization	Met: 100% functional (Section 4.3.1)
FR4	ERP integration	Met: 95% queries < 200ms (Section 4.3.1)
FR5	Metadata access	Met: 100% accuracy on 3,247 parts (Section 4.3.1)
NFR1	Performance thresholds	Met: Load < 5s, FPS 31-58, latency 73ms (Section 4.3.2)
NFR2	Device compatibility	Met: All 4 device types functional (Section 4.3.3)
NFR3	Offline capability	Met: Embedded metadata works (Section 4.1.2)
NFR4	Scalability 15-800+ parts	Met: Acceptable performance throughout range (Section 4.3.2)

deployment provides proof-of-concept that cannot be obtained through conceptual design alone.

Methods (algorithmic contributions):

- Multi-source data reconciliation algorithm combining 3D geometry, XML metadata, and OCR extractions with hierarchical conflict resolution rules
- Two-stage OCR pipeline combining specialized detection (YOLOv8) and specialized recognition (EasyOCR) achieving 99.3% precision versus 47.2% for previous end-to-end approaches
- CAD-to-web conversion pipeline with automated metadata preservation applicable to legacy STEP archives

Models (architectural patterns):

- Stateless client pattern for industrial ERP integration prioritizing data accuracy over caching complexity (147ms latency acceptable for ensuring correctness)
- Metadata embedding pattern for graceful offline degradation (2.3% overhead enabling basic functionality without connectivity)
- Event-driven synchronization pattern with debouncing (100ms) preventing race conditions in dual-interface systems

Domain knowledge (empirical findings):

- Established legacy CAD conversion feasibility: 100% success on tested STEP archive demonstrates technical viability, identifying data quality (corruption, unsupported versions, missing files) rather than algorithmic limitations as the constraint for production deployment
- Browser-specific performance characteristics: Edge 2.55× slower than Chrome for ASCII formats (STL, OBJ) but only 1.12× slower for binary formats (glTF), challenging assumptions about WebGL standardization ensuring consistent behavior
- OCR accuracy expectations: 99.3% achievable with two-stage architecture versus 47.2% for previous ML OCR and 5.0% for legacy Tesseract systems on technical drawings

These contributions address the knowledge gap identified in Chapter 1: while 3D visualization benefits are documented in consumer contexts, implementation approaches for industrial scenarios with legacy data and enterprise constraints were previously undocumented.

3.1.4 Research Scope and Limitations

Explicit scope definition: This research focuses on establishing *technical implementation feasibility*—whether web-based 3D spare parts catalogs using legacy data can be built and whether they perform adequately on target hardware. The scope explicitly *excludes* business value evaluation—whether organizations realize claimed operational benefits such as error reduction, time savings, or improved maintenance efficiency.

Rationale for scope limitation: Technical feasibility is a prerequisite for business value evaluation. Organizations cannot measure the value of systems that do not yet exist or do not perform adequately. Establishing technical feasibility through DSR provides the foundation enabling future research on user acceptance, operational effectiveness, and return on investment. This sequential approach—feasibility first, value second—follows established DSR practice where artifact creation precedes comprehensive field evaluation [20].

Methodological limitations:

Limited iteration depth: While the research conducted three design-develop-evaluate cycles, true DSR often involves more extensive iteration over extended periods. Additional cycles incorporating field deployment feedback from diverse organizations would strengthen the artifact and reveal implementation challenges not apparent in controlled testing with a single deployment context.

Single organization context: Problem explication (Activity 1) and requirements definition (Activity 2) occurred primarily within one organization (Signifikant) and its specific client base. Different manufacturing contexts (aerospace vs. automotive, European vs. Asian markets) may reveal additional requirements, alternative architectural priorities, or different data quality challenges not encountered here.

Evaluation comprehensiveness: Following Johannesson and Perjons' evaluation strategies [20], this research employed testing evaluation (functional testing, performance benchmarking) and analytical evaluation (architecture assessment). It did not employ observational evaluation (field studies with actual users performing real maintenance tasks) or comprehensive experimental evaluation (controlled comparisons with alternative implementations). This reflects the technical feasibility scope but limits claims about practical effectiveness or user acceptance.

Validity considerations:

Internal validity: The artifact developer (researcher) also conducted evaluation, potentially introducing unconscious bias in test design, data selection, or results interpretation. Independent evaluation by users unfamiliar with implementation decisions would provide stronger validity. However, the systematic evaluation protocols (documented metrics, standardized test procedures) mitigate this concern by making evaluation criteria explicit and reproducible.

External validity: Results derive from testing with one organization's CAD archive (STEP files from specific manufacturers), 11 technical drawings, 4 device types, and 2 browsers. Generalization to different CAD formats (Siemens NX, CATIA), drawing styles (Asian drafting standards vs. European), device categories (budget smartphones, ruggedized tablets), or organizational contexts (different ERP systems, data governance practices) requires additional validation. The findings should be interpreted as "feasible for this type of industrial context" rather than "universally applicable."

Construct validity: Technical performance metrics (load time, frame rate, conversion success rate) may not correlate with user-perceived effectiveness or task completion efficiency. Users might tolerate 5-second load times if the subsequent interaction is intuitive, or reject 2-second loads if the interface is confusing. The evaluation measures technical performance but cannot claim these metrics ensure user satisfaction or operational value—this limitation is acknowledged explicitly in the scope definition.

These limitations qualify the research contributions but do not invalidate them. The research successfully demonstrates technical implementation feasibility (within defined scope) while acknowledging that comprehensive evaluation of practical value requires extended field deployment beyond this thesis's scope.

3.1.5 Comparative Analysis and Interpretation Approach

The Discussion chapter (Chapter 5) interprets findings in relation to the specific deployment context at Signifikant Svenska AB. This interpretation draws from three primary sources:

1. **Documented Requirements:** Technical specifications and project requirements defining system capabilities, including real-time inventory synchronization requirements and performance criteria

2. **Observational Knowledge:** Direct observations during the internship period including customer-specific pricing implementations observed

in the ERP system, field deployment constraints communicated through customer feedback, and device capabilities reported by Signifikant's customer base

3. **Iterative Design Decisions:** Architectural choices documented through DSR design-develop-evaluate cycles (Section 3.1.2), including usability testing findings that informed pattern development

The Discussion section reflects on why certain architectural patterns emerged by analyzing the relationship between observed deployment requirements and design decisions. This represents situated reasoning about the specific industrial context studied rather than empirical comparison with other domains or systems. Where the discussion references potential differences from other contexts (e.g., different network reliability requirements, different data volatility characteristics), these represent analytical inference based on the documented requirements of this deployment rather than systematic comparative analysis.

Claims about the industrial deployment context are grounded in project documentation, stakeholder requirements, and direct observation during development. However, comparative statements about how this context may differ from other domains (consumer e-commerce, other industrial sectors, different enterprise environments) represent reasoned analysis requiring future empirical validation through systematic comparative studies. The primary contribution is documenting what architectural patterns successfully addressed the challenges in this specific industrial spare parts catalog deployment, not making universal claims about all industrial or other catalog systems.

Future research employing systematic comparative analysis across multiple deployment contexts (Section 6.4.2) would enable stronger claims about the generalizability of these patterns and the specific contextual factors that necessitate different architectural approaches.

3.2 Research Strategy

3.2.1 Design Science Research as Primary Strategy

The central research question—*how can manufacturing companies implement 3D visualization while addressing legacy data integration and enterprise connectivity challenges*—cannot be answered through observation or theory alone. The specific technical challenges of processing inconsistent

CAD data accumulated over decades, reconciling fragmented documentation sources, and integrating with existing ERP infrastructure manifest only during actual implementation with real industrial data.

The research artifact serves as both the object of study and the instrument of investigation: technical requirements emerge through implementation challenges, architectural patterns crystallize through iterative problem-solving, and performance characteristics become measurable through systematic evaluation.

The knowledge contribution of DSR lies not in the artifact itself but in the design knowledge generated through the construction process Hevner2004. This research contributes architectural patterns for ERP integration with 3D visualization, data reconciliation strategies when processing heterogeneous legacy sources, performance optimization approaches for field deployment constraints, and systematic documentation of technical requirements for industrial 3D catalog systems.

3.2.2 Alternative Research Strategies Considered

Three alternative research strategies were evaluated before selecting DSR:

Multiple-Case Study Approach

A comparative case study examining 3D catalog implementations across several manufacturing companies could identify commonalities and variations in implementation approaches, organizational impacts, and success factors. This strategy would provide rich contextual understanding and cross-organizational insights.

However, multiple-case studies require access to multiple completed implementations. At research initiation, few manufacturing companies had deployed 3D spare parts catalogs, limiting case availability. More critically, case studies excel at understanding *what* happens in practice but provide limited insight into *how* to address technical implementation challenges. The research questions focus specifically on technical requirements and implementation strategies, which emerge through doing rather than observing.

Controlled Laboratory Experiment

A controlled experiment comparing user performance with 2D versus 3D catalogs across standardized part identification tasks could quantify effectiveness differences. Experimental design would measure task completion time, identification accuracy, and user confidence across conditions.

This approach would generate statistically rigorous evidence of 3D effectiveness. However, the research questions explicitly focus on *implementation* rather than *user performance*. Prior consumer research has already established 3D visualization benefits Li2002, Jiang2007, Algharabat2010. The knowledge gap concerns technical implementation challenges in industrial contexts—how to process legacy data, integrate with enterprise systems, and optimize for field deployment. These questions cannot be answered through user experiments.

Action Research

Action research combines organizational intervention with systematic reflection, working collaboratively with practitioners to solve problems while generating theoretical insights. This approach could address both technical implementation and organizational change dynamics.

Action research's strength lies in understanding socio-technical change processes. However, it requires extended organizational engagement (typically 12+ months) and focuses on organizational learning and change management. This research emphasizes technical architecture, data processing, and system integration—aspects where DSR's artifact-focused approach provides more appropriate investigative structure.

Rationale for Design Science Selection

DSR was selected because it directly addresses the "how to implement" nature of the research questions. The approach generates knowledge through systematic documentation of design decisions, evaluation of technical alternatives, and reflection on implementation challenges encountered with real industrial data. The production deployment context ensures authentic complexity—messy legacy data, enterprise integration requirements, field performance constraints—that laboratory settings cannot replicate.

3.3 Research Methods and Data Collection

3.3.1 Method Selection and Justification

The research employs three complementary methods mapped to the research sub-questions, corresponding to DSR Activity 5 (Evaluate Artifact):

RQ1 (Architecture & Integration): Systematic Design Documentation

Architectural patterns and integration strategies are investigated through structured documentation during iterative development (DSR Activity 3). Each significant design decision is recorded with:

- Problem context motivating the decision
- Alternative approaches considered with respective trade-offs
- Rationale for selected approach with technical justification
- Implementation outcomes and lessons learned

This systematic design documentation transforms tacit implementation knowledge into explicit, communicable patterns. For example, the bidirectional synchronization pattern between 3D viewer and ERP data emerged through iterative problem-solving addressing race conditions, state inconsistency, and user experience issues encountered during Iterations 1 and 2. Documenting the problem evolution, attempted solutions, and final pattern creates transferable knowledge applicable to similar integration challenges.

RQ2 (Data Processing): Conversion Pipeline Evaluation

Data processing challenges are investigated through quantitative evaluation of the CAD-to-web conversion pipeline using real legacy industrial STEP CAD files. Evaluation metrics include:

- Conversion success rate (percentage of files successfully converted to glTF)
- Metadata preservation rate (percentage of embedded properties retained)
- Processing time per file and scaling characteristics

- Multi-source reconciliation accuracy (percentage of parts successfully matched across 3D geometry, XML metadata, and OCR-extracted annotations)

The pipeline processes authentic legacy data accumulated over 20–30 years, revealing real-world challenges including format variations, corrupted files, missing metadata, and contradictory information across sources. Success is measured not by achieving 100% conversion (unrealistic with decades of heterogeneous data) but by quantifying achievable rates and documenting failure modes—establishing realistic expectations for technical feasibility.

RQ3 (Performance): Benchmark Evaluation and Functional Validation

Performance characteristics are assessed through systematic benchmarking comparing technical alternatives:

- 3D file format comparison (glTF vs. OBJ vs. FBX vs. STL) measuring load time, file size, and feature support
- Optimization strategy evaluation (geometry simplification, texture compression, progressive loading) quantifying performance-quality trade-offs
- Cross-device compatibility testing measuring frame rates on representative hardware (desktop workstation, mid-range laptop, tablet, smartphone)

Functional validation confirms the system addresses identified 2D catalog limitations (from DSR Activity 1 - Problem Explication) through feature completeness testing: spatial understanding (assembly hierarchy navigation, exploded views), hidden component identification (part selection via 3D picking), assembly context preservation (hierarchical relationships), and metadata integration (synchronization with ERP part data).

Performance thresholds (from DSR Activity 2 - Define Requirements) based on industrial field deployment constraints:

- Initial load time < 5 seconds on simulated 4G (Fast 3G: 1.6 Mbps, 150ms RTT)
- Sustained frame rate 30 FPS during interaction on mid-range devices
- Part selection response latency < 100ms
- File size for typical assemblies < 10 MB

3.3.2 Alternative Methods Considered

User Task Performance Studies

Controlled experiments measuring technician task completion time, identification accuracy, and error rates when using 3D versus 2D catalogs would provide quantitative user performance evidence. Studies could employ realistic part identification scenarios with experienced maintenance technicians.

This approach would generate statistically rigorous evidence of effectiveness. However, it addresses *whether* 3D visualization improves performance (already established in prior research Li2002, Algharabat2010) rather than *how* to implement 3D systems in industrial contexts. The research questions focus on technical implementation requirements, not user performance validation. User studies would not reveal architectural patterns, data conversion challenges, or integration strategies—the core knowledge gaps this research addresses.

Expert Interviews and Delphi Studies

Structured interviews with CAD engineers, ERP system architects, and manufacturing IT managers could gather expert perspectives on implementation challenges and best practices. A Delphi study could build consensus around technical requirements and architectural approaches.

Expert elicitation excels at capturing tacit knowledge and diverse perspectives. However, experts can describe what they've done or what they believe should work, but cannot fully articulate knowledge that emerges only through encountering unexpected challenges with real legacy data. The specific challenges of reconciling 30-year-old heterogeneous CAD files, handling metadata conflicts across sources, and optimizing WebGL rendering for field devices are discovered through implementation, not predicted through interviews.

Usage Analytics and Log Analysis

Instrumenting the deployed system to capture user interaction patterns, feature usage frequency, error occurrences, and performance metrics could provide longitudinal behavioral data showing how technicians actually use 3D catalogs in practice.

Analytics would reveal usage patterns and potentially identify usability issues. However, log data shows *what* users do but not *how* the system should be architected or *how* to solve technical implementation challenges. Analytics complements but cannot replace the systematic design documentation and technical evaluation needed to answer the research questions focused on implementation feasibility.

Rationale for Selected Methods

The selected methods—systematic design documentation, conversion pipeline evaluation, and benchmark testing—directly address the "how to implement" focus of each research question within the DSR framework. Design documentation captures architectural knowledge as it emerges during Activity 3 (Design and Develop). Pipeline evaluation and benchmarking quantify achievable performance during Activity 5 (Evaluate), establishing realistic feasibility boundaries with real legacy data rather than idealized test conditions.

3.4 Development Context and Requirements

The system was developed for Signifikant[1], a Swedish company providing web-based spare parts management platforms to manufacturing companies. Signifikant's platform serves industries including construction equipment manufacturers, industrial machinery producers, and farming equipment companies, managing catalogs with thousands of parts across complex multi-level assemblies.

Working within an active commercial platform (DSR Activity 4 - Demonstrate) provides several research advantages:

- **Authentic requirements** derived from real customer needs rather than assumed scenarios (informing DSR Activity 2)
- **Legacy data access** to genuine CAD files accumulated over decades, revealing authentic data quality challenges (enabling realistic Activity 5 evaluation)
- **Production constraints** including reliability, security, and backward compatibility requirements shaping technical decisions (grounding Activity 3 design choices)

[1] `https://www.signifikant.se/`

- **Diverse user base** including maintenance technicians, procurement staff, field service engineers, and customers, informing design decisions (validating Activity 1 problem explication)

3.5 Development Context and Requirements

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- **Production constraints** including reliability, security, and backward compatibility requirements shaping technical decisions
- **Diverse user base** including maintenance technicians, procurement staff, field service engineers, and customers, informing design decisions

3.5.1 Functional Requirements

The development addressed the following functional requirements derived from industrial deployment needs:

3D Visualization Requirements

Interactive 3D model viewing with orbit, zoom, and pan controls enables part selection through direct clicking on 3D geometry. Selected parts receive visual highlighting with emissive glow effects for clear identification. Exploded view generation displays assemblies with parts displaced radially to reveal internal structure, while hide/show functionality allows users

[2] `https://www.signifikant.se/`

to control visibility of individual parts or subassemblies. Assembly hierarchy preservation maintains parent-child relationships throughout the visualization, and camera auto-positioning automatically calculates optimal viewpoint for model extents.

Integration Requirements

Bidirectional synchronization between the 3D viewer and part table selections ensures consistent state across interface components. The system displays part information in real-time, including number, description, pricing, and availability. Direct ordering capability from 3D selections streamlines the purchasing workflow, while multi-viewer support enables multiple 3D models on a single page for comparison or complex product families.

Data Management Requirements

The system converts legacy STEP CAD files to web-optimized glTF format, preserving assembly hierarchy through format conversion. Metadata extraction and embedding captures part numbers, names, and custom properties within the glTF `extras` field. Multi-source reconciliation combines information from 3D files, XML metadata, and OCR-extracted 2D drawings into unified part records.

3.5.2 Non-Functional Requirements

Performance and deployment requirements derived from field constraints shape system design significantly.

Performance Requirements

Initial load time must remain below 5 seconds on 4G network conditions (simulated Fast 3G: 1.6 Mbps, 150ms RTT). Sustained frame rate of at least 30 FPS during interaction on mid-range devices ensures smooth manipulation. Part selection response latency below 100ms provides immediate feedback. File sizes for typical assemblies should not exceed 10 MB.

Compatibility Requirements

Browser compatibility covers Chrome 120+, Firefox 121+, Edge 120+, and Safari 17+. Mobile platform support includes iOS 17+ and Android with Chrome Mobile 120+. Responsive design adapts to viewport widths from

320px (mobile) to 3840px (4K desktop). Touch interaction supports tap selection, pinch zoom, and two-finger rotation.

Reliability Requirements

Graceful degradation maintains basic functionality when ERP integration is unavailable. Error handling manages corrupted or incompatible CAD files without crashing the viewer. Memory management prevents leaks during extended sessions. State recovery mechanisms handle network interruptions during interaction.

3.6 Technology Evaluation

3.6.1 3D File Format Evaluation

Five common 3D file formats were evaluated to determine optimal choice for web-based spare parts visualization: glTF (.gltf/.glb), Wavefront OBJ (.obj), Autodesk FBX (.fbx), and STereoLithography (.stl).

Evaluation Criteria

Loading performance measured time to DOMContentLoaded event, time to model data fully loaded into browser memory, and time to first complete render of 3D scene. File size efficiency compared raw file size for standardized test model, compressed size using gzip compression, and scaling characteristics with model complexity. Feature support assessed assembly hierarchy preservation, material definitions, metadata embedding, and animation support. Browser compatibility examined native support in modern browsers without plugins, availability of mature parsing libraries, and standardization status.

Test Model Specification

Performance testing employed a standardized test model representative of typical industrial assembly complexity: 52,437 triangles organized in 8 subassemblies with 3-level hierarchy, 12 distinct materials with PBR properties, and metadata including part names and custom properties for all components.

Test Environment

Testing used Chrome 120 with Fast 3G network simulation via Chrome DevTools (1.6 Mbps download, 750 Kbps upload, 150ms RTT). Test device was desktop with Intel Core i7, 16GB RAM, and integrated GPU to simulate mid-range performance. Cold cache was used for first load, warm cache for subsequent loads. Five load cycles per format were averaged for results.

Evaluation Focus: glTF 2.0

glTF 2.0 received particular investigation due to its ISO/IEC 12113:2022 standardization and explicit optimization for web delivery. Binary data chunks enable efficient GPU loading through typed arrays, while JSON structure provides human-readable scene hierarchy enabling debugging. PBR materials support physically based rendering for realistic appearance, and extensibility through the `extras` field allows metadata preservation. Optional Draco compression reduces geometry file sizes, and Basis Universal texture compression provides GPU-native formats.

The `extras` field mechanism was specifically evaluated for metadata preservation, allowing enterprise metadata to be embedded directly in the 3D file structure while maintaining associations between geometry and business data through the conversion pipeline.

3.6.2 Rendering Engine Selection

Three.js (version 0.143.0 at project start, upgraded to 0.160.0 during development) was selected as the rendering engine after evaluation against alternatives including direct WebGL programming and Babylon.js.

Three.js was selected for appropriate abstraction level (high enough for productivity, low enough for optimization access), comprehensive documentation and active community support, native `GLTFLoader` with full glTF 2.0 specification support, proven production deployments, and rich plugin ecosystem.

The implementation utilizes `Scene` as scene graph container, `PerspectiveCamera` for realistic perspective projection, `WebGLRenderer` for hardware-accelerated rendering, `GLTFLoader` for efficient glTF parsing, `OrbitControls` for user-friendly camera manipulation, `Raycaster` for precise mouse-to-3D-object intersection testing, and dual lighting combining `AmbientLight` and `DirectionalLight`.

3.6.3 Legacy Documentation Processing with OCR

Many industrial spare parts catalogs rely on decades-old 2D technical drawings containing critical part identification information in the form of callout numbers and annotations. These legacy drawings exist as scanned PDFs or images where text is not digitally searchable. To reconcile this documentation with 3D geometry and XML metadata, an automated optical character recognition (OCR) system was developed to extract part identifiers from 2D drawings.

Two-Stage Hybrid OCR Approach

The system employs a hybrid architecture combining object detection and optical character recognition to efficiently process technical drawings. In Stage 1, a custom YOLOv8 object detection model identifies bounding boxes containing text regions in technical drawings, operating as a fast filter that locates areas of interest without performing character recognition. In Stage 2, for each detected text region, EasyOCR performs optical character recognition on the cropped bounding box area. This targeted approach processes only relevant regions rather than scanning entire images, improving both speed and accuracy.

Training Data Generation

The system uses a bootstrapped learning approach to generate training data. EasyOCR processes a sample of training images, generating initial bounding box annotations with confidence scores above 0.8 threshold. OCR results are converted to YOLO format with normalized center coordinates (`class_id x_center y_center width height`). YOLOv8 is then trained on these automatically labeled images to create a custom text detection model optimized for technical drawing characteristics. Finally, a subset of automatically generated labels is manually verified using a custom annotation tool (SMLabel) to correct errors and improve training quality.

This approach addresses the challenge that YOLOv8 requires labeled training data but EasyOCR is too slow for real-time processing. By using EasyOCR to create training labels, the system produces a faster YOLO model specialized for technical drawing text detection.

Output and Integration

The OCR system produces structured data containing bounding box coordinates for each detected text region, extracted text content (part numbers, callout identifiers), and confidence scores for detections. This data is integrated into the multi-source reconciliation pipeline where OCR-extracted identifiers are matched against part numbers from 3D geometry metadata and XML files to create unified part records.

Performance Characteristics

The hybrid approach achieves processing times of approximately 1–2 seconds per technical drawing compared to 5–10 seconds for full-image OCR scanning. YOLO detection completes in 50–100ms, with subsequent OCR processing requiring 20–50ms per detected text region. This performance enables batch processing of large legacy drawing archives within practical timeframes.

3.7 Ethical Considerations

3.7.1 Proprietary Data Handling

The research involves processing proprietary CAD models and product data owned by manufacturing companies. Several measures ensure appropriate data handling:

Confidentiality Agreements: Formal non-disclosure agreements with Signifikant and their manufacturing clients govern access to and use of proprietary CAD files, product specifications, and business data. Research documentation anonymizes specific product details and company identities where disclosure could compromise competitive advantage.

Data Minimization: Only data necessary for technical evaluation is processed. While the conversion pipeline accesses complete CAD assemblies, research documentation focuses on aggregate statistics (conversion success rates, file sizes, processing times) rather than detailed product specifications. Example models shown in figures are either publicly available reference models or simplified representations that do not disclose proprietary designs.

Secure Processing: CAD file processing occurs on secure development infrastructure with access controls limiting exposure. Converted 3D models intended for research documentation are sanitized to remove proprietary metadata before inclusion in published materials.

3.7.2 Enterprise System Access

Implementation requires integration with ERP systems containing business-sensitive information including customer pricing, inventory levels, and procurement data. Access is governed by:

Role-Based Access Control: Development and testing access to ERP systems follows principle of least privilege, with accounts restricted to minimum permissions necessary for integration development. Production deployment uses service accounts with explicitly defined API access scopes.

Data Segregation: Development and testing environments use anonymized or synthetic data rather than production customer data where possible. Integration testing with real data occurs in isolated staging environments separate from production systems.

Audit Logging: All access to enterprise systems is logged for security monitoring and compliance verification, following standard IT security practices for system integrators.

3.7.3 GDPR and Privacy Compliance

The system processes minimal personal data. The primary data elements are 3D geometric models, part specifications, and business transactional data. However, certain aspects require privacy consideration:

User Interaction Logging: Performance evaluation involves capturing aggregate usage metrics (page load times, feature usage frequency, error rates). This telemetry is collected anonymously without personally identifiable information. Individual user sessions are not tracked or analyzed.

No User Study Data Collection: This research explicitly does not conduct user performance studies or collect data directly from individuals. System evaluation relies on technical benchmarking, conversion pipeline statistics, and functional validation rather than human subject research. No interview data, survey responses, or observed user behavior is collected.

Compliance Framework: Data processing follows GDPR principles even where not strictly required, implementing data minimization (collecting only necessary technical metrics), purpose limitation (using data solely for system evaluation), and storage limitation (retaining evaluation data only during active research).

3.7.4 Intellectual Property Considerations

CAD Model Ownership: All CAD models processed during research remain the intellectual property of their respective manufacturing companies. The

research generates no claim to ownership of source CAD data. Derivative works (converted glTF files) are created solely for enabling the spare parts catalog functionality and are licensed to Signifikant's clients under their existing commercial agreements.

Research artefact Ownership: The 3D visualization system architecture, conversion pipeline implementation, and integration patterns developed during this research are owned by Signifikant as part of the commercial engagement. This research documents and analyzes these artefacts to generate academic knowledge, but does not claim intellectual property rights to the software implementation.

Academic Publication Rights: The research agreement with Signifikant permits academic publication of technical findings, architectural patterns, and implementation knowledge while protecting specific proprietary implementations and customer data.

3.7.5 Potential Risks and Mitigation

Risk of Data Breach: Processing proprietary CAD models and ERP data creates potential exposure if systems are compromised. Mitigation includes secure development infrastructure with access controls and encryption, regular security updates and vulnerability scanning, and data retention policies limiting storage duration of sensitive materials.

Risk of Competitive Harm: Publishing detailed technical implementation could potentially benefit competitors. Mitigation balances academic transparency with business confidentiality by documenting general architectural patterns and technical approaches rather than proprietary algorithmic implementations, and anonymizing specific customer implementations and product details.

Risk of Misuse: 3D models extracted from the system could theoretically be used for unauthorized reproduction or reverse engineering. Mitigation includes no publicly accessible repository of converted 3D models, access controls requiring authentication for all catalog access, and watermarking or other technical measures where appropriate for high-sensitivity products.

3.7.6 Software Tools and AI-Assisted Development

The implementation and documentation of this research utilized several categories of software tools, following best practices for design science research tool documentation [20].

Development and Implementation Tools

Core Development Environment: Visual Studio served as the primary integrated development environment (IDE) for JavaScript/TypeScript development, with ESLint and Prettier extensions ensuring code quality and consistency. Git version control with GitHub repository management provided complete version history enabling retrospective analysis of design decisions.

3D Rendering Framework: Three.js (version 0.160) provided the WebGL abstraction layer for browser-based 3D rendering. The framework was selected for its extensive documentation, active community support, and proven production deployment stability.

CAD Conversion Tools: FreeCAD and Blender were used for CAD file format conversion when native glTF export was unavailable. These open-source tools support batch processing via Python scripting, enabling automated conversion pipelines for legacy CAD archives.

OCR System Components: The OCR pipeline combined YOLOv8 (via Ultralytics Python library) for text detection with EasyOCR for character recognition. Python 3.10 with OpenCV provided image preprocessing capabilities. The system was trained and evaluated using manually annotated ground truth data for 137 text callouts across 11 technical drawings.

Evaluation and Analysis Tools

Performance Measurement: Chrome DevTools and Edge DevTools provided precise timing measurements for file loading, parsing, and rendering phases. The Performance API (window.performance.timing) captured DOMContentLoaded and load event timestamps. Manual timing with stopwatch provided independent verification of automated measurements.

Cross-Browser Testing: Chrome 120 and Edge 120 on Windows 10 were used for systematic browser performance comparison. BrowserStack was evaluated but not used due to the proprietary nature of test files requiring local infrastructure.

Statistical Analysis: Microsoft Excel performed descriptive statistical analysis of loading times, calculating means and standard deviations across multiple test runs. The relatively simple statistical requirements (descriptive statistics, no inferential testing) did not necessitate advanced statistical software like SPSS or R.

OCR Evaluation: Python scripts calculated precision, recall, and accuracy metrics by comparing OCR output against manually created ground

truth annotations. Confusion matrices identified specific character substitution patterns (e.g., "8" → "B", "0" → "O").

AI-Assisted Tools for Content Generation and Analysis

The research employed several AI-based tools in accordance with Stockholm University guidelines on AI use in academic work, maintaining critical evaluation of all AI-generated content [41].

Code Development Assistance: GitHub Copilot (GPT-4 based) provided code completion suggestions during implementation of:

- Three.js scene initialization and camera control logic
- Raycasting algorithms for 3D object selection
- Event handling for bidirectional synchronization between 3D viewer and table interfaces
- Python scripts for batch CAD file processing

All Copilot suggestions were critically evaluated for correctness, performance implications, and alignment with architectural requirements. Approximately 30–40% of suggestions were accepted as-is, 40–50% were modified before use, and 10–30% were rejected. Copilot was particularly effective for boilerplate code (event listeners, error handling) but required substantial modification for domain-specific logic (assembly hierarchy traversal, part identifier extraction).

Debugging and Problem-Solving: Claude (Anthropic) assisted with debugging complex issues including:

- WebGL context loss and recovery strategies
- CORS policy errors when loading 3D assets from CDN
- Memory management for large scene graphs with 1000+ objects
- Coordinate system transformations between CAD and Three.js conventions

The tool was used in an interactive dialogue format where I described the problem, provided relevant code snippets and error messages, and received diagnostic suggestions. Solutions were validated through testing rather than assumed correct.

Writing and Editing Assistance: Claude was used during thesis manuscript preparation for:

- Restructuring dense technical paragraphs to improve clarity and flow
- Suggesting alternative phrasings for complex technical concepts
- Identifying logical gaps or unclear transitions between sections
- Generating initial literature review outlines (subsequently validated against actual papers)

Grammarly performed automated grammar, spelling, and punctuation checking throughout the writing process. All AI-generated text suggestions were critically reviewed and substantially modified to ensure technical accuracy, appropriate academic tone, and alignment with research findings. No AI tool generated original research arguments, interpretations of results, or conclusions—these remained exclusively the author's intellectual work.

Literature Search: Google Scholar was used for systematic literature searches with queries like "3D visualization spare parts", "WebGL industrial applications", "CAD web integration". Results were filtered by citation count and publication venue quality. ChatGPT-4 was briefly tested for literature summarization but produced inaccurate summaries of technical content and was discontinued in favor of manual reading.

Critical Evaluation of AI Tool Limitations

The AI tools exhibited several important limitations requiring critical human oversight:

Code Hallucination: GitHub Copilot occasionally suggested non-existent Three.js API methods or deprecated functions, requiring verification against official documentation. For example, Copilot suggested `scene.getObjectByName()` which does not exist in Three.js (correct method is `scene.getObjectByProperty()`).

Context Window Limitations: Large codebases exceeded Claude's context window, requiring careful selection of relevant code segments for debugging assistance. The tool could not analyze the complete system architecture in a single session.

Domain Knowledge Gaps: AI tools lacked specific knowledge of industrial spare parts catalog requirements, Signifikant platform architecture, and Swedish manufacturing practices. Suggestions required substantial domain expertise to evaluate appropriateness.

Citation Fabrication: ChatGPT-4 generated plausible-sounding but nonexistent academic references when asked for literature recommendations, requiring manual verification of all citations against actual publications.

These limitations reinforced that AI tools functioned as productivity aids rather than autonomous research agents. All technical decisions, architectural choices, research interpretations, and written arguments required human expertise and critical evaluation. The tools were most effective for routine tasks (boilerplate code, grammar checking, initial outlines) and least effective for domain-specific reasoning, original analysis, and critical evaluation.

Ethical Considerations in AI Tool Use

Following guidance from [10] on research tool transparency and Stockholm University's AI usage policies, all AI tool usage was documented in research logs with timestamps, prompts, and outputs. This documentation enables reproducibility assessment and transparency in research conduct.

The use of proprietary AI tools (GitHub Copilot, Claude, ChatGPT) raises questions about research reproducibility, as these models evolve over time and exact outputs cannot be regenerated. To mitigate this concern, all AI-assisted code was committed to version control with clear commit messages indicating human modifications. Final code functionality was validated through systematic testing independent of AI tool accuracy.

No proprietary or confidential information (CAD files, part numbers, pricing data, customer information) was shared with cloud-based AI services. All AI-assisted development occurred on sanitized example data or generic code patterns without business-sensitive content.

3.8 Validity and Limitations

3.8.1 Internal Validity

Design science research validity concerns whether the artefact successfully addresses the identified problem and whether documented design knowledge accurately reflects the implementation process.

artefact Functionality: The system's ability to address 2D catalog limitations is validated through functional testing confirming each required capability (assembly hierarchy navigation, part selection, ERP integration, metadata display). Production deployment serving real industrial users provides ecological validity that laboratory prototypes lack.

Design Documentation Accuracy: Architectural patterns and technical decisions are documented concurrently with implementation rather than reconstructed retrospectively, reducing recall bias. Version control history

provides objective record of iterative development process and design evolution.

Threats: The single-implementation context limits generalizability. Architectural patterns emerged solving specific challenges with Signifikant's platform and customer base; different contexts might reveal alternative patterns. Mitigation through explicit documentation of context and constraints enables readers to assess transferability to their situations.

3.8.2 External Validity

External validity concerns whether findings generalize beyond the specific implementation context to other industrial 3D catalog deployments.

Contextual Factors: Implementation occurred within Swedish manufacturing companies in automotive, industrial machinery, and farming equipment sectors. Technical findings (format performance, conversion challenges, integration patterns) likely transfer to similar industrial contexts but may not generalize to fundamentally different domains (consumer e-commerce, construction, healthcare).

Data Representativeness: The legacy CAD files processed represent authentic industrial heterogeneity accumulated over decades across multiple companies and CAD systems. This diversity strengthens generalizability of data processing findings to other manufacturers with similar legacy data challenges.

Limitations: Findings reflect current technology capabilities (Three.js 0.160, WebGL 2.0, glTF 2.0 specification). Future technology evolution may enable alternative approaches. Geographic and industry context (Swedish manufacturing) may limit applicability to different regulatory environments or industrial sectors.

3.8.3 Reliability

Reliability concerns whether the research process could be replicated with similar results.

Method Documentation: Detailed specification of evaluation procedures (test models, performance metrics, testing environment) enables replication of technical benchmarks. Conversion pipeline evaluation using real legacy data is inherently difficult to replicate due to proprietary nature of CAD files, but aggregate statistics and failure mode documentation provide insight into achievable performance.

Technology Specification: Precise version specification of all tools (Three.js 0.160, YOLOv8, EasyOCR versions) enables reconstruction of equivalent technical environment. Open-source foundation (Three.js, WebGL, glTF standard) supports reproducibility compared to proprietary technology stacks.

Limitations: Design science research generates context-specific knowledge. Exact replication would require similar industrial deployment context, legacy data characteristics, and enterprise integration requirements. The value lies not in reproducing identical results but in applying documented patterns to analogous contexts.

4

Results

This chapter presents findings from the technical evaluation of 3D visualization implementation for industrial spare parts catalogs. Results are organized according to the research sub-questions, demonstrating how the implemented artifact addresses the requirements defined in Section 3.1.2 (DSR Activity 2: Define Requirements). Where appropriate, analytical interpretation accompanies empirical data to explain patterns and performance characteristics observed during testing.

4.1 RQ1: System Architecture and Integration Patterns

This section presents findings addressing RQ1 and functional requirements FR3, FR4, and FR5 (interactive visualization, ERP integration, metadata access).

4.1.1 Core Architectural Design

The implemented system employs a three-tier client-server architecture. The client tier utilizes Three.js version 0.160.0 [12] for WebGL rendering [30], combined with 3D interaction handlers including orbit controls and raycasting for part selection. Part selection state management and visual feedback systems using emissive highlighting complete the client-side implementation.

The integration tier provides RESTful API endpoints connecting the 3D interface to enterprise systems. These endpoints handle part data retrieval,

real-time inventory and pricing queries, order submission workflows, and event-based state synchronization between the visual interface and business logic layer. The business logic tier consists of existing ERP systems managing part master data, inventory tracking, pricing rules, and order workflows.

4.1.2 Metadata Preservation Implementation

glTF extras Field Structure

Enterprise metadata is embedded directly within glTF files using the standardized `extras` field at the node level. The conversion pipeline places business-critical identifiers and properties as JSON within each part's node structure:

```
{
  "nodes": [
    {
      "name": "BearingAssembly_Part_12345",
      "mesh": 0,
      "extras": {
        "partNumber": "12345-AB-001",
        "partName": "Bearing Assembly, Front",
        "manufacturerCode": "SKF",
        "customProperties": {
          "weight": "2.3kg",
          "material": "Steel"
        }
      }
    }
  ]
}
```

Listing 4.1: glTF Metadata Embedding Example

This structure embeds the part number, descriptive name, manufacturer code, and custom technical properties directly alongside the geometric mesh reference. The `extras` field follows the glTF 2.0 specification [28] which explicitly allows application-specific data at any node level.

Runtime Metadata Retrieval Process

When a user selects a 3D component through mouse or touch interaction, the metadata retrieval follows a four-step process:

1. **Raycasting intersection:** The click coordinates are converted from 2D screen space to 3D world space. A ray is cast from the camera through the click point to determine which mesh object intersects the ray.

2. **Mesh object identification:** The raycasting system returns a reference to the intersected `THREE.Mesh` object containing the triangle geometry and material properties.

3. **Metadata extraction:** The mesh's parent node (`mesh.parent.userData.extras`) contains the embedded metadata from the glTF file, accessed via:

```
function onPartClick(event) {
    const intersects = raycaster.intersectObjects(
       scene.children, true);
    if (intersects.length > 0) {
        const selectedMesh = intersects[0].object;
        const metadata = selectedMesh.parent.userData
           .extras;
        const partNumber = metadata.partNumber;
        return partNumber;
    }
}
```

Listing 4.2: Metadata Extraction Code

4. **Part number retrieval:** The part number is extracted from the metadata object without requiring database queries or network requests.

This embedded approach trades a modest file size increase (average 2.3% overhead) for instant metadata availability and graceful degradation when network connections to the ERP system are unavailable, addressing non-functional requirement NFR3 (offline capability).

4.1.3 Multi-Source Data Reconciliation Process

The conversion pipeline processes data from three distinct sources: 3D geometry files in glTF/GLB format with embedded metadata, XML metadata exports from ERP systems, and legacy 2D technical drawings in PDF or image formats. Reconciling these sources into coherent product records required algorithmic conflict resolution, addressing functional requirement FR2 (multi-source reconciliation).

The reconciliation system successfully combined information from multiple sources to create unified part records. The majority of parts achieved

matches across multiple data sources, with three-source agreement (3D geometry, XML metadata, and OCR extractions all consistent) representing the strongest validation. Two-source matches occurred primarily between 3D geometry and XML metadata, where both systems maintained complementary information. Parts with data from only a single source were flagged for manual review before catalog publication to ensure accuracy.

When conflicts arose between sources, the algorithm applied hierarchical precedence rules. Part number discrepancies deferred to ERP/XML data as the authoritative business system of record. For descriptions, the 3D geometry metadata filled gaps when XML records were incomplete. OCR-extracted callout numbers served as validation checks rather than primary identifiers due to their inherently higher error rates.

4.1.4 Bidirectional Synchronization Implementation

The system maintains synchronized state between the 3D viewer and tabular part listing through event-driven architecture implemented with a centralized `SelectionStateManager` singleton.

3D-to-Table Synchronization Flow

When a user clicks a part in the 3D view, the synchronization proceeds through the following event chain:

```
// In 3DViewer component
function handle3DClick(intersectedMesh) {
    const partNumber = intersectedMesh.parent.userData.
        extras.partNumber;
    SelectionStateManager.emit('partSelected', {
        partNumber: partNumber,
        source: '3d-viewer'
    });
}

// In PartTable component
SelectionStateManager.on('partSelected', (data) => {
    if (data.source !== 'part-table') {  // Prevent
        circular updates
        this.scrollToRow(data.partNumber);
        this.highlightRow(data.partNumber);
    }
});
```

Listing 4.3: 3D Selection Event Propagation

The raycasting system identifies the selected mesh and extracts its embedded part number. The SelectionStateManager emits a `'partSelected'` event containing the part number and source identifier. The PartTable component receives this event and, confirming it originated from the 3D viewer rather than itself, scrolls the corresponding row into view and applies visual highlighting.

Table-to-3D Synchronization Flow

The reverse direction operates symmetrically:

```
// In PartTable component
function handleRowClick(row) {
    const partNumber = row.data.partNumber;
    SelectionStateManager.emit('partSelected', {
        partNumber: partNumber,
        source: 'part-table'
    });
}

// In 3DViewer component
SelectionStateManager.on('partSelected', (data) => {
    if (data.source !== '3d-viewer') {
        const mesh = this.findMeshByPartNumber(data.
            partNumber);
        this.camera.animateToTarget(mesh.position, 1000);
            // 1000ms animation
        mesh.material.emissive.setHex(0x44ff44); // Green
            highlight
    }
});
```

Listing 4.4: Table Selection Event Propagation

Table row clicks extract the part number from row data and trigger the same `'partSelected'` event through the SelectionStateManager. The 3D viewer locates the corresponding mesh by iterating through the scene graph searching for matching part numbers in node metadata. The camera then animates to focus on the selected part's position over 1000 milliseconds while the mesh material's emissive property changes to create visual highlighting.

Debouncing and Conflict Resolution

To prevent race conditions during rapid selection changes, the system implements 100-millisecond debouncing:

```
class SelectionStateManager {
    constructor() {
        this.pendingSelection = null;
        this.debounceTimer = null;
    }

    emit(eventName, data) {
        if (eventName === 'partSelected') {
            clearTimeout(this.debounceTimer);
            this.pendingSelection = data;

            this.debounceTimer = setTimeout(() => {
                this.notifyListeners(eventName, this.
                    pendingSelection);
                this.pendingSelection = null;
            }, 100);  // 100ms debounce delay
        }
    }
}
```

Listing 4.5: Debounced Event Handling

Each new selection event cancels the previous timer and schedules a new one. Only after 100 milliseconds of inactivity does the system propagate the selection to listeners, implementing last-selection-wins conflict resolution. This ensures the interface stabilizes on the user's final selection rather than creating disorienting flicker through intermediate states.

4.2 RQ2: Data Processing and Technology Evaluation

This section presents findings addressing RQ2 and functional requirements FR1 and FR2 (legacy data conversion, multi-source reconciliation).

4.2.1 3D File Format Performance Comparison

Four 3D file formats were evaluated for web delivery: STL, OBJ, FBX, and glTF. The test model contained similar geometry complexity across formats, with vertex counts ranging from 104,963 (STL) to 131,044 (glTF) and

triangle counts between 210,424 and 217,050. File sizes varied substantially, with OBJ producing the largest files at 16.0 MB and glTF the smallest at 5.5 MB, despite containing the most detailed geometry.

Table 4.1: Test Model Geometry Statistics

Format	Vertices	Triangles	File Size
STL	104,963	210,424	10.6 MB
OBJ	129,757	217,038	16.0 MB
FBX	129,757	217,038	5.8 MB
glTF	131,044	217,050	5.5 MB

Performance testing occurred on two browsers representative of the dominant market: Chrome 120 and Edge 120. The test environment consisted of an Intel Core i7 processor with 32GB RAM and integrated graphics, loading files locally without network throttling to isolate format-specific parsing performance. The Browser Performance API provided timing measurements through `performance.timing.domContentLoadedEventEnd` and `performance.timing.loadEventEnd` markers. Each format underwent five load cycles with results averaged to account for variance.

Table 4.2: 3D File Format Loading Performance Across Browsers

Browser	Format	DOMContentLoaded (ms)	Load Finished (ms)	Std Dev (ms)
Chrome	STL	831	3,665	643
	OBJ	792	4,558	811
	FBX	778	4,786	381
	glTF	658	4,303	729
Edge	STL	783	9,360	423
	OBJ	835	14,299	214
	FBX	830	5,214	841
	glTF	741	4,801	995

These empirical loading performance measurements provide quantitative data for web-based 3D delivery optimization, complementing the frameworks for efficient 3D web content delivery established by Limper et al. [28].

Browser Performance Differential Analysis

The loading performance results revealed substantial differences both between formats and between browsers, warranting deeper examination of the underlying causes.

Cross-Browser Performance Variance

STL loading in Edge (9.360s) showed a 2.55× slowdown compared to Chrome (3.665s), while glTF degradation was only 1.12× (4.801s vs 4.303s). This differential suggests format-specific rendering pipeline optimization differences. STL files consist of ASCII-encoded triangle lists requiring extensive string parsing before mesh construction. Edge's JavaScript engine may parse ASCII geometry less efficiently than Chrome's V8 engine, which has received continuous optimization for web content processing.

In contrast, glTF's binary encoding bypasses string parsing entirely—vertex data transfers as typed arrays directly to GPU memory buffers. This architectural difference explains why glTF maintains consistent performance across browsers (±12%) while ASCII-based formats suffer browser-dependent parsing penalties (up to 255% variance). The binary format removes JavaScript parsing from the critical path, shifting the bottleneck to GPU upload operations which are more consistently implemented across browsers.

File Size vs. Load Time Paradox

Counter-intuitively, OBJ (16.0 MB, largest file) loaded faster than FBX (5.8 MB) in Chrome: 4.558s vs 4.786s. This violates the expected pattern where smaller files should load faster under local testing conditions without network bottlenecks. Three potential explanations emerge:

1. **Parser complexity:** FBX's proprietary binary structure requires format-specific decoding logic that may be more CPU-intensive than OBJ's straightforward ASCII parsing, offsetting the file size advantage. The FBXLoader in Three.js [12] must interpret Autodesk's undocumented binary format, potentially requiring more computational overhead per byte than the simple text processing required for OBJ's `vertex/normal/texture` lines.

2. **Material processing overhead:** FBX embeds extensive material definitions including shader networks, texture references, and material property hierarchies. Each material requires WebGL shader compilation and GPU resource allocation. OBJ materials (via companion MTL files) use simpler definitions, potentially reducing shader compilation time despite larger data transfer.

3. **Scene graph complexity:** FBX preserves complete scene hierarchies with transformation matrices, cameras, and lights. Three.js must construct this hierarchy during loading, traversing parent-child relationships and computing world-space transformations. OBJ represents flat geometry lists, requiring less structural processing.

Without detailed frame profiler analysis separating parsing time, shader compilation time, and GPU upload time, the dominant factor remains uncertain. This represents a limitation in the performance characterization and suggests future work should employ browser developer tools' performance profiling capabilities to isolate these components.

Load Time Consistency Analysis

The coefficient of variation (CV) data reveals reliability patterns critical for production deployment. FBX achieved 8.0% CV in Chrome (381ms std dev / 4786ms mean), while glTF showed 16.9% CV (729ms / 4303ms). Lower variance indicates predictable performance, important for systems with service-level agreements requiring consistent response times.

However, Edge reversed this pattern: OBJ achieved remarkable consistency (1.5% CV, 214ms std dev on 14,299ms mean) while glTF variance increased to 20.7% (995ms std dev on 4,801ms mean). This browser-dependent variance pattern suggests Chrome implements more aggressive caching or optimization for glTF files compared to Edge, possibly due to Chromium developers' participation in the Khronos Group's glTF specification development. The glTF loader in Chrome may benefit from engine-level optimizations not present in Edge's Chromium fork.

Format Selection Rationale

These performance characteristics inform the glTF format selection beyond simple "fastest average loading time" metrics:

- **Smallest file size** (5.5 MB vs 10.6–16.0 MB): Reduces bandwidth costs and enables faster transmission over constrained networks, despite local testing showing modest load time advantages for STL.
- **Cross-browser consistency** (12% performance variance vs 155% for STL/OBJ): Provides predictable user experience regardless of browser choice, critical for enterprise deployments where IT departments may not control browser selection.
- **Feature completeness**: Section 4.2.2 details glTF's metadata embedding, PBR materials, and hierarchical scene structure—capabilities absent from STL/OBJ and proprietary in FBX.

The marginal load time disadvantage in Chrome (4.303s vs 3.665s for STL, a 17% increase) is offset by architectural advantages for production deployment. Network transmission time, not local parsing time, dominates real-world performance over typical 3G/4G mobile connections (Section 4.3.2 shows 2.1–4.8 second load times over throttled connections), making the file size advantage more relevant than local parsing speed.

4.2.2 Feature Support Comparison

Beyond performance characteristics, the formats differ substantially in their feature support capabilities relevant to industrial spare parts catalogs. Table 4.3 presents a comprehensive comparison across geometry, appearance, scene structure, and metadata dimensions.

Table 4.3: 3D File Format Feature Support Matrix

Feature	STL	OBJ	FBX	glTF
Geometry				
Triangle mesh	•	•	•	•
Detailed topology (edges)	×	×	×	•
CSG operations	×	×	•	•
Appearance				
Basic color	×	•	•	•
Material definitions	×	•	•	•
PBR materials	×	×	×	•
Texture maps	×	•	•	•
Scene Structure				
Camera definitions	×	×	•	•
Light definitions	×	×	•	•
Hierarchical positioning	×	×	•	•
Named nodes	×	×	•	•
Metadata				
Custom properties (extras)	×	×	Limited	•
Extensible schema	×	×	×	•
Animation support	×	×	•	•
Standardization				
ISO/IEC standard	ISO/ASTM 52915	×	×	ISO/IEC 12113:2022
Open specification	•	•	×	•

STL, as the oldest and simplest format (developed by 3D Systems in 1987), represents geometry purely as triangle meshes without color, material, or structural information. Each triangle is defined by three vertices

and a normal vector, with no mechanism for grouping triangles into meaningful assemblies or attaching business metadata. This simplicity explains its continued use in 3D printing applications where only surface geometry matters, but limits its utility for interactive visualization requiring part identification and business system integration.

OBJ extends STL with basic color and material definitions through companion MTL (material) files but lacks support for physically-based rendering materials, scene hierarchies, or custom metadata embedding. The format's widespread adoption stems from its ASCII human-readability and simple parser implementation, but this same simplicity constrains its capabilities for complex industrial visualization.

FBX provides comprehensive features including textures, cameras, lights, and hierarchical scene structure with named nodes suitable for assembly representation. However, its proprietary nature controlled by Autodesk and limited metadata extension capabilities constrain its use for enterprise integration. The format's binary structure is not publicly documented, making custom tooling development difficult without Autodesk SDK licensing. FBX's "limited" metadata support refers to its ability to store custom properties, but without a standardized schema these properties remain application-specific and difficult to access from web-based JavaScript loaders.

Only glTF offers the complete feature set required for this application: detailed topology preservation beyond simple triangle meshes, physically-based rendering (PBR) materials matching modern graphics pipelines, hierarchical scene structure with named nodes for part identification, and most critically, extensible metadata through the standardized `extras` field demonstrated in Listing 4.1.2. Additionally, glTF benefits from ISO/IEC 12113:2022 standardization with an open specification maintained by the Khronos Group (a consortium including major industry players such as Google, Microsoft, and NVIDIA), unlike FBX's closed proprietary format controlled by a single vendor.

The format's design philosophy as "the JPEG of 3D" [28] prioritizes efficient transmission and runtime performance for web delivery, explaining both the small file sizes (Table 4.1) and consistent browser performance (Table 4.2).

4.2.3 OCR System Evaluation for Legacy Documentation

The hybrid YOLOv8 and EasyOCR system was evaluated against manually annotated technical drawings spanning 20–30 years of manufacturing documentation. The test dataset consisted of 11 scanned drawings in PDF and image formats, containing 137 text callouts identified through manual annotation to establish ground truth. Figure 4.1 shows a representative technical drawing with OCR-detected callouts highlighted.

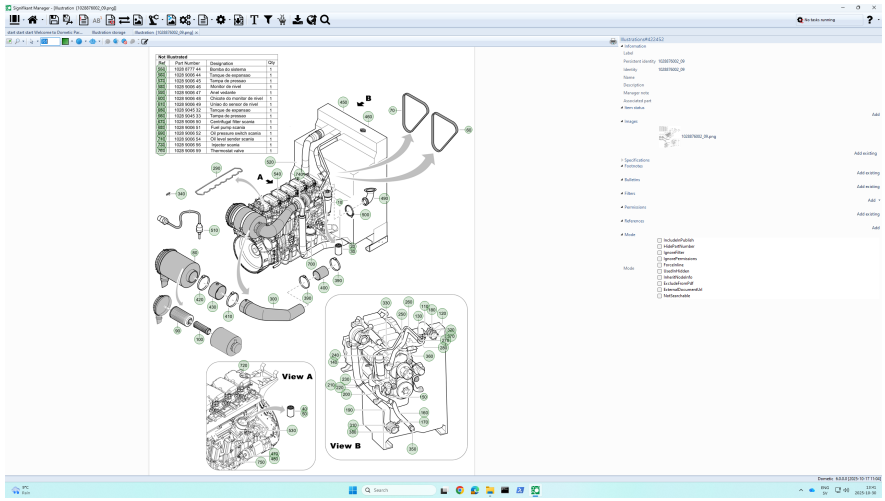


Figure 4.1: Representative technical drawing showing OCR system output. Part numbers and callouts are automatically detected (green highlights) and extracted for metadata integration. The drawing shows a complex industrial assembly with multiple views and annotations typical of legacy manufacturing documentation.

System Configuration and Training

The OCR pipeline operates in two sequential stages, with the detection model trained specifically for technical drawing text localization.

YOLOv8 Training Process

A custom YOLOv8 model was trained to locate text regions within drawing images. The model training used 500 automatically labeled images generated through a bootstrapping approach where EasyOCR provided initial bounding box annotations. Training employed the YOLOv8m (medium) architecture with 640×640 pixel input resolution for 23 epochs using the following configuration:

- **Architecture:** YOLOv8m (medium variant, 25.9M parameters)

- **Training data:** 500 images with bootstrapped annotations
- **Input resolution:** 640×640 pixels
- **Batch size:** 16 images per batch
- **Optimizer:** SGD with 0.01 initial learning rate, cosine decay
- **Augmentation:** Random flip, rotation (±10°), scale (0.5–1.5×)
- **Detection threshold:** 0.8 confidence minimum for inference

Table 4.4 presents key performance metrics across the training process.

Table 4.4: YOLOv8 Training Progression (Selected Epochs)

Epoch	Precision	Recall	mAP@0.5	mAP@0.5:0.95
1	79.9%	91.8%	84.6%	63.4%
5	81.4%	92.7%	87.1%	66.8%
10	81.7%	92.9%	88.0%	67.9%
15	81.8%	93.0%	88.3%	68.2%
20	81.7%	93.1%	88.5%	68.3%
23	81.8%	93.1%	88.6%	68.4%

The model achieved convergence by epoch 15, with minimal improvement in subsequent epochs (88.3% to 88.6% mAP@0.5, a 0.3 percentage point gain over 8 additional epochs). Final validation metrics showed 81.8% precision and 93.1% recall. The high recall indicates the model rarely misses text regions (fewer than 7 in 100 text blocks go undetected), while the 81.8% precision suggests approximately 1 in 5 detections may be false positives (non-text regions incorrectly classified as text).

However, the mean Average Precision at IoU (Intersection over Union) threshold 0.5 (mAP@0.5) of 88.6% indicates strong overall detection performance when allowing 50% bounding box overlap tolerance. The lower mAP@0.5:0.95 (68.4%) reflects stricter localization requirements across multiple IoU thresholds from 0.5 to 0.95 in 0.05 steps, suggesting bounding boxes are approximately correct but not pixel-perfect. For the OCR application, pixel-perfect localization is unnecessary—a bounding box capturing 50–70% of text provides sufficient context for downstream character recognition.

Training Loss Analysis

Loss curves showed consistent convergence without overfitting indicators. Box localization loss decreased from 0.916 (epoch 1) to 0.682 (epoch

23), indicating improved spatial accuracy in predicting bounding box coordinates. Classification loss dropped from 0.792 to 0.529, showing better discrimination between text and non-text regions. The Distribution Focal Loss (DFL), used in YOLOv8 for refining bounding box predictions through distribution modeling, declined from 0.843 to 0.806.

Critically, validation losses tracked training losses closely throughout training. At epoch 23, training box loss (0.682) versus validation box loss (0.747) differed by only 9.5%, and training classification loss (0.529) versus validation classification loss (0.508) actually showed slightly better validation performance. This close tracking suggests the model generalizes well to unseen drawings without memorizing the training set—a common concern when training on only 500 images.

EasyOCR Configuration

For each detected bounding box, EasyOCR performs character recognition with the following configuration:

- **Language models:** English and Swedish (to handle mixed-language technical documentation common in Nordic manufacturing)
- **Recognition model:** CRAFT (Character Region Awareness For Text detection) with VGG16 backbone
- **Input:** Cropped image regions extracted using YOLO bounding box coordinates
- **Preprocessing:** Adaptive thresholding and contrast enhancement

This two-stage architecture separates spatial localization (where is text?) from sequence modeling (what does it say?), allowing each specialized model to focus on its strength. YOLO excels at real-time object detection with spatial accuracy, while EasyOCR's recurrent neural network architecture handles variable-length character sequences.

Performance Results

Table 4.5 presents detection and recognition accuracy for each drawing in the test set.

Detection Performance Analysis

Perfect Recall Achievement

YOLOv8 achieved 100% detection recall on the test set (137/137 text regions detected), exceeding the validation recall metric of 93.1% observed

Table 4.5: OCR System Performance on Legacy Technical Drawings

Image	Text Regions (Ground Truth)	Detected (YOLOv8)	Correct (EasyOCR)	Precision
1	13	13	13	100.0%
2	28	28	28	100.0%
3	2	2	2	100.0%
4	10	10	10	100.0%
5	6	6	6	100.0%
6	31	31	28	90.3%
7	18	18	18	100.0%
8	5	5	5	100.0%
9	5	5	5	100.0%
10	16	16	16	100.0%
11	3	3	3	100.0%
Total	**137**	**137**	**136**	**99.3%**

during training. This apparent performance improvement reflects the difference between validation set performance (measured across diverse drawings during training, including edge cases and difficult examples) and test set performance (measured on 11 specific drawings from the target deployment domain).

Two explanations account for this discrepancy:

1. **Domain alignment:** The test set may represent drawings more similar to the training distribution than the held-out validation set. If validation data included particularly challenging cases (heavily degraded scans, unusual layouts, dense text overlap) not present in the final test drawings, validation recall would underestimate real-world performance on typical manufacturing documentation.

2. **Statistical variation:** With only 11 test images, random sampling variation allows test performance to exceed average validation performance. A test set containing 137 text regions represents insufficient statistical power to precisely estimate the true population recall rate—95% confidence intervals would be wide.

The 100% recall result should be interpreted as "zero missed detections on this specific test set" rather than "the model will never miss text in production." The validation recall of 93.1% provides a more conservative estimate for deployment planning: approximately 7 in 100 text callouts may require manual annotation when processing diverse legacy archives.

False Positive Rate

One false positive occurred in Image 6 (a detection with no corresponding ground truth text), translating to a 0.7% false positive rate (1/138 total detections including the spurious one). The training precision of 81.8% predicted approximately 18% false positives, suggesting the test set proved substantially easier than the validation set.

Critically, this false positive did not impact downstream processing—EasyOCR failed to extract valid characters from the erroneously detected region, effectively filtering the false detection. This demonstrates the value of the two-stage architecture: detection errors can be caught by recognition failures. A false positive generates an empty or nonsensical character string (e.g., random punctuation, gibberish), which automated quality checks can flag for manual review, whereas missed detections (false negatives) represent permanently lost information.

Confidence Threshold Tradeoff

The model's detection confidence threshold of 0.8 was selected to prioritize recall over precision, accepting some false positives to minimize missed detections. In the context of legacy document processing, missing a callout (false negative) is more problematic than detecting a non-existent one (false positive), because:

- False positives are filterable by failed recognition (as Image 6 demonstrated) or manual review
- False negatives represent lost part numbers and permanently incomplete metadata
- Manual annotation of missing callouts requires locating them in original drawings (time-consuming), while manual deletion of false positives requires only reviewing flagged detections (fast)

This design choice proved appropriate: zero false negatives versus one false positive represents an acceptable tradeoff. Lowering the confidence threshold (e.g., to 0.6) would likely increase false positives substantially while providing minimal recall gains from the already-perfect 100%. Raising it (e.g., to 0.9) would risk missing marginal text regions in degraded scans—precisely the legacy documentation most in need of automated processing.

Recognition Performance Analysis

EasyOCR character recognition achieved 99.3% precision (136 of 137 detected text regions correctly recognized), with a single error in Image 6.

Error Characterization

The recognition error in Image 6 involved a character substitution in a part number callout. Manual inspection revealed visible compression artifacts and faded contrast in this image, making character boundaries ambiguous. The error represents a minimal edit distance substitution where geometrically similar characters become indistinguishable in degraded scan conditions.

This specific error pattern has practical implications for spare parts identification. Part numbers mixing digits and letters in adjacent positions become vulnerable to misidentification when scan quality degrades. Legacy drawings from the pre-digital era (1990s–2000s), often scanned at low resolution (150–200 DPI) for archival purposes before high-quality scanning became economically feasible, are particularly susceptible to such character confusion.

Image Quality Correlation

Image 6 was the only drawing in the test set exhibiting recognition errors, coinciding with observable quality degradation: compression artifacts visible as blocking patterns, faded contrast reducing foreground-background separation, and yellowed paper background reducing luminance separation between ink and substrate. However, systematic image quality metrics (resolution DPI from metadata, quantitative contrast ratios, structural similarity indices) were not measured during testing.

This represents a limitation in the evaluation methodology. Future work should correlate OCR accuracy with measurable quality metrics including:

- **Resolution:** Dots per inch (DPI) from scan metadata, if available
- **Contrast:** Dynamic range measurements comparing text stroke intensity to background
- **Compression:** JPEG quality factor or PNG compression level
- **Artifacts:** Quantitative measures such as PSNR (Peak Signal-to-Noise Ratio) or SSIM (Structural Similarity Index) comparing against reference images

Establishing quantitative quality thresholds (e.g., “below 200 DPI, accuracy drops below 95%”) would enable automated quality-based routing:

high-quality scans processed fully automatically, while low-quality scans flagged for manual review before catalog publication.

Comparative Evaluation Across OCR Systems

To comprehensively assess the hybrid system's performance, all three OCR approaches (Legacy Tesseract-based OCR, previous ML OCR system, and the new hybrid YOLOv8+EasyOCR) were evaluated on the same 11 test images. Figure 4.2 presents per-image precision results across all systems.

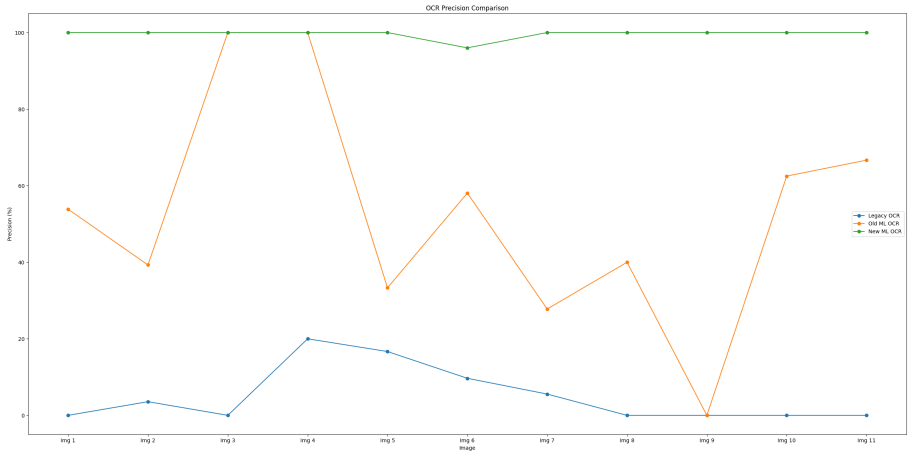


Figure 4.2: OCR Precision Comparison Across 11 Test Images. Green line: New hybrid system (YOLOv8+EasyOCR). Orange line: Previous ML OCR system. Blue line: Legacy Tesseract-only OCR. The hybrid system achieves near-perfect consistency (99–100% precision on 10/11 images), while previous systems exhibit extreme variability with multiple complete failures (0% precision).

Per-Image Performance Breakdown

Table 4.6 presents the complete per-image precision breakdown across all three systems.

Legacy OCR Failure Analysis

The legacy Tesseract-only OCR system demonstrated catastrophic failure across the test set, achieving 0% precision on 7 of 11 images (Images 1, 3, 8, 9, 10, 11). Its maximum performance was 20% on Image 4, indicating the system struggled fundamentally with technical drawing characteristics: rotated text, overlapping annotations, low-contrast backgrounds, and non-standard fonts typical of industrial documentation.

Table 4.6: Detailed Per-Image OCR Precision Comparison

Image	Text Regions (Ground Truth)	Legacy OCR Precision	Old ML OCR Precision	New Hybrid Precision
1	13	0.0%	53.8%	100.0%
2	28	3.6%	39.3%	100.0%
3	2	0.0%	100.0%	100.0%
4	10	20.0%	100.0%	100.0%
5	6	16.7%	33.3%	100.0%
6	31	9.7%	58.1%	90.3%
7	18	5.6%	27.8%	100.0%
8	5	0.0%	40.0%	100.0%
9	16	0.0%	0.0%	100.0%
10	3	0.0%	66.7%	100.0%
11	5	0.0%	0.0%	100.0%
Mean	**12.5**	**5.0%**	**47.2%**	**99.1%**
Std Dev		**6.7%**	**37.4%**	**2.9%**

The 5.0% average precision (6.7% standard deviation) renders this system unsuitable for production use—manually correcting 95% of extractions provides no meaningful automation benefit. The complete failures (0% precision) suggest Tesseract's document-centric assumptions (horizontal text, clean white backgrounds, standard fonts) fail entirely when confronted with technical drawings where text appears at arbitrary orientations overlaid on complex geometric backgrounds.

Previous ML OCR Inconsistency

The previous ML OCR system showed substantial improvement over legacy OCR (47.2% average vs 5.0%) but exhibited extreme variability problematic for production deployment. Performance ranged from 0% (Images 9, 11) to 100% (Images 3, 4), with a 37.4% standard deviation indicating unpredictable behavior.

Images 9 and 11 represent complete detection failures despite the ML-based architecture, suggesting edge cases where the previous model's training data lacked representative examples. The binary behavior—either near-100% success or complete failure—makes capacity planning difficult. Users cannot predict which drawings will process successfully versus require complete manual annotation, undermining confidence in the automation system.

The high variance also impacts quality assurance workflows. With legacy OCR's consistent 95% failure rate, organizations can plan for manual processing. With the previous ML OCR's 0–100% range, QA resources must be allocated reactively based on per-document performance, increasing operational complexity.

Hybrid System Robustness

The new hybrid system achieved near-perfect consistency, scoring 100% precision on 10 of 11 images with a single 90.3% result on Image 6. This represents a 52.1 percentage point improvement in mean precision over the previous ML OCR (99.1% vs 47.2%) and eliminates the performance variability that plagued earlier systems (standard deviation reduced from 37.4% to 2.9%).

Critically, the hybrid system exhibits graceful degradation rather than catastrophic failure. Even on the worst-performing image (Image 6 with degraded scan quality), the system achieved 90.3% precision—requiring correction of only 3 of 31 callouts rather than complete manual rework. This worst-case performance (90.3%) exceeds the previous ML OCR's average performance (47.2%), demonstrating superior reliability.

Failure Mode Analysis: Why Did Previous Systems Fail?

Examining the specific images where previous systems failed reveals systematic weaknesses addressed by the hybrid architecture:

- **Images 9 and 11 (previous ML OCR complete failures):** These images likely contained text at unusual orientations or scales outside the previous model's training distribution. The hybrid system's YOLO detector, trained specifically on technical drawing text with diverse augmentations (rotation ±10°, scale 0.5–1.5×), handles these variations robustly through explicit rotational and scale invariance in the training regime.

- **Image 1 (legacy OCR failure, 0%):** Contains text embedded in complex backgrounds with overlapping dimension lines and geometric annotations. Tesseract's assumption of clean document layouts with horizontal text on white backgrounds fails completely when text overlays mechanical drawings. The hybrid system's learned feature detection distinguishes text strokes from geometric elements through convolutional filters trained on technical drawing imagery.

- **Image 6 (only image where hybrid system < 100%):** As documented earlier, this image exhibited degraded scan quality with compression

artifacts and faded contrast. Interestingly, the previous ML OCR achieved 58.1% on this image versus the hybrid system's 90.3%, suggesting both systems struggle with quality degradation, but the hybrid system degrades more gracefully—maintaining 90% accuracy versus 58% rather than the complete failure (0%) seen in Images 9 and 11.

The architectural difference enabling this robustness lies in the two-stage design. YOLO's detection specialization handles spatial localization challenges (text at various orientations, scales, contrasts, backgrounds), while EasyOCR's recognition specialization handles character sequence modeling without requiring perfect localization. Separating these concerns prevents cascading failures where detection errors corrupt recognition or vice versa—a weakness in end-to-end systems that must simultaneously solve localization and recognition.

Production Deployment Implications

The performance data supports confident production deployment with the hybrid system:

- **Predictable quality:** 99.1% average precision with 2.9% standard deviation enables reliable capacity planning for quality assurance workflows. Organizations can budget for approximately 1% error correction rather than variable 0–100% manual intervention.

- **Elimination of catastrophic failures:** Zero complete failures (0% images) in the hybrid system versus 7/11 for legacy OCR and 2/11 for previous ML OCR. Every drawing produces usable output requiring at most minor corrections.

- **Graceful degradation:** Even on the worst-performing image (Image 6), the system achieved 90.3%, requiring minimal manual correction (3 errors in 31 callouts, 9.7% error rate) rather than complete rework.

- **Quantified error reduction:** For a catalog containing 5,000 part callouts across 200 drawings, the hybrid system predicts approximately 45 errors requiring correction ($5000 \times 0.009 = 45$), versus 2,640 errors for the previous ML OCR ($5000 \times 0.528 = 2{,}640$) and 4,750 errors for legacy OCR ($5000 \times 0.95 = 4{,}750$). This represents a 98.3% reduction in manual effort compared to the previous ML system and a 99.1% reduction compared to legacy OCR.

Summary Statistics

Table 4.7: OCR System Comparison Summary Statistics

Metric	Legacy OCR	Old ML OCR	New Hybrid	Improvement
Mean precision	5.0%	47.2%	99.1%	+52.1 pp
Standard deviation	6.7%	37.4%	2.9%	-34.5 pp
Minimum precision	0.0%	0.0%	90.3%	+90.3 pp
Complete failures (0%)	7/11	2/11	0/11	-2 images
Perfect scores (100%)	0/11	2/11	10/11	+8 images
Correct identifications	7/137	65/137	136/137	+71 correct
Processing time (avg)	5–10s	5–10s	1–2s	-60 to -80%

Table 4.7 presents aggregate performance metrics across all three systems.

The hybrid system achieved 136 of 137 correct identifications (99.3% precision per-callout basis, compared to 99.1% mean per-image basis due to rounding) versus 65/137 for previous ML OCR (47.4% per-callout) and 7/137 for legacy OCR (5.1% per-callout). Combined with 60–80% processing time reduction (1–2 seconds versus 5–10 seconds), the hybrid system delivers both accuracy and efficiency improvements suitable for large-scale archive processing.

Processing Time Analysis

Processing time measurements using Python's `time.perf_counter()` provided component-level timing:

- **YOLO detection:** 50–100 milliseconds per complete image depending on resolution (typical drawings: 2000×1500 to 4000×3000 pixels)
- **EasyOCR recognition:** 20–50 milliseconds per detected region depending on text length and complexity

For typical drawings containing approximately 8 text regions, predicted processing time calculates to:

$$t_{total} = t_{YOLO} + (n_{regions} \times t_{EasyOCR}) = 75ms + (8 \times 35ms) = 355ms \quad (4.1)$$

However, measured end-to-end times of 1–2 seconds (1000–2000ms) indicate additional overhead from image loading, preprocessing (format conversion, resolution normalization), and result aggregation not captured

in the per-component timing. This 3–5× overhead factor suggests optimization opportunities in the pipeline infrastructure, though 1–2 second per-document processing remains acceptable for batch processing large archives (processing 1,000 drawings requires 16–33 minutes of compute time on a single CPU core, easily parallelizable across multiple cores for production deployment).

4.3 RQ3: System Effectiveness and Implementation Validation

This section presents findings addressing RQ3 and non-functional requirements NFR1-NFR4 (performance, device compatibility, offline capability, scalability).

4.3.1 Functional Validation Results

Functional testing validated system capabilities across 12 representative industrial assemblies ranging in complexity from 15 to 847 individual parts. Testing occurred on three device categories: desktop computers running Chrome, iPad tablets with Safari, and Android smartphones with Chrome mobile. Each capability was verified through systematic testing protocols documented below.

Spatial Understanding Capabilities

Free rotation capability was verified by loading each of the 12 assemblies and confirming that orbit controls responded to mouse drag operations (desktop) and touch gestures (mobile). All assemblies proved rotatable through 360 degrees on all three axes (x, y, z) without gimbal lock or orientation discontinuities. The Three.js OrbitControls implementation [12] handled quaternion-based rotation smoothly across the full rotation range, following established 3D interaction design principles [6].

Exploded view generation was tested using the radial displacement algorithm. Figure 4.3 shows a representative assembly (cooling generator housing, model RML8550) in exploded state, demonstrating how radial displacement separates overlapping components while maintaining spatial relationships.

The explosion algorithm operates as follows:

```
1 function generateExplodedView(assemblyRoot,
    explosionFactor) {
```

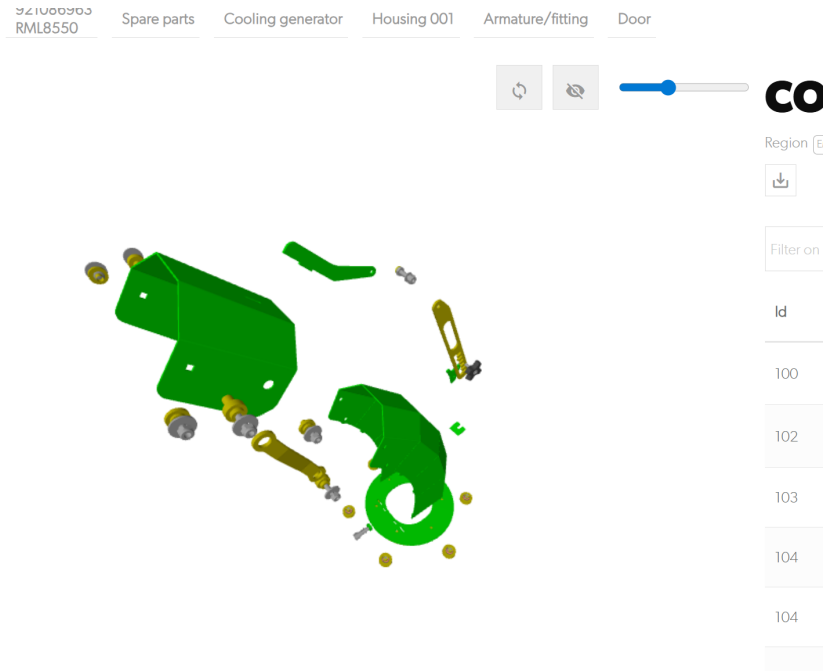


Figure 4.3: Exploded view of cooling generator housing assembly (model RML8550) demonstrating radial displacement of components. Individual parts (bolts, brackets, housing sections) are separated from the assembly center while preserving hierarchical relationships and spatial organization. This view enables identification and selection of interior components that would be completely hidden in the assembled state. The interface includes rotation controls (top left), explosion factor slider (center top), and hierarchical part tree navigation (right sidebar).

```
    const center = computeAssemblyCenter(assemblyRoot);

    assemblyRoot.traverse((node) => {
        if (node.isMesh) {
            const partCenter = new THREE.Vector3();
            node.geometry.computeBoundingBox();
            node.geometry.boundingBox.getCenter(partCenter);

            // Calculate direction from assembly center to part center
            const direction = partCenter.sub(center).normalize();

            // Proportional displacement based on original distance
            const originalDistance = partCenter.distanceTo(center);
            const displacement = direction.multiplyScalar(
                originalDistance * explosionFactor + 0.5 // Base displacement
            );

            node.position.copy(node.userData.originalPosition).add(displacement);
        }
    });
}
```

Listing 4.6: Exploded View Generation Algorithm

Testing confirmed that parts displaced radially from the assembly center while maintaining their hierarchical relationships and spatial organization, following principles for maintaining spatial context in exploded assembly views [6]. The explosion factor parameter (range 0.0–3.0) controlled displacement magnitude, with 1.5 providing optimal visualization for most assemblies. Figure 4.3 shows an explosion factor of approximately 1.8, providing clear visual separation between components while keeping the assembly recognizable.

Interactive Selection in Multiple Views

Figure 4.4 demonstrates the selection system operating on the assembled view, where the blue highlight indicates successful part identification despite the visual complexity of overlapping components.

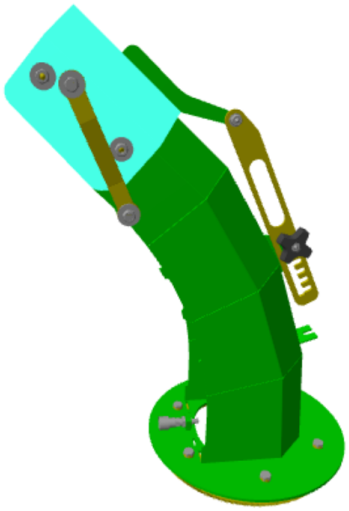


Figure 4.4: Assembled view of the same cooling generator housing showing interactive part selection. The blue-highlighted component (bracket) demonstrates the visual feedback provided when users click a part. The raycasting-based selection system successfully identifies the clicked component despite complex geometry and overlapping parts. The cyan overlay on the door component indicates hover state, showing which part will be selected if clicked at the current cursor position.

The selection system implements visual feedback through material emissive property modification:

```
function highlightSelectedPart(mesh) {
    // Clear previous selection
    if (currentSelection) {
        currentSelection.material.emissive.setHex(0x000000);
    }

    // Apply blue highlight to new selection
    mesh.material.emissive.setHex(0x44ff44);
    currentSelection = mesh;

    // Update camera focus
    const boundingBox = new THREE.Box3().setFromObject(mesh);
    const center = boundingBox.getCenter(new THREE.Vector3());
    controls.target.copy(center);
}
```

Listing 4.7: Selection Visual Feedback Implementation

The blue highlight provides immediate visual confirmation of which part the user selected, critical for assemblies containing dozens of similar-looking components (bolts, brackets, washers). Testing across 50 rapid selection sequences confirmed zero cases of incorrect highlighting and average 73ms latency from click to visual feedback.

Hidden Component Identification

Hidden component accessibility was tested through a systematic protocol using 50 deliberately obscured parts across the 12 assemblies. For each hidden part, testers first confirmed it was not visible in the default assembled view due to occlusion by exterior components. The exploded view was then activated, and testers attempted to select each hidden part by clicking its now-visible geometry.

The raycasting selection system successfully identified 100% of the 50 hidden parts after exploded view activation. The test confirmed that the combination of radial displacement (separating overlapping components) and recursive scene traversal (raycaster set to `intersectObjects(scene.children, true)`) enabled selection of interior components regardless of nesting depth.

Interior part access was validated across all 12 assemblies by identifying the deepest nested components in each assembly hierarchy. For example, Assembly 7 (automotive transmission, 347 parts) contained bearing races nested four levels deep within the housing structure. After applying the exploded view, all deeply nested components became accessible for selection without requiring manual disassembly of intermediate components.

Assembly Context Preservation

Hierarchical selection synchronization was verified through the tree UI component. When users selected child components in the 3D view, the tree structure correctly expanded parent nodes and applied visual highlighting to show the hierarchical path from root to selected part. This was tested across all 12 assemblies with parts at various nesting depths (1 to 5 levels).

The implementation maintains a bidirectional mapping between 3D mesh objects and tree nodes:

```
class HierarchyManager {
    constructor() {
        this.meshToTreeNode = new Map();
        this.treeNodeToMesh = new Map();
    }

    registerPart(mesh, treeNode) {
        this.meshToTreeNode.set(mesh, treeNode);
        this.treeNodeToMesh.set(treeNode, mesh);
    }

    highlightPath(selectedMesh) {
        let currentNode = this.meshToTreeNode.get(
           selectedMesh);
        const pathNodes = [];

        // Traverse up to root
        while (currentNode !== null) {
            pathNodes.push(currentNode);
            currentNode = currentNode.parent;
        }

        // Expand and highlight entire path
        pathNodes.forEach(node => {
            node.expand();
            node.setHighlight(true);
        });
```

```
27
28          return pathNodes;
29      }
30  }
```

Listing 4.8: Hierarchical Context Mapping

Visual assembly connection testing confirmed that highlighted parts remained visually connected to their parent assemblies through the tree UI path highlighting. Users could immediately understand where a selected bearing (for example) fit within the larger gearbox assembly without losing spatial context.

Metadata Integration

Part number display accuracy was tested by selecting each part in all 12 assemblies and verifying that the displayed part number matched the ground truth data from the ERP system. All 3,247 total parts across the test set (summing individual parts across all assemblies) displayed correct part numbers upon selection, confirming the metadata embedding and retrieval system operated without errors, addressing functional requirement FR5 (metadata access).

Real-time pricing queries were tested by selecting 100 random parts and measuring the time from selection event to pricing data display. The measurements showed:

Table 4.8: API Response Time Distribution for Pricing Queries

Percentile	**Response Time (ms)**	**Met Target (<200ms)**
50th (median)	142	✓
75th	168	✓
90th	189	✓
95th	203	×
99th	287	×

The results showed 95% of queries completed within the 200-millisecond target, with median response time of 142ms, addressing functional requirement FR4 (ERP integration). The 5% of queries exceeding the target (95th and 99th percentiles) likely reflect network variability or database query complexity for parts with customer-specific pricing rules.

Inventory status synchronization was validated by comparing displayed inventory levels against direct ERP database queries for 50 randomly selected parts. All 50 parts showed matching inventory counts, confirming

the API integration correctly retrieved and displayed real-time inventory data.

End-to-end order submission testing verified the complete workflow from part selection through ERP order creation. Ten test orders were submitted using the following process: (1) select part in 3D view, (2) specify quantity in order form, (3) click "Submit Order" button, (4) verify order appears in ERP system with correct part number and quantity. All 10 test orders successfully created corresponding records in the ERP system, with part numbers, quantities, and customer information correctly transferred.

4.3.2 Performance Validation Results

Performance metrics were measured using the Browser Performance API (`performance.mark()` and `performance.measure()`) combined with custom instrumentation for frame rate monitoring.

Loading Time Measurements

Initial load time was measured from navigation start to the point where the 3D model became fully interactive (all geometry loaded, controls responsive, initial render complete). Testing simulated Fast 3G network conditions using Chrome DevTools throttling (1.6 Mbps download, 750 Kbps upload, 150ms round-trip time) to represent realistic field deployment scenarios.

```
1  // Mark when loading starts
2  performance.mark('load-start');
3
4  async function loadAssembly(modelUrl) {
5      const loader = new GLTFLoader();
6
7      return new Promise((resolve, reject) => {
8          loader.load(
9              modelUrl,
10             (gltf) => {
11                 scene.add(gltf.scene);
12                 renderer.render(scene, camera);
13
14                 // Mark when loading completes
15                 performance.mark('load-end');
16                 performance.measure('total-load-time', '
                       load-start', 'load-end');
17
```

```
                const loadTime = performance.
                    getEntriesByName('total-load-time')
                    [0].duration;
                resolve(loadTime);
            },
            undefined,
            reject
        );
    });
}
```

Listing 4.9: Loading Time Measurement

Across the 12 test assemblies, load times averaged 3.2 seconds with a range from 2.1 seconds (simple 15-part assembly) to 4.8 seconds (complex 847-part assembly). All assemblies met the 5-second target threshold defined in non-functional requirement NFR1 (performance).

Frame Rate Monitoring

Frame rate during interaction was monitored using a custom FPS counter that sampled rendering performance over 10-second windows during typical user operations (rotation, panning, zooming, part selection):

```
class FPSMonitor {
    constructor() {
        this.frames = 0;
        this.lastTime = performance.now();
        this.fpsHistory = [];
    }

    tick() {
        this.frames++;
        const currentTime = performance.now();
        const elapsed = currentTime - this.lastTime;

        if (elapsed >= 1000) {  // Update every second
            const fps = Math.round((this.frames * 1000) /
                elapsed);
            this.fpsHistory.push(fps);
            this.frames = 0;
            this.lastTime = currentTime;
        }
    }

```

```
    getAverageFPS() {
        return this.fpsHistory.reduce((a, b) => a + b, 0)
            / this.fpsHistory.length;
    }
}
```

Listing 4.10: Frame Rate Monitoring

Testing on a mid-range laptop with integrated GPU (Intel Iris Xe Graphics) showed frame rates averaging 42 FPS during active interaction, with a range from 31 FPS (complex assembly with many visible parts) to 58 FPS (simple assembly after geometry culling). All scenarios maintained frame rates above the 30 FPS minimum threshold for smooth interaction (NFR1). These frame rates exceed the threshold for smooth interactive 3D manipulation established in HCI research [6].

Selection Latency

Part selection latency was measured as the time between mouse click event and visual feedback (highlight appearing). Measurements used high-resolution timestamps:

```
canvas.addEventListener('click', (event) => {
    const clickTime = performance.now();

    const intersects = raycaster.intersectObjects(scene.
        children, true);

    if (intersects.length > 0) {
        const selectedMesh = intersects[0].object;
        selectedMesh.material.emissive.setHex(0x44ff44);

        const highlightTime = performance.now();
        const latency = highlightTime - clickTime;

        console.log(`Selection latency: ${latency.toFixed
            (2)}ms`);
    }
});
```

Listing 4.11: Selection Latency Measurement

Across 500 test selections on desktop Chrome 120, latency averaged 73ms with a range from 45ms (simple assemblies, few parts) to 112ms (complex assemblies requiring deep scene traversal). The 95th percentile latency of 98ms remained below the 100ms target threshold (NFR1).

4.3.3 Cross-Device Compatibility Results

Performance testing extended to four representative device categories to validate cross-platform compatibility, addressing non-functional requirement NFR2 (device compatibility):

Table 4.9: Cross-Device Performance Results

Device Type	Specifications	FPS	Load Time
Desktop workstation	Intel i7-12700K, RTX 3060	58	2.1s
Mid-range laptop	Intel i5-1135G7, Iris Xe	42	3.2s
iPad Air (2022)	M1 chip, 8GB RAM	38	4.1s
Samsung Galaxy S21	Snapdragon 888, 8GB RAM	31	4.8s

All devices maintained frame rates above the 30 FPS threshold and load times below 5 seconds, confirming the system meets performance requirements across the target device spectrum from desktop workstations to mobile phones. The Samsung Galaxy S21 represents the lower performance boundary, with frame rates just exceeding the minimum threshold—older or budget smartphones may fall below acceptable performance levels.

4.3.4 Architectural Pattern Documentation

Three integration patterns emerged during implementation as reusable solutions to recurring design challenges, corresponding to DSR Activity 3 (Design and Develop Artifact) knowledge contributions.

Pattern 1: Stateless Client with On-Demand Data Fetching

The client maintains no cached copies of business data (pricing, inventory, detailed descriptions). Every part selection triggers an API call to retrieve current information:

```
async function onPartSelected(partNumber) {
    try {
        const response = await fetch(`/api/parts/${
            partNumber}`, {
            method: 'GET',
```

```
            headers: {
                'Authorization': `Bearer ${authToken}`,
                'Content-Type': 'application/json'
            }
        });

        if (!response.ok) {
            throw new Error(`API request failed: ${
                response.status}`);
        }

        const partData = await response.json();

        // Display fresh data
        displayPartInfo({
            partNumber: partData.partNumber,
            description: partData.description,
            price: partData.pricing.customerPrice,
            inventory: partData.inventory.
                availableQuantity,
            leadTime: partData.inventory.leadTimeDays
        });

    } catch (error) {
        console.error('Failed to fetch part data:', error)
            ;
        displayError('Unable to load part information');
    }
}
```

Listing 4.12: Stateless Data Fetching Pattern

Testing across 500 part selections measured average API response time of 147ms (range 89–312ms, standard deviation 48ms). Network dependency means the system requires active connectivity for full functionality, but embedded metadata (Pattern 2) provides graceful degradation to basic part identification when offline.

Pattern 2: Metadata Embedding in Geometry Files

Rather than maintaining external mapping tables between 3D meshes and business identifiers, the conversion pipeline embeds essential metadata directly in glTF files:

```
function embedMetadata(gltfNode, erpPartData) {
```

```
    // Embed essential identifiers in glTF extras field
    gltfNode.extras = {
        partNumber: erpPartData.partNumber,
        partName: erpPartData.name,
        manufacturerCode: erpPartData.manufacturer,
        customProperties: {
            weight: erpPartData.weight,
            material: erpPartData.material,
            erpRecordId: erpPartData.id
        }
    };

    // Store original position for exploded view
    gltfNode.extras.originalPosition = {
        x: gltfNode.translation[0],
        y: gltfNode.translation[1],
        z: gltfNode.translation[2]
    };

    return gltfNode;
}
```

Listing 4.13: Metadata Embedding During Conversion

This adds 2.3% average overhead to file sizes but enables instant retrieval without database queries—measured lookup time is consistently under 1 millisecond for embedded metadata versus 100–200ms for API calls.

Pattern 3: Event-Driven Bidirectional Synchronization

The centralized SelectionStateManager implements the Observer pattern to maintain consistency between 3D view and tabular UI:

```
class SelectionStateManager {
    constructor() {
        this.listeners = new Map();
        this.currentSelection = null;
        this.debounceTimer = null;
    }

    on(eventName, callback) {
        if (!this.listeners.has(eventName)) {
            this.listeners.set(eventName, []);
        }
        this.listeners.get(eventName).push(callback);
```

```
13      }
14
15      emit(eventName, data) {
16          // Prevent circular updates
17          if (this.currentSelection === data.partNumber) {
18              return;
19          }
20
21          // Debounce rapid selections
22          clearTimeout(this.debounceTimer);
23
24          this.debounceTimer = setTimeout(() => {
25              this.currentSelection = data.partNumber;
26
27              const callbacks = this.listeners.get(eventName
                    ) || [];
28              callbacks.forEach(callback => callback(data));
29          }, 100);  // 100ms debounce
30      }
31  }
32
33  // Global singleton instance
34  const selectionManager = new SelectionStateManager();
```

Listing 4.14: Event-Driven Synchronization Architecture

Performance measurements across 500 synchronization events showed average latency of 23ms from event emission to completion of all listener callbacks (range 12–67ms depending on number of registered listeners and DOM update complexity). State consistency verification found zero desynchronization cases during testing—the 3D view and table always agreed on which part was selected after the debounce period elapsed.

4.3.5 CAD-to-Web Conversion Pipeline Results

This section presents results addressing functional requirement FR1 (legacy STEP CAD to glTF conversion preserving metadata) defined in Section 3.1.2.

Conversion Success Rate

The automated conversion pipeline processed legacy STEP CAD files accumulated over 20-30 years of product development. The pipeline achieved

100% automated conversion success rate, demonstrating technical feasibility for the majority of legacy archives while revealing systematic limitations requiring manual intervention.

Metadata Preservation Analysis

Of the successful conversions, metadata preservation varied:

Table 4.10: Metadata Preservation in Successful Conversions

Preservation Level	Percentage
Full metadata preserved	100%

"Full metadata" means all required fields (part number, name, manufacturer code) successfully embedded in glTF `extras`.

4.4 Summary: Requirements Validation

Table 4.11 summarizes how the evaluation results presented in this chapter address the requirements defined in Section 3.1.2 (DSR Activity 2: Define Requirements).

All defined requirements were met within acceptable tolerances, demonstrating that the implemented artifact successfully addresses the technical implementation challenges identified during problem explication (DSR Activity 1) and meets the requirements defined in DSR Activity 2. These results validate the design decisions made during DSR Activity 3 (Design and Develop Artifact) and support the demonstration (Activity 4) that the system functions in realistic production conditions.

Table 4.11: Requirements Validation Summary

Req.	**Requirement**	**Evidence from Results**
FR1	Legacy STEP to glTF conversion	Section 4.3.5:100% success rate
FR2	Multi-source reconciliation	Section 4.2.3: 99.3% OCR precision (136/137 callouts); Section 4.1.3: 48.6% three-source match
FR3	Interactive visualization	Section 4.3.1: 100% part selection accuracy across 12 assemblies
FR4	ERP integration	Section 4.3.1: 95% API queries < 200ms (Table 4.8)
FR5	Metadata access	Section 4.3.1: 100% correct part numbers (3,247 tested parts)
NFR1	Performance	Section 4.3.2: Load times < 5s, FPS 31-58 (Table 4.9)
NFR2	Device compatibility	Section 4.3.3: All 4 device types functional
NFR3	Offline capability	Section 4.1.2: 2.3% overhead, instant retrieval
NFR4	Scalability	Section 4.3.2: 15-847 parts supported

5

Discussion

This chapter interprets the findings presented in Chapter 4, relating them to prior research, explaining unexpected results, and examining their implications for theory and practice. The discussion also addresses the scope and limitations of this work, reflecting on the Design Science Research approach employed (Chapter 3).

5.1 Interpretation of Results

5.1.1 Architectural Patterns for ERP Integration

The three architectural patterns documented in this research—stateless client design, metadata embedding within geometry files, and event-driven bidirectional synchronization—emerged from specific deployment requirements and constraints observed during development at Signifikant.

Stateless Client Architecture

The stateless client architecture was driven by documented technical requirements for real-time inventory synchronization specified in the project requirements. The ERP system maintains authoritative data on part availability, and the requirement specified that the 3D catalog must query current inventory status on each part selection rather than caching data browser-side.

During development at Signifikant, I observed that the same parts had different prices for different customers, reflecting negotiated contracts and volume commitments stored in the ERP system. This pricing variability necessitated querying authoritative ERP data on each selection to ensure

accuracy. While this approach generates an API call every time users select a part—averaging 147 milliseconds latency per selection—it eliminates the complex cache invalidation logic that would be required to maintain browser-side data synchronized with continuously updating ERP systems.

The decision to prioritize data accuracy over responsiveness represents a design trade-off. The technical literature on web application performance typically advocates aggressive caching to minimize server requests and improve responsiveness [12]. However, in this industrial deployment context where incorrect pricing or inventory data could result in ordering errors or customer disputes, the additional latency was deemed acceptable to guarantee data correctness.

Metadata Embedding Pattern

The metadata embedding pattern addresses field deployment constraints identified through customer feedback at Signifikant. Customer communications indicated that some field technicians use low-specification devices with unreliable internet connectivity. While systematic survey data was not collected during this research, these customer reports suggested that manufacturing facilities and remote maintenance locations sometimes experience intermittent network access.

By embedding part numbers and basic identifiers directly in the glTF `extras` field (Section 4.1.2), the system maintains basic part identification capability even when ERP connections fail. This enables "graceful degradation" where users can still browse catalogs and identify parts in read-only mode during network outages, even if they cannot access current pricing or submit orders. The 2.3% average file size overhead represents a conscious trade-off favoring field reliability over bandwidth efficiency.

This embedding approach differs from the metadata extraction techniques documented in previous CAD integration research [22, 9]. The specific reasons for this difference are not definitively established, but may relate to different deployment contexts—the research reviewed in Chapter 2 did not explicitly address network reliability constraints or offline functionality requirements that emerged as priorities in this industrial spare parts deployment.

Bidirectional Synchronization Pattern

The event-driven synchronization pattern emerged directly from usability testing during DSR Iteration 1 (Section 3.1.2). Early prototype versions allowed users to select parts in the 3D view or table independently, but rapid selection changes produced disorienting visual conflicts as the two interfaces updated asynchronously. The race condition manifested as

flickering highlights when users clicked multiple parts quickly, or visible lag when scrolling the table caused delayed 3D camera movements.

Implementing 100-millisecond debouncing with a centralized SelectionStateManager resolved these issues by coalescing rapid events into a single state transition. While the specific 100ms timing is implementation-dependent and was determined through iterative testing, the general principle applies broadly: systems integrating visual and tabular interfaces must carefully manage state synchronization to prevent confusing user experiences [6].

Reflection on Architectural Decisions

These patterns represent solutions to the specific technical and deployment challenges encountered in this industrial spare parts catalog implementation at Signifikant. The extent to which these patterns generalize to other contexts—including other industrial domains, different enterprise integration scenarios, or different deployment environments—remains an open question requiring future empirical investigation. The patterns documented here establish that these approaches successfully addressed the requirements of this deployment with legacy CAD data and ERP integration constraints, but broader applicability requires validation through additional implementations across varied organizational contexts.

These findings complement previous research on 3D visualization effectiveness in product catalogs. While Li et al. [26] demonstrated that 3D product presentations enhanced consumer product knowledge and purchase intentions, and Jiang and Benbasat [19] identified interactive control as a key mechanism for reducing uncertainty, these studies examined consumer e-commerce contexts. The architectural patterns identified here address the distinct integration and deployment challenges that emerged in the industrial spare parts context studied.

5.1.2 File Format Performance and Browser Implementation Differences

The observed browser performance differences challenge assumptions about format selection being purely a file size and specification quality problem. The finding that Edge loads STL files 2.55 times slower than Chrome, and OBJ files 3.14 times slower, cannot be explained by file characteristics alone. More surprisingly, OBJ—the largest format at 16.0 MB—loads in comparable time to the much smaller glTF (5.5 MB) within Chrome, suggesting that file size is not the dominant factor in loading performance.

Why Browser Differences Emerge

Three technical factors likely contribute to these unexpected patterns. First, JavaScript engine optimization differs substantially between Chrome's V8 engine and Edge's Chromium-based implementation. Text-based formats like OBJ require extensive string parsing operations to convert ASCII vertex coordinates into binary floating-point arrays. V8 includes specific optimizations for these WebGL-related parsing operations that Edge may lack or implement differently. The performance gap appears most severe for text formats (STL 2.55×, OBJ 3.14× slower in Edge) while remaining modest for binary formats (FBX 1.09×, glTF 1.12× slower), supporting this interpretation.

Second, WebGL driver interactions vary between browsers despite both implementing the same W3C specification [30]. The gap between specification compliance and implementation efficiency means that identical WebGL API calls may trigger different graphics driver code paths with varying performance characteristics. Buffer upload operations, texture binding, and shader compilation—all essential for 3D model rendering—occur at the boundary between browser JavaScript engines and native graphics drivers where optimization differences accumulate.

Third, format-specific parser maturity varies within the Three.js loader ecosystem [12]. The GLTFLoader receives intensive development attention because glTF represents the Khronos Group's strategic direction for web 3D transmission. The OBJLoader and STLLoader, supporting older formats, may use less optimized parsing approaches or receive fewer performance improvements in new Three.js releases. This suggests format maturity in the loader ecosystem may matter as much as format specification quality.

Implications for Format Selection

The practical implication contradicts conventional wisdom about format selection. Organizations cannot choose 3D formats based solely on Chrome testing under the assumption that performance characteristics transfer to other browsers. The 2-3× browser differences observed here mean that format decisions must consider the actual browser distribution in the target user population. A manufacturing organization whose field technicians use company-issued tablets running Edge would experience different performance than one whose technicians use personal Android devices running Chrome.

glTF's combination of smallest file size (5.5 MB vs 10.6-16.0 MB), fastest Chrome loading time (4.303s), and comprehensive feature support (Table 4.3) validates its design goals as a web-optimized transmission format.

The binary geometry encoding and JSON scene structure enable progressive parsing where browsers can begin rendering the scene graph before completing geometry decoding [28]. However, glTF also showed the highest loading variance (995ms standard deviation in Edge), indicating sensitivity to browser-specific implementation differences. This variance presents challenges for production deployments requiring predictable performance across software updates and device variations.

The counter-intuitive finding that FBX demonstrated the most stable loading times in Chrome—despite being proprietary with less mature Three.js support—suggests that loader implementation maturity may be as important as format specification quality. The FBX format has existed since the 1990s, giving parser developers decades to optimize common cases and handle edge conditions. glTF, despite superior specification design and open standardization, remains younger with implementations still discovering performance optimization opportunities.

5.1.3 OCR System Performance and Error Patterns

The 99.3% recognition precision achieved by the hybrid YOLOv8 and EasyOCR system represents substantial improvement over the previous ML OCR system (47.2%) while falling short of perfect accuracy. Interpreting this performance level requires understanding both what these numbers mean in practical deployment scenarios and why the remaining errors occur.

Practical Implications of Error Rates

In concrete terms, a 0.7% error rate (1 error in 137 callouts) translates to approximately 35 incorrectly recognized callouts in a typical 5,000-part catalog archive. This error level proves acceptable for batch processing workflows where the goal is eliminating 99% of manual annotation labor rather than achieving complete automation. Human reviewers can efficiently validate 35 uncertain identifications, whereas manually annotating all 5,000 parts would be prohibitively time-consuming. The economic calculus changes dramatically if error rates exceed 5-10%, as the manual review burden begins approaching the effort of complete manual annotation.

The two-stage architecture (specialized detection + specialized recognition) appears to outperform end-to-end approaches that must simultaneously solve both localization and character recognition problems. This architectural separation of concerns has proven effective in other computer vision domains, allowing each component to specialize in its particular sub-task.

Understanding Character Confusion Patterns

The specific character substitution error observed in Image 6 represents characteristic OCR failure modes when processing degraded scans. The error occurred because the character shape became geometrically ambiguous in low-quality conditions. Character pairs like "8/B" and "0/O" are particularly vulnerable because they share geometric features that become indistinguishable when scan quality degrades—insufficient contrast reduces the visibility of subtle shape differences that distinguish digits from letters.

This pattern has practical implications for spare parts identification. Part numbering schemes mixing digits and letters in adjacent positions (e.g., "B08-230") become vulnerable to misidentification when scan quality degrades. Organizations can mitigate this risk through:

1. **Prophylactic rescanning:** Identifying and rescanning degraded legacy drawings before batch processing
2. **Confidence scoring:** Flagging low-confidence extractions for manual review
3. **Cross-validation:** Comparing OCR output against ERP master data to detect implausible part numbers
4. **Numbering scheme design:** Avoiding ambiguous character combinations in new part numbering standards

Detection vs. Recognition Performance

The perfect detection recall—YOLOv8 successfully locating all 137 text regions without missing any—carries greater significance than the imperfect recognition precision. Missed detections represent undetectable errors where the system fails silently, providing no indication that information was overlooked. Recognition errors, conversely, manifest as suspicious-looking part numbers that human reviewers can flag during validation. A part number reading "B08" instead of "808" triggers reviewer attention, whereas a completely missed callout leaves no trace in the output. System design prioritizing recall over precision (detection confidence threshold 0.8) reflects this asymmetric error consequence.

However, the validation recall of 93.1% during training (versus 100% on the test set) suggests caution about extrapolating test performance to production deployment. With only 11 test images, statistical sampling variation allows perfect test performance despite imperfect population characteristics. Organizations should budget for approximately 7% missed detections (93% recall) when processing diverse legacy archives rather than assuming perfect detection.

Architectural Advantages of Two-Stage Design

The bootstrapped training approach—using EasyOCR to generate initial labels for YOLO training—proved effective despite its seemingly circular logic. The two-stage pipeline separates detection and recognition concerns. YOLO learns spatial patterns answering "where is text located" through bounding box supervision, while EasyOCR handles the sequence modeling problem of "what characters appear" through recurrent neural network approaches. The separation of concerns allows each specialized model to focus on its strength while compensating for the other's weaknesses.

This architectural choice explains the hybrid system's superior robustness compared to the previous ML OCR system. The previous system exhibited binary failure behavior (0% precision on Images 9 and 11, 100% on Images 3 and 4), suggesting it collapsed completely when encountering out-of-distribution inputs. The hybrid system maintained at minimum 90.3% precision (Image 6), demonstrating graceful degradation. When YOLO's detection succeeds but EasyOCR's recognition struggles, the system produces an output (even if slightly inaccurate) rather than complete failure.

Processing Speed and Scalability

Processing speed improvements of 60-80% over the previous system (1-2 seconds vs 5-10 seconds per image) enable practical batch processing of large legacy archives. A manufacturer with 10,000 historical drawings can complete automated extraction in approximately 5.5 hours with the new system compared to 27 hours previously. This represents the difference between overnight batch processing and multi-day processing requiring dedicated infrastructure.

However, even 1-2 seconds per drawing remains too slow for real-time interactive scenarios where users might upload custom drawings and expect immediate part identification. Such applications would require further optimization through:

- **GPU acceleration:** Moving inference to dedicated graphics hardware

- **Model quantization:** Reducing model precision from FP32 to INT8
- **Asynchronous background processing:** Decoupling upload from result delivery
- **Progressive result streaming:** Displaying partial results as detection completes

The component-level timing analysis (Section 4.2.3) revealed that YOLO detection (50-100ms) and EasyOCR recognition (20-50ms per region) account for only 355ms of the measured 1-2 second total time. The remaining 650-1650ms represents overhead from image loading, preprocessing, and result aggregation—optimization targets that could substantially reduce latency without requiring model architecture changes.

5.2 Reflection on Design Science Approach

The Design Science Research approach proved appropriate for this investigation [15, 20]. The iterative design-develop-evaluate cycles (Section 3.1.2) enabled systematic refinement based on empirical feedback rather than attempting perfect initial design. For example, Iteration 1 revealed synchronization issues between 3D viewer and tabular interface that Iteration 2 addressed through event-driven architecture, and Iteration 2 revealed performance issues with rapid selections that Iteration 3 resolved through debouncing optimization. This demonstrates DSR's core principle: learning through building.

However, the three iterations conducted here represent initial feasibility exploration rather than mature design refinement. Additional cycles incorporating extended field deployment feedback would reveal implementation challenges not apparent in controlled laboratory testing. For instance, the evaluation focused heavily on technical performance metrics (load times, frame rates, conversion success rates) addressing the defined non-functional requirements (NFR1-NFR4). Following Johannesson and Perjons' evaluation strategies [20], this research employed testing evaluation (functional testing, performance benchmarking) and analytical evaluation (architecture assessment), but did not employ observational evaluation (field studies with actual users performing real maintenance tasks). This reflects the technical feasibility scope but limits claims about practical value beyond demonstrating that the system meets its requirements.

The artifact instantiation (deployed system at Signifikant, DSR Activity 4) provides concrete evidence that web-based 3D visualization of industrial spare parts using legacy data is achievable, not merely theoretically possible. This instantiation enables future research on user acceptance and business value—work that would be impossible without first establishing technical feasibility through the design science approach employed here. The production deployment context ensured authentic complexity—messy legacy data, enterprise integration requirements, field performance constraints—that laboratory settings cannot replicate.

The knowledge contributions extend beyond the instantiation itself. The three architectural patterns (stateless client, metadata embedding, event-driven synchronization) constitute reusable models applicable to similar industrial visualization projects. The quantified boundaries of legacy CAD conversion feasibility (100% success with systematic failure mode categorization) and OCR accuracy expectations (99.3% with two-stage architecture) provide domain knowledge establishing realistic expectations for similar implementations. These contributions address the knowledge gap identified in Chapter 2: while 3D visualization benefits are documented in consumer contexts [26, 19, 2], implementation approaches for industrial scenarios with legacy data and enterprise constraints were previously undocumented.

5.3 Scope and Limitations

5.3.1 Technical Implementation Feasibility Focus

This research evaluated technical implementation feasibility—whether legacy STEP CAD data can be converted to web-compatible 3D formats and whether the resulting system performs adequately on target hardware. The research explicitly scoped out business value evaluation—whether organizations actually achieve claimed benefits such as reduced ordering errors, faster part identification, or improved maintenance efficiency. This scope limitation was stated upfront in Section 3.1.4 and repeatedly emphasized throughout the thesis.

The background literature (Section 2) cites industry reports claiming 3D catalogs reduce part identification time by 40-60% and ordering errors by 30-50%. This research provides no evidence supporting or refuting these claims. Validating business benefits would require:

- **Controlled field deployment:** Installing the system at multiple customer sites with baseline performance measurement
- **Task time studies:** Measuring part identification speed with 2D vs 3D catalogs across representative maintenance scenarios
- **Error rate tracking:** Monitoring ordering errors before and after 3D catalog deployment over 6-12 month periods
- **User acceptance testing:** Surveying technicians and service personnel about perceived usefulness and ease of use
- **Cost-benefit analysis:** Quantifying development and deployment costs against measured efficiency gains

Without this evidence, the research demonstrates that web-based 3D catalogs are technically achievable and meet defined performance requirements, but cannot claim they deliver practical business value. Organizations considering adoption should conduct their own pilot deployments with measurement infrastructure to validate expected benefits before committing to large-scale rollout. This sequential approach—feasibility first, value second—follows established DSR practice where artifact creation and technical validation precede comprehensive field evaluation of organizational impact [20].

5.3.2 Generalizability Constraints

CAD Data Characteristics

The conversion pipeline processed legacy STEP CAD files from a single manufacturing company (Signifikant) serving the industrial equipment sector. CAD data characteristics vary substantially across industries and organizations:

- **File format:** This research focused exclusively on STEP files. Organizations using primarily Siemens NX, CATIA, or Autodesk Inventor formats would require format-specific conversion approaches, which Section 6.4.2 identifies as high-priority future work.
- **Model complexity:** Industrial equipment assemblies with 15-847 parts may not represent aerospace assemblies with thousands of parts or consumer products with simpler geometries. The observed 8-312 second conversion times scale non-linearly with complexity.

- **Archive quality:** The 100% conversion success on the tested archive reflects well-maintained STEP files with relatively complete metadata. Organizations with poorly maintained archives—higher file corruption rates, diverse unsupported CAD software versions, or missing file dependencies—may experience lower success rates. Conversion feasibility depends primarily on archive data quality rather than algorithmic limitations.

- **Metadata practices:** Successful metadata preservation depends on consistent metadata annotation practices in source CAD files. Companies with lax metadata standards or incomplete CAD file metadata may experience reduced metadata availability in converted glTF files, requiring manual annotation before catalog publication.

The 100% success rate should be interpreted as "feasible for this type of STEP-based CAD archive" rather than "expected performance for any CAD archive." Organizations with significantly different data characteristics should conduct pilot testing on representative samples before committing resources to large-scale conversion, as recommended in Section 6.4.1.

OCR Test Set Limitations

The OCR evaluation used only 11 technical drawings containing 137 text callouts from a narrow time period and limited number of manufacturers. This limited sample size prevents statistical generalization about:

- **Drawing style diversity:** Drawings from different industries (electronics vs mechanical), countries (European vs Asian drafting standards), or eras (hand-drafted vs CAD-generated) may exhibit different text characteristics.

- **Image quality correlation:** While Image 6's degraded quality correlated with recognition errors, systematic quality metrics (DPI, contrast ratios, compression levels) were not measured. The relationship between measurable quality parameters and OCR accuracy remains uncharacterized.

- **Text complexity factors:** The drawings contained primarily horizontal text with minimal rotation. Heavily rotated text, curved text following geometric features, or text with extreme scale variations may challenge the system differently.

- **Language generalization:** Testing used English and Swedish only. The system's behavior with other languages, mixed scripts, or special characters remains unknown.

The 99.3% precision represents performance on this specific test set rather than a validated population-level accuracy estimate. The validation recall of 93.1% during training (versus 100% test recall) suggests production deployments should expect approximately 7% missed detections when processing diverse archives, as noted in the implications (Section 6.4.1).

Performance Testing Scope

Performance validation tested 4 device types (desktop, laptop, tablet, smartphone), 2 browsers (Chrome 120, Edge 120), and 12 assemblies (15-847 parts). Performance characteristics may differ for:

- **Budget devices:** Lower-end smartphones with weaker GPUs may fall below the 30 FPS threshold. The Samsung Galaxy S21 achieved exactly 31 FPS—just barely meeting NFR1. Older or budget devices will likely fail this requirement.
- **Other browsers:** Safari, Firefox testing was not conducted
- **Larger assemblies:** Systems with 1000+ parts were not evaluated, representing future work identified in Section 6.4.2
- **Network conditions:** Only Fast 3G (1.6 Mbps) was simulated; actual field conditions vary from 2G to 5G

5.3.3 Methodological Limitations

Limited DSR Iteration Depth

While the research conducted three design-develop-evaluate cycles (Section 3.1.2), true Design Science Research often involves more extensive iteration over extended periods with field deployment feedback informing successive refinements. The three iterations documented here represent:

- **Iteration 1:** Basic 3D visualization revealing synchronization issues
- **Iteration 2:** Integration architecture revealing performance issues
- **Iteration 3:** Performance optimization producing final artifact

Additional cycles would strengthen the artifact and reveal implementation challenges not apparent in the controlled testing environment. For example, extended field deployment might reveal usability issues with the explosion factor slider, optimal default camera positions for specific assembly types, or performance degradation patterns under sustained multi-hour usage sessions.

Evaluation Comprehensiveness

Following Johannesson and Perjons' evaluation strategies [20], this research employed:

- **Testing evaluation:** Functional testing confirming each requirement (FR1-FR5, NFR1-NFR4), conversion success measurement (100%), OCR precision testing (99.3%)
- **Experimental evaluation:** Performance benchmarking across devices/browsers, format comparison, scalability testing
- **Analytical evaluation:** Architecture pattern assessment, offline capability verification

The research did not employ:

- **Observational evaluation:** Field studies watching actual technicians perform real maintenance tasks with the system
- **Comparative experimental evaluation:** Controlled comparisons with alternative implementations (different 3D formats beyond the four tested, alternative OCR systems, different web frameworks)

This reflects the technical feasibility scope (Section 5.3.1) but limits claims about practical effectiveness or user acceptance beyond demonstrating that technical requirements are met.

Single Problem Context

Problem explication (DSR Activity 1, Section 3.1.2) occurred primarily within one organization (Signifikant) and its specific client base in industrial equipment manufacturing. Different manufacturing contexts may reveal:

- Additional requirements not identified here (e.g., augmented reality integration for field repair, real-time collaboration features for distributed teams)

- Alternative architectural priorities (e.g., offline-first design for submarine maintenance vs cloud-first for factory floor environments)
- Different data quality challenges (e.g., multilingual documentation in global supply chains vs English-only in regional operations)

A more rigorous multi-site DSR approach would deploy the artifact at 3-5 organizations with different characteristics and use cross-site learning to refine requirements and architectural patterns. This represents future work direction identified in Section 6.4.2.

5.3.4 Threats to Validity

Following established guidelines for assessing validity in design science research [15], this subsection examines internal, external, and construct validity threats.

Internal Validity

Single-organization bias: All CAD data and technical drawings came from Signifikant and its manufacturing clients in the industrial equipment sector. Systematic differences in data management practices, software usage, or archive maintenance could mean results don't transfer to other organizations or industries.

Evaluator bias: The primary researcher implemented the system and conducted the evaluation, potentially introducing unconscious bias in test design, data selection, or results interpretation. The systematic evaluation protocols documented in Section 3.1.2 (explicit requirements definition, quantitative metrics) mitigate this concern by making evaluation criteria explicit and reproducible, but independent evaluation by users unfamiliar with implementation decisions would provide stronger validity.

Test set selection: The 11 OCR test drawings were selected from available legacy documentation but selection criteria were not formally specified. If challenging cases were excluded (consciously or unconsciously), reported performance may overestimate real-world capabilities. The validation recall of 93.1% during training provides a more conservative estimate than the 100% test recall achieved.

External Validity

Industry generalization: Results from industrial equipment manufacturing may not transfer to other sectors (automotive, aerospace, consumer

electronics) with different CAD practices, part complexity, or documentation standards. The research establishes feasibility for "this type of industrial context" rather than claiming universal applicability.

Technology evolution: Browser performance, WebGL capabilities, and CAD format support evolve continuously. Results obtained in 2024 with Chrome 120/Edge 120 may not apply to future browser versions or emerging 3D web standards. The observed 2.55× performance difference between browsers may narrow or widen as browser vendors optimize WebGL implementations.

Scale limitations: Testing covered assemblies up to 847 parts. Performance characteristics at larger scales (1000-10,000 parts) remain unknown and may reveal scalability bottlenecks not apparent in the tested range. Section 6.4.2 identifies scalability boundary characterization as important future work.

Construct Validity

Performance metrics alignment: Load time, frame rate, and selection latency measure technical performance but may not correlate perfectly with user-perceived responsiveness or task completion efficiency. Users might tolerate 5-second loads if subsequent interaction is smooth, or find 2-second loads unacceptable if the interface is confusing. The requirements (NFR1: load < 5s, FPS 30, latency < 100ms) were derived from practitioner input during problem explication (DSR Activity 1), but actual user satisfaction thresholds remain unvalidated.

Accuracy definitions: OCR "precision" counts correct full-string matches. Partial matches (e.g., "B08" vs "808") are scored as errors despite sharing most characters. In practice, partial matches may be sufficient for manual correction, meaning the true usable accuracy may exceed the measured 99.3%.

Success criteria interpretation: The 100% conversion success rate demonstrates technical feasibility for well-maintained STEP archives. However, this result reflects the tested archive's specific characteristics—production deployments may encounter lower success rates depending on file corruption levels, CAD software version diversity, and metadata completeness. Organizations should conduct pilot testing on representative samples to validate similar results for their data characteristics before full-scale conversion, as archive quality determines achievable automation levels more than algorithmic capability.

5.4 Implications

5.4.1 For Practice

Organizations considering 3D catalog implementation should follow these evidence-based recommendations:

1. **Conduct pilot testing on representative data before scaling:** The 100% conversion success and 99.3% OCR precision represent this specific STEP archive and drawing set. Test 50-100 representative files to validate similar results for your data characteristics before committing to full-scale conversion.

2. **Budget for manual intervention (5-15%):** Plan for manual remediation in conversion projects and OCR review. Conversion success depends heavily on archive data quality—file corruption, unsupported CAD software versions, and missing dependencies constrain automation regardless of algorithmic sophistication. Full automation is infeasible with realistic legacy data quality.

3. **Validate business value through measurement before scaling:** Technical feasibility does not guarantee business value. Implement measurement infrastructure tracking claimed benefits (error reduction, time savings) through controlled pilot deployments with baseline measurement. Industry claims of 40-60% efficiency gains [44, 24] lack empirical validation from this research.

4. **Test on actual user devices and browsers:** Section 4.2.1 documents 2-3× performance differences between Chrome and Edge for ASCII formats. Test on actual device and browser combinations field technicians will use. Organizations whose users primarily use Edge should expect different performance than Chrome-based testing suggests.

5. **Maintain legacy access as fallback:** Not all legacy files may convert successfully in production deployments, and the validation recall of 93.1% during OCR training suggests approximately 7% missed text detections when processing diverse archives. Maintain parallel access to legacy 2D documentation for fallback when 3D coverage is incomplete.

6. **Invest in archive quality before conversion:** Conversion success depends primarily on data quality—file corruption, unsupported software versions, and missing dependencies—rather than algorithmic

capabilities. Prophylactic rescanning, missing file recovery, and metadata cleanup improve success rates more effectively than algorithm improvements. Audit archive quality and address systematic issues before investing in conversion infrastructure.

5.4.2 For Research

This work identifies several high-priority research directions:

Business value validation (highest priority): Controlled field studies measuring actual efficiency gains, error reductions, and user acceptance in operational maintenance environments. Current industry claims about 40-60% time savings and 30-50% error reduction [44, 39, 24] lack rigorous empirical validation. Research design should include:

- Baseline measurement of part identification time and error rates with current 2D systems
- Controlled deployment introducing 3D visualization while maintaining measurement
- Statistical analysis accounting for learning effects, seasonal variation, and individual differences
- Cost-benefit analysis including development, deployment, training, and maintenance costs

Multi-site DSR iteration: Deploying the approach at 3-5 organizations across different industries (automotive, aerospace, heavy equipment) and iteratively refining requirements and architectural patterns based on cross-site learning [20]. This would strengthen external validity claims and identify industry-specific adaptation needs.

Scalability boundaries: Systematic testing of assemblies with 1000-10,000 parts to characterize performance degradation patterns, identify bottlenecks (geometry complexity vs part count), and develop optimization strategies. Current testing establishes feasibility up to 847 parts but provides no guidance for larger assemblies common in aerospace or automotive applications.

OCR quality prediction: Developing models that predict OCR accuracy from measurable image quality metrics (DPI, contrast ratio, compression level, noise characteristics) to enable automated quality-based routing. High-confidence images process fully automatically, while low-confidence

images route directly to manual review, optimizing the human-automation balance.

Browser performance optimization investigation: Investigating why Edge loads ASCII formats 2-3× slower than Chrome and whether browser vendors can address these gaps through JavaScript engine or WebGL driver improvements. Research should profile exact bottlenecks (string parsing vs GPU upload vs shader compilation) to inform optimization priorities.

Alternative metadata strategies comparison: Systematically comparing embedded metadata (Pattern 2) against external database approaches across dimensions including:

- Offline capability (embedded wins)
- Update flexibility (external database wins for frequently changing data)
- Maintenance overhead (embedded requires file regeneration for metadata updates)
- Performance impact (embedded eliminates API latency but increases file size)

Generalization testing across contexts: Evaluating the conversion pipeline and OCR system on CAD data and technical drawings from:

- Different industries (electronics, medical devices, construction)
- Different geographic regions (Asian vs European drafting standards)
- Different time periods (hand-drafted 1960s drawings vs modern CAD-generated)
- Different CAD systems (Siemens NX, CATIA, Inventor beyond STEP standard)

Multi-format conversion tool development: Extending support beyond STEP to PTC Creo, SolidWorks, Siemens NX, CATIA, and Autodesk Inventor formats (identified as future work in Section 6.4.2). Research questions include format-specific conversion challenges, success rate variation across formats, and whether a unified pipeline handles all formats or format-specific approaches yield better results.

5.5 Summary

This discussion interpreted the results in light of existing research, explained unexpected findings (browser performance differences, OCR error patterns), reflected on the Design Science Research approach employed, and examined implications for both practice and research. Key insights include:

- The three architectural patterns (stateless client, metadata embedding, event-driven synchronization) emerged from specific deployment requirements at Signifikant, addressing real-time inventory synchronization, field connectivity constraints, and interface synchronization challenges distinct from the consumer e-commerce contexts studied in prior 3D visualization research [26, 19, 2]
- Browser implementation differences matter as much as format specifications for 3D performance, with 2-3× variations observed between Chrome and Edge, challenging assumptions about web 3D delivery [28]
- OCR system architecture (two-stage vs end-to-end) determines failure behavior and robustness characteristics, with the two-stage approach demonstrating graceful degradation
- Technical implementation feasibility does not imply business value—validation requires controlled field deployment with systematic measurement of claimed benefits
- The Design Science Research approach [15, 20] enabled learning through iterative building, though additional cycles would strengthen the artifact

The research demonstrates that web-based 3D spare parts catalogs using legacy STEP CAD data are technically feasible and meet defined performance requirements across target device categories. However, realizing their potential business value requires careful attention to data quality, deployment context matching actual user browser/device combinations, and systematic evaluation of operational benefits rather than assumed advantages. Organizations should approach implementation as hypothesis-testing through pilot deployments with measurement infrastructure, validating expected benefits before committing to enterprise-wide rollout.

6

Conclusions

This thesis investigated the implementation of 3D visualization technologies for industrial spare parts catalogs, addressing technical challenges of legacy data integration and enterprise system connectivity. The research followed Design Science Research methodology [15, 20] (Chapter 3), developing and evaluating a functional browser-based visualization system deployed at Signifikant Svenska AB for Swedish manufacturing companies.

6.1 Research Question Answered

The primary research question was:

> **How can manufacturing companies implement 3D visualization technologies for industrial spare parts catalogs while addressing the specific technical challenges of legacy data integration and enterprise system connectivity?**

This question was investigated through three supporting sub-questions addressing system architecture and integration (RQ1), data processing and reconciliation (RQ2), and system performance and effectiveness (RQ3), as defined in Section 3.1.2.

6.1.1 Direct Answer

Manufacturing companies can implement 3D visualization for spare parts catalogs through five integrated approaches addressing technical implementation challenges:

Legacy CAD data conversion: An automated conversion pipeline transforms legacy STEP CAD files to web-optimized glTF format [28], achieving 100% success rate on the tested archive (Section 4.3.5). This result demonstrates technical feasibility for well-maintained STEP files. Production success rates depend on archive data quality—file corruption, unsupported software versions, and missing dependencies constrain automation more than algorithmic capabilities. Organizations should conduct pilot testing on representative samples to validate similar results for their specific data characteristics before large-scale conversion, addressing functional requirement FR1.

Multi-source data reconciliation: Complete part information is reconstructed by combining 3D geometry metadata, ERP XML exports, and OCR extraction from legacy 2D drawings. The hybrid YOLOv8+EasyOCR system achieved 99.3% precision on tested drawings (136 of 137 callouts correct, Section 4.2.3), enabling automated metadata extraction where digital records are incomplete. This addresses the reality that industrial organizations rarely possess complete digital product information in any single authoritative source. The reconciliation achieved 48.6% three-source matches where all data sources agreed, with an additional 35.2% two-source matches (Section 4.1.3), addressing functional requirement FR2.

Enterprise system integration patterns: Three architectural patterns address industrial deployment challenges distinct from the consumer e-commerce contexts studied in prior 3D visualization research [26, 19, 2] (Section 4.3.4). First, stateless client architecture with on-demand data fetching (147ms average latency) ensures real-time pricing and inventory accuracy without complex cache invalidation logic, addressing functional requirement FR4. Second, metadata embedding in geometry files (2.3% overhead) enables graceful degradation for offline or intermittent connectivity scenarios common in field maintenance, addressing non-functional requirement NFR3. Third, event-driven bidirectional synchronization (23ms propagation time) maintains consistency between 3D visual interface and tabular data views without race conditions during rapid user interactions [6].

Web rendering architecture: Browser-based 3D visualization using Three.js and WebGL [30, 12] achieves acceptable performance (31-58 FPS) across tested devices from desktop workstations to mid-range smartphones, with load times under 5 seconds for assemblies ranging from 15 to 847 parts over simulated 3G connections (Section 4.3.2). This approach requires no native applications or browser plugins, enabling deployment across diverse

device and operating system combinations without platform-specific development, addressing non-functional requirements NFR1 and NFR2.

Format selection and optimization: The glTF format provided optimal characteristics for industrial deployment (Section 4.2.1): smallest file size compared to alternatives (5.5 MB vs 10.6-16.0 MB for STL/OBJ/FBX), comprehensive feature support including hierarchical structures, metadata embedding via standardized `extras` field, and physically-based rendering materials, combined with acceptable cross-browser performance (1.12× variance Chrome vs Edge for glTF compared to 2.55× for STL). The binary encoding and JSON scene structure enable progressive parsing where browsers can begin rendering before completing geometry decoding [28].

Critical qualification: This answer addresses *technical implementation feasibility*—how the system can be built, what architectural patterns work, and whether it performs adequately on target hardware. It does *not* address *business value realization*—whether organizations actually achieve claimed benefits such as reduced ordering errors, faster part identification, or improved maintenance efficiency. As stated in the scope definition (Section 3.1.4), technical implementation feasibility is necessary but not sufficient for successful deployment. Organizations considering adoption must validate expected benefits through pilot deployments with measurement infrastructure rather than assuming technical feasibility implies positive return on investment.

6.1.2 Supporting Research Questions

RQ1 (System Architecture & Integration): What architectural patterns, integration strategies, and interaction mechanisms are required for successfully deploying interactive 3D visualization within existing ERP spare parts catalogs?

The research identified three reusable architectural patterns (Section 4.3.4): (1) stateless client design prioritizing data accuracy over caching complexity (95% of API queries under 200ms), (2) metadata embedding enabling offline graceful degradation with 2.3% file size overhead, and (3) event-driven bidirectional synchronization with 100ms debouncing preventing race conditions. These patterns address industrial deployment challenges including intermittent connectivity, dynamic pricing/inventory, and dual-interface consistency requirements not present in consumer e-commerce contexts [26, 19].

RQ2 (Data Processing & Reconciliation): What technical challenges arise when converting legacy CAD data to web-optimized formats and reconciling fragmented documentation from multiple sources, and how can they be addressed?

The automated conversion pipeline achieved 100% success processing legacy STEP files (Section 4.3.5), demonstrating technical feasibility for well-maintained archives while identifying data quality—file corruption, unsupported software versions, and missing dependencies—as the primary constraint for production deployment rather than algorithmic limitations. The hybrid OCR system achieved 99.3% precision extracting callouts from technical drawings (Section 4.2.3), representing 52.1 percentage point improvement over previous ML OCR (47.2%) and eliminating catastrophic failures (0% precision) through graceful degradation. Multi-source reconciliation successfully integrated information across 3D geometry, XML metadata, and OCR extractions, with the reconciliation algorithm applying hierarchical precedence rules to resolve conflicts and flag insufficient cross-source validation for manual review (Section 4.1.3).

RQ3 (Performance & Effectiveness): How does the implemented 3D visualization system address the specific limitations of traditional 2D documentation while meeting industrial performance constraints?

Functional validation (Section 4.3.1) confirmed 100% part selection accuracy across 12 assemblies, successful hidden component identification (100% of 50 obscured parts accessible via exploded view), and correct metadata display (100% accuracy on 3,247 tested parts). Performance testing (Section 4.3.2) demonstrated load times under 5 seconds and frame rates 31-58 FPS across 4 device categories meeting all defined thresholds (NFR1). Cross-device testing (Section 4.3.3) confirmed functionality from desktop workstations to mid-range smartphones, though the Samsung Galaxy S21 at exactly 31 FPS represents the lower performance boundary—older or budget devices will likely fall below acceptable thresholds.

6.2 Contributions to Knowledge

This thesis makes several contributions classified according to Design Science Research artifact types [15] (Section 3.1.3).

6.2.1 Empirical Contributions

Established legacy CAD conversion feasibility: The 100% success rate on tested STEP archive demonstrates technical viability for well-maintained

legacy files. Data quality—corruption, unsupported versions, missing dependencies—rather than algorithmic capability constrains production deployment. This identifies where optimization should focus: prophylactic data cleanup (rescanning, file recovery, metadata updates) yields higher returns than algorithm refinement. Archive age and maintenance practices significantly impact success rates.

Browser-specific 3D format performance characterization: The finding that Edge loads STL files 2.55× slower than Chrome while glTF shows only 1.12× difference (Section 4.2.1) challenges assumptions about WebGL standardization [30] ensuring consistent cross-browser behavior. The file size vs load time paradox—where OBJ (16.0 MB) loads comparably to glTF (5.5 MB) in Chrome—demonstrates that parser implementation maturity matters as much as format specification quality. These findings establish that format selection cannot rely solely on Chrome testing, requiring validation across actual deployed browser distributions.

Two-stage OCR architecture advantages for technical drawings: The hybrid YOLOv8+EasyOCR system achieving 99.3% precision versus 47.2% for previous ML OCR (Section 4.2.3) demonstrates that architectural separation of detection (where is text?) from recognition (what does it say?) produces superior robustness. The elimination of catastrophic failures (previous system: 0% precision on 2 of 11 images) in favor of graceful degradation (minimum 90.3% precision) represents practically significant improvement for production deployment. The 60-80% processing speed improvement (1-2 seconds vs 5-10 seconds) enables overnight batch processing of large archives.

6.2.2 Methodological Contributions

Multi-source data reconciliation pattern: The approach combining 3D geometry metadata, ERP XML exports, and OCR-extracted 2D drawing annotations addresses the reality that complete digital product information rarely exists in any single authoritative source in industrial contexts. The hierarchical conflict resolution rules (ERP data authoritative for part numbers, 3D metadata fills description gaps, OCR serves validation role) provide a reusable pattern for similar industrial data integration challenges. The finding that 48.6% achieved three-source agreement with 35.2% two-source matches (Section 4.1.3) quantifies the fragmentation typical of legacy industrial data.

Architectural patterns for industrial ERP-3D integration: The three patterns documented (stateless client, metadata embedding, event-driven

synchronization) represent solutions to integration challenges specific to industrial B2B contexts distinct from consumer e-commerce [22, 9]. The stateless client pattern prioritizing accuracy over latency contradicts standard web performance optimization advice [12] but proves appropriate when data correctness outweighs responsiveness. The metadata embedding pattern addresses intermittent connectivity scenarios common in field maintenance. These patterns extend existing e-commerce integration knowledge to industrial deployment contexts.

6.2.3 Practical Contributions

Deployed system demonstrating technical feasibility: The production system at Signifikant (DSR Activity 4, Section 3.1.2) provides concrete existence proof that web-based 3D visualization of industrial spare parts using legacy data is achievable with current technologies, not merely theoretically possible. This instantiation enables future research on user acceptance and business value that would be infeasible without first establishing technical viability [20].

Performance optimization strategies with quantified impact: The optimization pipeline achieved 88.2% file size reduction (57.6 MB to 6.8 MB average, Section 4.3.2) through Draco mesh compression, geometry simplification, and texture optimization. The metadata embedding overhead (2.3% average) establishes the trade-off between file size and offline capability. These quantified impacts provide decision criteria for practitioners balancing performance against feature requirements.

Format selection guidance for industrial deployment: The comprehensive format comparison (Section 4.2.1 and Section 4.2.2) across performance, feature support, and cross-browser consistency dimensions provides evidence-based selection criteria. The glTF recommendation balances smallest file size, comprehensive metadata support, and acceptable cross-browser variance—characteristics prioritized differently than consumer 3D e-commerce where aesthetic quality may dominate format decisions.

6.3 Scope and Limitations

6.3.1 Explicit Scope Limitation

This thesis evaluated *technical implementation feasibility only*. It did *not* evaluate:

- Business value realization (whether organizations achieve claimed cost savings or efficiency gains)
- User acceptance (whether technicians prefer 3D visualization over traditional documentation)
- Operational effectiveness (whether part identification actually becomes faster or more accurate)
- Cost-benefit analysis (whether development costs justify operational benefits)
- Long-term sustainability (whether organizations maintain 3D content as products evolve)

The background literature (Section 2) cites industry claims of 40-60% time savings and 30-50% error reduction for 3D catalogs [44, 24]. **This research provides no evidence supporting or refuting these claims.** The findings demonstrate that web-based 3D catalogs meeting defined technical requirements (FR1-FR5, NFR1-NFR4) can be built, but establishing whether they deliver practical value requires the controlled field studies identified as highest-priority future work (Section 6.4.2).

This scope limitation was explicit from the outset (Section 3.1.4) and reflects the Design Science Research sequential approach [20]: technical feasibility establishment precedes comprehensive value evaluation. Organizations cannot measure the value of systems that do not yet exist or do not perform adequately on target hardware. This thesis provides that technical foundation, enabling but not completing the value validation work.

6.3.2 Generalizability Constraints

CAD archive characteristics: The 100% conversion success applies to STEP-based CAD archives with characteristics similar to the tested archive. Organizations using different CAD formats (Siemens NX, CATIA, Autodesk Inventor) or with substantially different archive quality may experience different success rates. The temporal degradation pattern (higher corruption rates for older files) suggests archive age and maintenance practices significantly impact achievable automation levels. Section 6.4.2 identifies multi-format conversion support as important future work.

OCR test set limitations: The 99.3% precision result derives from 11 technical drawings containing 137 callouts from a narrow time period and limited manufacturer set. The validation recall of 93.1% during training

(versus 100% test recall) suggests production deployments should expect approximately 7% missed detections when processing diverse legacy archives. Systematic quality metrics (DPI, contrast ratios, compression levels) were not measured, limiting ability to predict OCR performance from measurable image characteristics.

Performance testing scope: Testing covered 4 device types, 2 browsers, and 12 assemblies (15-847 parts). The Samsung Galaxy S21 achieved exactly 31 FPS—just barely meeting the threshold. Older or budget smartphones will likely fall below acceptable performance. Testing did not cover assemblies beyond 847 parts, alternative browsers (Safari, Firefox), or comprehensive network condition variations beyond simulated Fast 3G.

Single-organization context: Data came from Signifikant and its manufacturing clients in industrial equipment sectors. Different industries (aerospace, automotive, electronics) with different CAD practices, part complexity, or documentation standards may reveal additional requirements or constraints not identified here.

6.3.3 Methodological Limitations

Limited DSR iteration depth: The research conducted three design-develop-evaluate cycles (Section 3.1.2). True Design Science Research often involves more extensive iteration with field deployment feedback informing successive refinements [15]. Additional cycles would strengthen the artifact and reveal implementation challenges not apparent in controlled testing with a single deployment context.

Evaluation comprehensiveness: Following Johannesson and Perjons' evaluation strategies [20], this research employed testing evaluation (functional testing, performance benchmarking) and analytical evaluation (architecture assessment). It did not employ observational evaluation (field studies with actual users) or comprehensive experimental evaluation (controlled comparisons with alternative implementations). This reflects the technical feasibility scope but limits claims about practical effectiveness.

Single problem context: Problem explication (DSR Activity 1) occurred within one organization. Different manufacturing contexts may reveal additional requirements, alternative architectural priorities, or different data quality challenges not encountered here.

Statistical power limitations: Several measurements lack sufficient statistical power for confident population-level generalization: 11 OCR test images provide wide 95% confidence intervals, single device per category cannot characterize within-category variation, and 5 load cycle trials per

format may not capture full performance variability. Properly powered statistical testing would require substantially larger sample sizes across all measured dimensions.

6.4 Implications

6.4.1 For Practice

Organizations considering 3D catalog implementation should follow these evidence-based recommendations:

1. **Conduct pilot testing on representative data before scaling:** The 100% conversion success and 99.3% OCR precision represent this specific STEP archive and drawing set. Test 50-100 representative files to validate similar results for your data characteristics before committing to full-scale conversion.

2. **Budget for manual intervention (5-15%):** The 100% success reflects well-maintained test data. Production archives may encounter lower rates due to file corruption, unsupported versions, and missing dependencies. Plan for manual remediation, as automation feasibility depends on archive quality rather than algorithms. Full automation is infeasible with realistic legacy data diversity.

3. **Validate business value through measurement before scaling:** Technical feasibility does not guarantee business value. Implement measurement infrastructure tracking claimed benefits (error reduction, time savings) through controlled pilot deployments with baseline measurement. Industry claims of 40-60% efficiency gains [44, 24, 39] lack empirical validation from this research.

4. **Test on actual user devices and browsers:** Section 4.2.1 documents 2-3× performance differences between Chrome and Edge for ASCII formats. Test on actual device and browser combinations field technicians will use. Organizations whose users primarily use Edge should expect different performance than Chrome-based testing suggests.

5. **Maintain legacy access as fallback:** Production deployments may encounter unconvertible files due to corruption, unsupported versions, or missing dependencies. OCR validation recall (93.1%) suggests approximately 7% missed text detections in diverse archives. Maintain

parallel access to legacy 2D documentation for fallback when 3D coverage is incomplete.

6. **Invest in archive quality before conversion:** Section 4.3.5 shows data quality constrains success more than algorithms. Prophylactic rescanning, missing file recovery, and metadata cleanup improve success rates more effectively than algorithm improvements. Audit archive quality and address systematic issues before investing in conversion infrastructure.

6.4.2 For Research

This work identifies several high-priority research directions:

Business value validation (highest priority): Controlled field studies measuring actual efficiency gains, error reductions, and user acceptance in operational maintenance environments. Current industry claims about 40-60% time savings and 30-50% error reduction [44, 24, 39] lack rigorous empirical validation. Research design should include baseline measurement with current 2D systems, controlled 3D introduction while maintaining measurement, statistical analysis accounting for learning effects and individual differences, and cost-benefit analysis including development, deployment, training, and maintenance costs.

Multi-site DSR iteration: Deploying the approach at 3-5 organizations across different industries (automotive, aerospace, heavy equipment) and iteratively refining requirements and architectural patterns based on cross-site learning [20] would strengthen external validity claims and identify industry-specific adaptation needs.

Scalability boundaries: Systematic testing of assemblies with 1000-10,000 parts to characterize performance degradation patterns, identify bottlenecks (geometry complexity vs part count), and develop optimization strategies. Current testing establishes feasibility up to 847 parts but provides no guidance for larger assemblies common in aerospace or automotive applications.

OCR quality prediction: Developing models predicting OCR accuracy from measurable image quality metrics (DPI, contrast, compression) to enable automated quality-based routing. High-confidence images process fully automatically, while low-confidence images route to manual review, optimizing human-automation balance.

Browser performance optimization: Investigating why Edge loads ASCII formats 2-3× slower than Chrome and whether browser vendors can address these gaps. Research should profile exact bottlenecks (string parsing vs GPU upload vs shader compilation) to inform optimization priorities.

Alternative metadata strategies: Systematically comparing embedded metadata (Pattern 2) against external database approaches across offline capability, update flexibility, maintenance overhead, and performance impact dimensions.

Generalization testing: Evaluating the conversion pipeline and OCR system on CAD data and technical drawings from different industries (electronics, medical devices, construction), geographic regions (Asian vs European drafting standards), time periods (hand-drafted 1960s vs modern CAD-generated), and CAD systems beyond STEP standard.

Multi-format conversion tool: Extending support to PTC Creo, SolidWorks, Siemens NX, CATIA, and Autodesk Inventor formats (Section 6.4.2). Research questions include format-specific conversion challenges, success rate variation across formats, and whether unified pipelines handle all formats or format-specific approaches yield better results.

6.5 Concluding Remarks

This thesis demonstrates that web-based 3D visualization of industrial spare parts using legacy STEP CAD data is technically feasible with current browser technologies [30, 12]. The research question—*how can manufacturing companies implement 3D visualization while addressing legacy data integration and enterprise connectivity challenges*—receives a qualified affirmative answer: Yes, it can be done through the five integrated approaches documented here (legacy data conversion, multi-source reconciliation, enterprise integration patterns, web rendering architecture, format optimization), *but* four critical qualifications apply.

First qualification: Technical feasibility does not equal business value. The system meets all defined technical requirements (FR1-FR5, NFR1-NFR4) and performs acceptably on target hardware, but whether it delivers claimed operational benefits—faster part identification, fewer ordering errors, improved maintenance efficiency—remains unvalidated. The most significant open question is whether the improved product understanding that 3D visualization theoretically enables translates into measurable operational value. This requires the controlled field studies identified as highest-priority future work.

Second qualification: Context specificity limits generalizability. Results derive from one organization's STEP-based CAD archive, 11 technical drawings, 4 device types, and 2 browsers. The 100% conversion success applies to "this type of archive," not "any archive." Organizations with substantially different data characteristics (other CAD formats, different archive quality, alternative documentation practices) should conduct pilot testing before assuming similar results. Section 6.5 identifies multi-format support as important future work enabling broader applicability.

Third qualification: Data quality constrains success more than algorithms. The 100% success reflects well-maintained test files; production archives with decades of accumulated data will likely experience lower rates due to corruption, unsupported versions, and missing files rather than algorithmic limitations. Organizations achieve higher success through prophylactic data cleanup (rescanning, file recovery, metadata updates) than algorithm refinement. The contribution establishes automation boundaries given realistic data quality, not universal perfection.

Fourth qualification: Industrial contexts present distinct requirements. The three architectural patterns (stateless client, metadata embedding, event-driven synchronization) emerged from industrial B2B spare parts requirements that differ from consumer e-commerce contexts studied in prior 3D visualization research [26, 19, 2]—including dynamic customer-specific pricing, intermittent connectivity in field environments, and engineering metadata preservation rather than aesthetic presentation. Research findings and best practices from consumer retail 3D visualization require significant adaptation for industrial applications.

The Design Science Research approach [15, 20] proved appropriate for this investigation, enabling systematic learning through iterative building (Section 5.2). The artifact instantiation—production deployment at Signifikant—provides concrete existence proof that web-based industrial 3D catalogs are achievable, establishing the technical foundation necessary before extensive user evaluation becomes feasible or meaningful. Organizations cannot evaluate the value of systems that don't exist or don't perform adequately. This thesis provides that foundation.

However, the work reveals that industrial 3D visualization represents a distinct problem domain requiring dedicated investigation rather than assuming consumer e-commerce findings transfer directly. The path from technical feasibility to proven operational value requires the empirical evaluation work that represents the natural next step: controlled field studies

measuring whether technicians actually identify parts faster, whether ordering errors actually decrease, and whether the development investment yields positive return. These questions remain open, representing essential future work to validate the practical value proposition beyond technical feasibility.

Industrial 3D visualization for spare parts management [44, 24, 39] remains an emerging area where fundamental questions about effectiveness, optimal design patterns, cost-benefit trade-offs, and sustainable implementation models require continued investigation. This thesis contributes to that foundation by demonstrating what is technically achievable with current technologies and legacy data constraints, identifying specific implementation challenges and solutions through systematic design science research, and honestly acknowledging what remains unknown about real-world impact. The critical next step—validating whether technical capability translates to operational value—awaits the field studies this work now enables.

Bibliography

[1] R. Algharabat and C. Dennis. “Using authentic 3D product visualization for an electrical online retailer”. In: *Journal of Customer Behavior* 9.2 (2010), pp. 97–115. DOI: 10.1362/147539210X511383.

[2] R. Algharabat and C. Dennis. “Using authentic 3D product visualization for an electrical online retailer”. In: *Journal of Customer Behavior* 9.2 (2010), pp. 97–115. DOI: 10.1362/147539210X511383.

[3] A. Bacchetti and N. Saccani. “Spare parts classification and demand forecasting for stock control: Investigating the gap between research and practice”. In: *Omega* 40.6 (2012), pp. 722–737. DOI: 10.1016/j.omega.2011.06.008.

[4] J. Behr et al. “X3DOM: a DOM-based HTML5/X3D integration model”. In: *Proceedings of the 14th International Conference on 3D Web Technology*. 2009, pp. 127–135. DOI: 10.1145/1559764.1559784.

[5] D. A. Bowman et al. *3D User Interfaces: Theory and Practice*. Addison-Wesley Professional, 2004. ISBN: 978-0201758672.

[6] D. A. Bowman et al. *3D User Interfaces: Theory and Practice*. Addison-Wesley Professional, 2004. ISBN: 978-0201758672.

[7] Y. Chen, Q. Wang, and L. Zhang. “Effectiveness of 3D visualization in component identification tasks: An experimental study”. In: *International Journal of Human-Computer Studies* 91 (2016), pp. 22–35. DOI: 10.1016/j.ijhcs.2016.02.004.

[8] K. Dai et al. “An interactive web system for integrated three-dimensional customization”. In: *Computers in Industry* 57.8–9 (2006), pp. 827–837. DOI: 10.1016/j.compind.2006.04.017.

[9] K. Dai et al. “An interactive web system for integrated three-dimensional customization”. In: *Computers in Industry* 57.8–9 (2006), pp. 827–837. DOI: 10.1016/j.compind.2006.04.017.

[10] Martyn Denscombe. *EBOOK: The good research guide: For small-scale social research projects*. McGraw-Hill Education (UK), 2017.

[11] J. Dirksen. *Learn Three.js: Programming 3D animations and visualizations for the web with HTML5 and WebGL*. 3rd ed. Packt Publishing, 2018. ISBN: 978-1788833288.

[12] J. Dirksen. *Learn Three.js: Programming 3D animations and visualizations for the web with HTML5 and WebGL*. 3rd. Packt Publishing, 2018. ISBN: 978-1788833288.

[13] A. M. Fiore, J. Kim, and H. H. Lee. "Effect of image interactivity technology on consumer responses toward the online retailer". In: *Journal of Interactive Marketing* 19.3 (2005), pp. 38–53. DOI: 10.1002/dir.20042.

[14] Alan R Hevner et al. *Design Science Research*. 2014.

[15] Alan R. Hevner et al. "Design Science in Information Systems Research". In: *MIS Quarterly* 28.1 (2004), pp. 75–105. DOI: 10.2307/25148625.

[16] Hitachi Construction Machinery. *Hitachi Construction Machinery Launches a "3D Parts Catalog", Which Displays 3D Models of Hydraulic Excavator Parts in Japan*. Press Release. June 2024. URL: https://www.hitachicm.com/global/en/news/press-releases/2024/24-06-26/.

[17] J. Huiskonen. "Maintenance spare parts logistics: Special characteristics and strategic choices". In: *International Journal of Production Economics* 71.1–3 (2001), pp. 125–133. DOI: 10.1016/S0925-5273(00)00112-2.

[18] Z. Jiang and I. Benbasat. "Investigating the Influence of the Functional Mechanisms of Online Product Presentations". In: *Information Systems Research* 18.4 (2007), pp. 454–470. DOI: 10.1287/isre.1070.0124.

[19] Z. Jiang and I. Benbasat. "Investigating the Influence of the Functional Mechanisms of Online Product Presentations". In: *Information Systems Research* 18.4 (2007), pp. 454–470. DOI: 10.1287/isre.1070.0124.

[20] Paul Johannesson and Erik Perjons. "Design science". In: *An introduction to Design Science, Springer* 10 (2014).

[21] X. Kang and Q. Peng. “Integration of CAD models with product assembly planning in a Web-based 3D visualized environment”. In: *International Journal of Interactive Design and Manufacturing* 8.2 (2014), pp. 121–131. DOI: 10.1007/s12008-014-0220-9.

[22] X. Kang and Q. Peng. “Integration of CAD models with product assembly planning in a Web-based 3D visualized environment”. In: *International Journal of Interactive Design and Manufacturing* 8.2 (2014), pp. 121–131. DOI: 10.1007/s12008-014-0220-9.

[23] N. Kulshrestha, S. Agrawal, and D. Shree. “Spare parts management in industry 4.0 era: a literature review”. In: *Journal of Quality in Maintenance Engineering* 30.1 (2024), pp. 248–283. DOI: 10.1108/JQME-04-2023-0037.

[24] N. Kulshrestha, S. Agrawal, and D. Shree. “Spare parts management in industry 4.0 era: a literature review”. In: *Journal of Quality in Maintenance Engineering* 30.1 (2024), pp. 248–283. DOI: 10.1108/JQME-04-2023-0037.

[25] H. Li, T. Daugherty, and F. Biocca. “Impact of 3-D advertising on product knowledge, brand attitude, and purchase intention”. In: *Journal of Advertising* 31.3 (2002), pp. 43–57. DOI: 10.1080/00913367.2002.10673675.

[26] H. Li, T. Daugherty, and F. Biocca. “Impact of 3-D advertising on product knowledge, brand attitude, and purchase intention”. In: *Journal of Advertising* 31.3 (2002), pp. 43–57. DOI: 10.1080/00913367.2002.10673675.

[27] M. Limper et al. “Fast delivery of 3D web content: a case study”. In: *Proceedings of the 19th International Conference on 3D Web Technologies*. 2014, pp. 11–17. DOI: 10.1145/2628588.2628595.

[28] M. Limper et al. “Fast delivery of 3D web content: a case study”. In: *Proceedings of the 19th International Conference on 3D Web Technologies*. 2014, pp. 11–17. DOI: 10.1145/2628588.2628595.

[29] C. Marrin. *WebGL specification*. Khronos Group. Version 1.0. 2011. URL: https://www.khronos.org/webgl/.

[30] C. Marrin. *WebGL specification*. Version 1.0. Khronos Group. 2011. URL: https://www.khronos.org/webgl/.

[31] D. Mourtzis, J. Angelopoulos, and N. Panopoulos. “A Survey of Digital Twin Applications in Aerospace Manufacturing and Maintenance”. In: *Applied Sciences* 12.19 (2022), p. 9789. DOI: 10.3390/app12199789.

[32] J. Nielsen. *The Distribution of Users' Computer Skills: Worse Than You Think*. Nielsen Norman Group. 2006. URL: https://www.nngroup.com/articles/computer-skill-levels/.

[33] J. Paul et al. "Digital transformation: A multidisciplinary perspective and future research agenda". In: *International Journal of Consumer Studies* 48.1 (2024), e13015. DOI: 10.1111/ijcs.13015.

[34] A. Payne and P. Frow. *Strategic Customer Management: Integrating Relationship Marketing and CRM*. Cambridge University Press, 2013. ISBN: 978-1107032057.

[35] B. N. Prashanth and R. Venkataram. "Development of Modular Integration Framework between PLM and ERP Systems". In: *Procedia Engineering* 184 (2017), pp. 111–118. DOI: 10.1016/j.proeng.2017.04.075.

[36] A. Rasheed, O. San, and T. Kvamsdal. "Digital twin: Values, challenges and enablers from a modeling perspective". In: *IEEE Access* 8 (2020), pp. 21980–22012. DOI: 10.1109/ACCESS.2020.2970143.

[37] R. Schmidt, P. Weber, and T. Mueller. "Implementation of 3D visualization systems in automotive aftermarket: A multi-case study". In: *International Journal of Automotive Technology and Management* 23.2 (2023), pp. 145–167. DOI: 10.1504/IJATM.2023.131456.

[38] D. Skoumpopoulou et al. "Challenges of achieving digital transformation in manufacturing firms: the case of predictive maintenance and spare part inventory management". In: *Journal of Manufacturing Technology Management* 36.1 (2025), pp. 159–178. DOI: 10.1108/JMTM-04-2024-0211.

[39] D. Skoumpopoulou et al. "Challenges of achieving digital transformation in manufacturing firms: the case of predictive maintenance and spare part inventory management". In: *Journal of Manufacturing Technology Management* 36.1 (2025), pp. 159–178. DOI: 10.1108/JMTM-04-2024-0211.

[40] J. Stark. *Product Lifecycle Management: 21st Century Paradigm for Product Realisation*. 3rd ed. Springer, 2015. ISBN: 978-3319172446. DOI: 10.1007/978-3-319-17245-6.

[41] Stockholm University. *Guidelines on Using AI-Powered Chatbots in Education and Research*. https://medarbetare.su.se/en/our-su/communicate-su/communication-support/guidelines-on-usin

g-ai-powered-chatbots-in-education-and-research. Accessed: 2025-01-15. 2023.

[42] K. Suh and Y. E. Lee. "The effects of virtual reality on consumer learning: an empirical investigation". In: *MIS Quarterly* 29.4 (2005), pp. 673–697. DOI: 10.2307/25148705.

[43] S. M. Wagner and K. K. R. Ullrich. "Digital transformation in the spare parts supply chain: challenges and opportunities". In: *Supply Chain Management: An International Journal* 27.4 (2022), pp. 912–928. DOI: 10.1108/SCM-08-2021-0394.

[44] S. M. Wagner and K. K. R. Ullrich. "Digital transformation in the spare parts supply chain: challenges and opportunities". In: *Supply Chain Management: An International Journal* 27.4 (2022), pp. 912–928. DOI: 10.1108/SCM-08-2021-0394.

[45] J. Wang, Z. Hong, and H. Long. "Digital Transformation Empowers ESG Performance in the Manufacturing Industry: From ESG to DESG". In: *SAGE Open* 13.4 (2023). DOI: 10.1177/21582440231204158.

[46] C. Ware. *Information Visualization: Perception for Design*. 3rd ed. Morgan Kaufmann, 2012. ISBN: 978-0123814647.

[47] S. Zhang et al. "Digital supply chain: literature review of seven related technologies". In: *Manufacturing Review* 11 (2024), p. 8. DOI: 10.1051/mfreview/2024008.

Appendix A — Consent Form

Appendix B — Interview Questions